

Revision Book

MATH1821 & MATH2822:
Mathematical Methods for Actuarial Science I&II
2024-2025

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Chapter 1

Introduction

This revision book is a collection of materials for all revision classes in 2024-2025 1st and 2nd semester of the course MATH1821 and MATH2822. Students may use this book for knowledge review or practice in the future. The suggested solutions are attached right after the questions.

Some of you may use this book as a reference for your further learning in other mathematics courses (to fulfill major or minor requirements).

Chapter 2

Revision Notes of MATH1821

2.1 Chapter 1: Basic Concepts

2.1.1 Functions

Definition 2.1.1 (Function). For any sets X and Y , we say that f is a **function** from X to Y , denoted $f : X \rightarrow Y$, if it provides a rule which assigns to every element $x \in X$ **exactly one** element $y \in Y$.

A function is also called a **mapping**.

Definition 2.1.2 (Domain, codomain and range). In defining a function f , theoretically we have to specify two sets X and Y . They are called the **domain** ($\text{dom } f$) and **codomain** of f respectively.

Natural domain: the largest subset of $\mathbb{R}$ for which the function formula is well defined.

Restricted domain: if there is a story (or some restrictions) on the function, then the restricted domain will be different from the natural domain.

Range: the range of a function $f : X \rightarrow Y$ is the set $\{f(x) : x \in X\}$, which is the set of all possible outputs.

Usually, the range of a function is a subset of its codomain, not necessary to be a real subset.

Different classification of functions:

- **piecewise-defined functions** (e.g. absolute value function)
- **odd functions** ($f(-x) = -f(x)$ for all x) and **even functions** ($f(-x) = f(x)$ for all x)
- **increasing functions** (if $a > b$ implies $f(a) \geq f(b)$) and **decreasing functions** (if $a > b$ implies $f(a) \leq f(b)$)
- **strictly increasing functions** (if $a > b$ implies $f(a) > f(b)$) and **strictly decreasing functions** (if $a > b$ implies $f(a) < f(b)$)
- **random variable:** frequently used in probability and statistics, map from the sample space, which is the set of all possible outcomes of the experiment, to $\mathbb{R}$

Definition 2.1.3 (Image and preimage). Let $f : X \rightarrow Y$ be a function. For $S \subseteq X$, the **image** of S under f , denoted by $f(S)$, is the set $\{f(x) : x \in S\}$.

For $T \subseteq Y$, the **preimage** of T under f , denoted by $f^{-1}(T)$, is the set $\{x \in X : f(x) \in T\}$.

Definition 2.1.4 (Injective and surjective). Let $f : X \rightarrow Y$ be a function. We say that f is

- **injective** if for any $x_1 \neq x_2$ we have $f(x_1) \neq f(x_2)$.
- **surjective** if for any $y \in Y$ there exists $x \in X$ for which $f(x) = y$.
- **bijective** if the function is injective and surjective. A bijective function leads to a **one-to-one correspondence** between elements of X and Y .

Definition 2.1.5 (Composition of functions). Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be functions. This naturally corresponds to a function from X to Z . Such a function is called the **composition** of g and f . Notation: $g \circ f$, normally, we have $(g \circ f)(x) = g(f(x))$.

Definition 2.1.6 (Inverse functions). Let $f : X \rightarrow Y$ be a function. We say that a function $g : Y \rightarrow X$ is the **inverse** of f , denoted $g = f^{-1}$, if

$$(g \circ f)(x) = x \text{ for all } x \in X \text{ and } (f \circ g)(y) = y \text{ for all } y \in Y.$$

Remark. Do not confuse the notation of inverse function ($f^{-1}(y)$ for $y \in Y$) and preimage ($f^{-1}(T)$ for $T \subseteq Y$).

Theorem 2.1.1. Let $f : X \rightarrow Y$ be a function. Then the inverse function f^{-1} exists **if and only if** f is **bijective**.

2.1.2 Trigonometric Functions

Definition 2.1.7 (Radian measure). A central angle subtended by an arc of length equal to the radius of a circle is said to have **radian measure 1**, written 1 rad. In general, if the central angle θ is subtended by an arc of length s , then

$$s : r = \theta : 1,$$

and hence

$$s = r\theta.$$

Similarly, the area of a sector with central angle θ is

$$\frac{1}{2}r^2\theta.$$

Definition 2.1.8 (Cotangent, secant and cosecant).

$$\cot \theta = \frac{\text{adjacent side}}{\text{opposite side}}, \quad \sec \theta = \frac{\text{hypotenuse}}{\text{adjacent side}}, \quad \csc \theta = \frac{\text{hypotenuse}}{\text{opposite side}}.$$

Definition 2.1.9 (Inverse trigonometric functions). For $x \in [-1, 1]$, we define the **arcsine** of x , denoted by $\sin^{-1}$ or $\arcsin x$, to be the (unique) value of $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ such that $\sin \theta = x$. Similarly, we can define the functions:

- $\cos^{-1} : [-1, 1] \rightarrow [0, \pi]$
- $\tan^{-1} : \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$

Compound angle formulas:

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

Double and triple angle formulas:

- $\sin 2A = 2 \sin A \cos A$
- $\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$
- $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
- $\sin 3A = 3 \sin A - 4 \sin^3 A$
- $\cos 3A = 4 \cos^3 A - 3 \cos A$
- $\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$

Half angle formulas:

- $\sin^2 \frac{A}{2} = \frac{1 - \cos A}{2}$
- $\cos^2 \frac{A}{2} = \frac{1 + \cos A}{2}$
- $\tan^2 \frac{A}{2} = \frac{1 - \cos A}{1 + \cos A}$

Product-to-sum formulas:

- $\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$
- $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$
- $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$

Sum-to-product formulas:

- $\sin A + \sin B = 2 \sin\left(\frac{A + B}{2}\right) \cos\left(\frac{A - B}{2}\right)$
- $\sin A - \sin B = 2 \cos\left(\frac{A + B}{2}\right) \sin\left(\frac{A - B}{2}\right)$
- $\cos A + \cos B = 2 \cos\left(\frac{A + B}{2}\right) \cos\left(\frac{A - B}{2}\right)$
- $\cos A - \cos B = -2 \sin\left(\frac{A + B}{2}\right) \sin\left(\frac{A - B}{2}\right)$

2.1.3 Complex Numbers

Definition 2.1.10 (Complex Numbers). The set of complex numbers is $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$, where i is the **imaginary unit** satisfying $i^2 = -1$.

Definition 2.1.11 (Conjugates). For a complex number of the form $z = a + bi$, we define $\bar{z} = a - bi$ to be the **conjugate** of z .

Properties of conjugates

- $\overline{z \pm z'} = \bar{z} \pm \bar{z}'$
- $\overline{zz'} = \bar{z} \cdot \bar{z}'$
- $\frac{\bar{z}}{z'} = \frac{\bar{z}}{\bar{z}'}$ provided that $z' \neq 0$
- $z + \bar{z}$ is real
- $z - \bar{z}$ is purely imaginary
- $z\bar{z} = |z|^2$, where $|z|$ denotes the **modulus** of z

Theorem 2.1.2. Let $f(x)$ be a polynomial with real coefficients. If $f(z) = 0$ for some $z \in \mathbb{C}$, then $f(\bar{z}) = 0$.

Definition 2.1.12 (Polar form of complex numbers). We can change the expression of a complex number from rectangular coordinates to **polar coordinates** (r, θ) .

$$z = a + bi = r \cos \theta + i \sin \theta = r \operatorname{cis} \theta \quad \text{where } r = |z|.$$

Computing with polar form **can simplify computations in complex numbers**.

- $z_1 z_2 = r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2)$;
- $\frac{z_1}{z_2} = \frac{r_1}{r_2} \operatorname{cis}(\theta_1 - \theta_2)$ provided that $z_2 \neq 0$.

Theorem 2.1.3 (n-th roots of complex number). Let $\alpha = r \operatorname{cis} \theta$ where $r > 0$ and $\theta \in [0, 2\pi)$. Then for any $n \in \mathbb{N} \setminus \{1\}$, the equation $z^n - \alpha = 0$ has n distinct roots

$$z_k = r^{1/n} \operatorname{cis} \frac{\theta + 2k\pi}{n}, \quad k = 0, 1, \dots, n-1.$$

Theorem 2.1.4 (Euler's formula). For $\theta \in \mathbb{R}$, we define

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Theorem 2.1.5 (Index laws of complex numbers). For any $z \in \mathbb{C}$ with $z = a + bi$, we define $e^z = e^{a+bi} = e^a e^{bi} = e^a (\cos b + i \sin b)$. It can be checked that such definition preserves the index laws

$$e^{z+z'} = e^z \cdot e^{z'} \text{ for any } z, z' \in \mathbb{C}.$$

2.1.4 Mathematical Induction

Principle of mathematical induction:

To prove that a statement $S(n)$ holds for all positive integers n , we can proceed as follows:

1. Prove that $S(1)$ is true.
2. Assume that $S(k)$ is true for some positive integer k .
3. Using the [induction hypothesis](#), prove that $S(k+1)$ is true.

Remark. There are a lot of variations of mathematical inductions, you will learn them systematically in course MATH2012. In general, We firstly show that the $S(n)$ is true when n is some small number. Then we assume $S(n)$ is true for some $n = k$, based on this, we try to show that $S(n)$ is true when n is greater than k or in some situations that will occur after $n = k$. See Exercise 1F, example (d), (e) in lecture notes.

2.2 Chapter 2: Limits and Continuity

Definition 2.2.1 (Limit). Intuitively, we write

$$\lim_{x \rightarrow c} f(x) = L$$

to mean the value of $f(x)$ gets closer and closer to L as x gets closer and closer to c .

Remark. This is **not** a rigorous definition for limit, you will learn more in course MATH2241.

Limits may not exist:

- limit is taken at $\pm\infty$
- the value of $f(x)$ [oscillates](#) as $x \rightarrow c$
- $f(x)$ approaches different values from the left and from the right

Definition 2.2.2 (One-sided limits). We say that the **left-hand limit** of $f(x)$ as x tends to c is equal to L , denoted by $\lim_{x \rightarrow c^-} f(x) = L$, if $f(x)$ gets closer and closer to L as x gets closer and closer to c from the left.

Similarly, we say that the **right-hand limit** of $f(x)$ as x tends to c is equal to L , denoted by $\lim_{x \rightarrow c^+} f(x) = L$, if $f(x)$ gets closer and closer to L as x gets closer and closer to c from the right.

Theorem 2.2.1 (Sandwich theorem). Suppose that I is an interval with $c \in I$, and that $g(x) \leq f(x) \leq h(x)$ for all $x \in I \setminus \{c\}$. If $\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L$ then $\lim_{x \rightarrow c} f(x) = L$ as well.

Theorem 2.2.2 (Divergence to infinity). Suppose that I is an interval with $c \in I$, and that $g(x) \leq f(x)$ for all $x \in I \setminus \{c\}$. If $\lim_{x \rightarrow c} g(x) = \infty$, then $\lim_{x \rightarrow c} f(x) = \infty$ as well.

Remark. We can replace ∞ by $-\infty$; True for one-sided limits; We also have a similar result for limits at infinity instead of limit at c .

Special limits:

- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
- $\lim_{x \rightarrow \infty} \frac{e^x}{x} = \infty$
- $\lim_{x \rightarrow 0^+} \frac{\ln x}{x} = -\infty$

Definition 2.2.3 (Continuous). Let f be a function and c be an **interior point** of $\text{dom } f$ (i.e. there exists an open interval I containing c such that $I \subseteq \text{dom } f$). We say that f is **continuous at** c if $\lim_{x \rightarrow c} f(x)$ exists and is equal to $f(c)$.

Definition 2.2.4 (Boundary and isolated points). X is the domain of a function f .

- We call $c \in \mathbb{R}$ a **boundary point** of X if any open interval I containing c intersects both X and $\mathbb{R} \setminus X$.
- We call $c \in X$ an **isolated point** of X if there exists an open interval I containing c such that $I \cap X = \{c\}$.

Definition 2.2.5 (Continuity at boundary and isolated points). Suppose f is a function with domain X .

- If c is a boundary point of X , then we say that f is continuous at c if either $\lim_{x \rightarrow c^-} f(x)$ or $\lim_{x \rightarrow c^+} f(x)$ exists and is equal to $f(c)$ where the choice of one-sided limit depends on the side of c on which f is defined.
- By definition f is continuous at an isolated point.

Definition 2.2.6 (Continuity of a function). We can say that f is continuous on a set $S \subseteq \text{dom } f$ if it is continuous at every point in S , or simply f is continuous if it is continuous at every point of its domain.

Types of discontinuities:

- Removable discontinuity
- Jump discontinuity
- Infinite discontinuity
- Oscillating discontinuity

Properties of continuous functions:

Suppose f and g are two functions, both of which are continuous at c , and k be a constant, Then the following functions are also continuous at c :

- $f \pm g$
- kf
- fg
- $\frac{f}{g}$ provided that $g \neq 0$
- f^k provided that it is defined on an interval containing c

It follows that all polynomials, rational functions, exponential, logarithmic, and trigonometric functions are continuous.

Theorem 2.2.3 (Continuity of composite functions). Suppose $\lim_{x \rightarrow c} f(x) = L$. If g is a function which is continuous at L , then

$$\lim_{x \rightarrow c} g(f(x)) = g(\lim_{x \rightarrow c} f(x)) = g(L).$$

If f is continuous at c , then $f(c) = L$ and the above result can be rephrased as if f is continuous at c and g is continuous at $f(c)$, then $g \circ f$ is continuous at c .

Theorem 2.2.4 (Extreme value theorem). Suppose that the function f is continuous on the interval $[a, b]$. Then f attains a maximum and a minimum value in this interval.

Theorem 2.2.5 (Intermediate value theorem). Suppose that the function f is continuous on the interval $[a, b]$. Suppose $f(a) \neq f(b)$ and y is a real number lying strictly between $f(a)$ and $f(b)$ (i.e. either $f(a) \leq y \leq f(b)$ or $f(a) \geq y \geq f(b)$). Then there exists $c \in (a, b)$ such that $f(c) = y$.

2.3 Chapter 3: Differentiation

2.3.1 Derivatives

Definition 2.3.1 (Derivative). A function f is said to be **differentiable** at c if the limit

$$\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

exists. The value of the limit is known as the **derivative** of f at c , denoted by $f'(c)$. The derivative of a function is a function as well, given by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

wherever it is well-defined.

Remark. $f'(x)$ is a function, hence it will have its own domain and codomain, which are not necessary to be the same as $f(x)$.

Theorem 2.3.1 (Differentiable and Continuous). *If f is differentiable at c , then it must be continuous at c .*

*Consider the **contrapositive** statement of this statement, we can conclude that if a function f is not continuous at c , then it must be not differentiable at c .*

Remark. For a conditional statement, we have hypothesis and conclusion. In this statement, f is differentiable at c is the hypothesis, and f is continuous at c is the conclusion. The **contrapositive** of a conditional statement both swaps the hypothesis and the conclusion and negates both the hypothesis and the conclusion.

If a conditional statement is true, then its contrapositive must be true. If a conditional statement is false, then its contrapositive must be false. This can be shown by truth table.

However, we need to notice that the **converse** of this statement may not be true. If f is continuous at c , f may not be differentiable at c . A trivial example is $f(x) = |x|$ is continuous at $x = 0$, but is not differentiable at $x = 0$.

Definition 2.3.2 (Higher order derivatives). Recall that the derivative of a function f is also a function, when we take the derivative of the derivative of f , we will have the **second order derivative**. Similarly, we will have higher order derivatives if we continue taking derivatives.

Remember to check the **differentiability** of the function.

Definition 2.3.3 (Implicit Differentiation). When we face some difficulties to express y in terms of x , we can use the technique of **implicit differentiation**.

We differentiate both sides of the equality $f(x, y) = 0$ with respect to x to find $\frac{dy}{dx}$.

Derivatives of Trigonometric Functions:

- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(\cot x) = -\csc^2 x$
- $\frac{d}{dx}(\sec x) = \sec x \tan x$
- $\frac{d}{dx}(\csc x) = -\csc x \cot x$

Theorem 2.3.2 (Derivative of Inverse Function). *Suppose f has an inverse function g . If f is differentiable at (c, d) with $f'(c) \neq 0$, then g is differentiable at d and*

$$g'(d) = g'(f(c)) = \frac{1}{f'(c)}.$$

Theorem 2.3.3 (Recall the Chain Rule). Suppose $h(x) = f(g(x))$ for every x , that is, function h is the composition function of f and g , then the derivative of $h(x)$ is

$$h'(x) = (f(g(x)))' = f'(g(x))g'(x).$$

Derivatives of Inverse Trigonometric Functions:

- $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

2.3.2 Monotonicity and Concavity

Definition 2.3.4 (Derivatives and Monotonicity). If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is increasing in (a, b) ; If $f'(x) \leq 0$ for all $x \in (a, b)$, then f is decreasing in (a, b) .

Definition 2.3.5 (Stationary Points and Extrema). **Stationary points** are points where $f'(x) = 0$ and are potential candidates for extremum points.

Remark. A **global** extremum must be a **local** extremum, while the converse may not be true.

Stationary points are only the potential candidates for extremum points. A stationary point may not be an extremum point. Consider the point $(0, 0)$ on the function $f(x) = x^3$. It is a stationary point but not an extremum point.

Definition 2.3.6 (Critical Points and Extrema). We define **critical points** of a function f to be points at which $f'(x)$ is either **zero** or **undefined**.

Theorem 2.3.4 (Extremum). Let f be a continuous function on $[a, b]$. Then the maximum and minimum of f occur at either **critical points** or **boundary points**.

Definition 2.3.7 (Convex and Concave). A function is said to be **convex** in an interval I if

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2) \text{ for all } x_1, x_2 \in I \text{ and } t \in (0, 1).$$

A function is said to be **concave** in an interval I if

$$f(tx_1 + (1-t)x_2) \geq tf(x_1) + (1-t)f(x_2) \text{ for all } x_1, x_2 \in I \text{ and } t \in (0, 1).$$

Remark. If a function f is twice differentiable on an interval I , then being convex (concave) on I is equivalent to $f''(x) \geq 0$ ($f''(x) \leq 0$) for all $x \in I$.

A **point of inflection** of a function f is a point at which $f''(x)$ changes sign. For nice functions, this must be a point at which $f''(x)$ is either zero or undefined.

2.3.3 Asymptotes

Definition 2.3.8 (Asymptotes). The graph of $y = f(x)$ has a **vertical asymptote** $x = c$ if either

$$\lim_{x \rightarrow c^-} f(x) = \pm\infty \text{ or } \lim_{x \rightarrow c^+} f(x) = \pm\infty.$$

The graph of $y = f(x)$ has a **horizontal asymptote** $y = d$ if either

$$\lim_{x \rightarrow \infty} f(x) = d \text{ or } \lim_{x \rightarrow -\infty} f(x) = d.$$

The graph of $y = f(x)$ has an **oblique asymptote** $y = ax + b$ if either

$$\lim_{x \rightarrow \infty} [f(x) - (ax + b)] = 0 \text{ or } \lim_{x \rightarrow -\infty} [f(x) - (ax + b)] = 0.$$

How to find the value of a and b of an oblique asymptote $y = ax + b$?
 Compute the values of a and b by evaluating the following limits:

$$a = \lim_{x \rightarrow \infty} \frac{f(x)}{x} \text{ and } b = \lim_{x \rightarrow \infty} [f(x) - ax].$$

and then replace ∞ by $-\infty$ and repeat the calculations.

Theorem 2.3.5 (Rolle's Theorem). *Suppose the function f is continuous on $[a, b]$ and differentiable on (a, b) . If $f(a) = f(b)$, then*

$$f'(c) = 0 \text{ for some } c \in (a, b).$$

Theorem 2.3.6 (Mean Value Theorem). *Suppose the function f is continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Theorem 2.3.7 (L'Hôpital's rule). *Let f and g be differentiable functions such that either*

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = 0 \text{ or } \lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = \pm\infty.$$

Then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

provided that the latter limit is finite or is $\pm\infty$.

Remark. The same result holds if $x \rightarrow c$ is replaced by a one-sided limit or a limit at infinity.

2.4 Chapter 4: Integration

2.4.1 Integration Techniques

In this chapter, we mainly focused on the calculations of integration of different functions. We also learned several techniques which are normally applied to commonly-seen functions.

- Integration by substitution
- Integration of trigonometric functions
- Integration by partial fraction decomposition
- Integration by parts (reduction formula)

Integration by substitution

Recall the chain rule in derivatives, for differentiable functions F and g , we have

$$\frac{d}{dx}[F(g(x))] = F'(g(x)) \cdot g'(x).$$

Then we have

$$\int F'(g(x)) \cdot g'(x) dx = F(g(x)) + C.$$

If $f(u) = F'(u)$, then we have

$$\int f(g(x)) \cdot g'(x) dx = F(g(x)) + C.$$

Generalized Result for Some Integrations:

$$\int \frac{1}{ax + b} = \frac{1}{a} \ln |ax + b| + C$$

$$\int \frac{dx}{(x + a)^n} = \begin{cases} -\frac{1}{n-1} \cdot \frac{1}{(x + a)^{n-1}} + C & \text{if } n \in \mathbb{N} \setminus \{1\} \\ \ln |x + a| + C & \text{if } n = 1 \end{cases}$$

$$\int \frac{x}{(x^2 + a^2)^n} dx = \begin{cases} \frac{1}{2} \ln(x^2 + a^2) + C & \text{if } n = 1 \\ \frac{1}{2(1-n)} \cdot \frac{1}{(x^2 + a^2)^{n-1}} & \text{if } n \in \mathbb{N} \setminus \{1\} \end{cases}$$

Integrating Trigonometric Functions

Basic formulas:

- $\int \sin x \, dx = -\cos x + C$
- $\int \cos x \, dx = \sin x + C$
- $\int \sec^2 x \, dx = \tan x + C$

Commonly used trigonometric techniques:

- $\sin^2 x + \cos^2 x = 1$
- Double/Triple angle formula
- Half angle formula
- Sum-to-product and Product-to-sum formula

Integration by Partial Fraction Decomposition

Definition 2.4.1 (Partial Fraction). A **partial fraction** is a fraction of one of the following two forms:

$$\frac{p}{(x-k)^n} \text{ or } \frac{qx+r}{(ax^2+bx+c)^n}$$

where a, b, c, k, p, q, r are constants with $b^2 - 4ac < 0$ and n is a positive integer.

Definition 2.4.2 (Partial Fraction Decomposition). Given a proper rational function $f(x)$ with real coefficients, i.e.

$$f(x) = \frac{p(x)}{q(x)}$$

where $p(x)$ and $q(x)$ are polynomials with real coefficients and $\deg p(x) < \deg q(x)$, a **partial fraction decomposition** of $f(x)$ means rewriting $f(x)$ as a sum of partial fractions.

Remark. Notice that every proper rational function with real coefficients can be expressed as a sum of partial fractions. Just take it for granted now, this can be proved by some algebra techniques.

Method of Partial Fraction Decomposition:

$$\frac{5x-4}{x^2-x-2} = \frac{A}{x+1} + \frac{B}{x-2}$$

$$5x-4 \equiv A(x-2) + B(x+1)$$

Method 1 - Plugging in appropriate values of x

- We plug in $x = 2$, then we will have

$$6 = 3B$$

$$B = 2$$

- We plug in $x = -1$, then we will have

$$-9 = -3A$$

$$A = 3$$

Method 2 - Comparing coefficients

$$A(x - 2) + B(x + 1) = (A + B)x + (-2A + B) \equiv 5x - 4$$

$$A + B = 5$$

$$-2A + B = -4$$

Solve the linear equations system, we have $A = 3$ and $B = 2$.

Method of Partial Fraction Decomposition - Repeated Factor

$$\frac{12x^2 + 1}{4x^3 - 4x^2 + x} = \frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{(2x - 1)^2}$$

Integration by Parts

We have

$$\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx$$

or in more compact notation,

$$\int u dv = uv - \int v du$$

Applying to definite integrals

$$\int_a^b u dv = u(b)v(b) - u(a)v(a) - \int_x^b v du$$

Integration by Parts - Reduction Formula

Summary

Integration by substitution	Not easy to integrate directly	$\int \frac{1}{1-2x} dx \int 2x\sqrt{1+x^2} dx$
Integration of trigonometric functions	See some trigonometric function terms or some terms can be substituted by trigonometric functions	$\int \cos^2 3x dx \int \frac{dx}{x^2+4}$
Integration by partial fraction decomposition	Integrator is a proper rational function	$\int \frac{12x^2+1}{4x^3-4x^2+x} dx \int \frac{5x-4}{x^2-x-2} dx$
Integration by parts	Integrator is a product of some commonly-seen functions	$\int x \sec^2 x dx \int \sin(\ln x) dx$

2.4.2 Improper Integrals

Definition 2.4.3 (Improper Integrals). Usually, a definite integral can be interpreted as the area of a region that is normally **bounded**. The integral becomes an **improper integral** if it represents the area of a region that is **unbounded**. There are generally two types of unboundedness: Horizontally unbounded and vertically unbounded.

Definition 2.4.4 (Horizontal Unboundedness). If a function f is continuous on $[a, \infty)$, we define

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$$

provided that the limit on the right hand side exists. In this case, we say that the improper integral on the left hand side is **convergent**.

If the limit on the right hand side does not exist, then we say that the improper integral on the left hand side is **divergent**.

If f is continuous on $(-\infty, b]$, we define

$$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx.$$

Definition 2.4.5 (Horizontal Unboundedness). If the function f is continuous on $\mathbb{R}$, then we define

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx$$

if we can find a real number c for which both improper integrals on the right hand side converge.

Definition 2.4.6 (Vertical Unboundedness).

- If the function f is continuous on $[a, b)$ and $x = b$ is a vertical asymptote of f , we define

$$\int_a^b f(x) dx = \lim_{c \rightarrow b} \int_a^c f(x) dx$$

- If f is continuous on $(a, b]$ and $x = a$ is a vertical asymptote of f , we define

$$\int_a^b f(x) dx = \lim_{c \rightarrow a} \int_c^b f(x) dx$$

- If f is continuous on $[a, b]$ except that it has a vertical asymptote at $x = c$ where $c \in (a, b)$. Then we define

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

2.4.3 Applications

Application in Probability Theory

Properties of Probability Density Function

- $f(x) \geq 0$ for any x
- $\int_{-\infty}^{\infty} f(x) dx = 1$ for any probability density function $f(x)$ no matter for discrete random variable or continuous random variable
- We use the area between $f(x)$ and the x-axis to denote the probability of the occurrence of an event.
- For the probability density function $f(x)$ for a **continuous random variable**, when we input some value $x = a$, the output of the function $f(x) = f(a)$ is not the probability when $x = a$, **the probability at every single point is always 0** for a continuous random variable.

Definition 2.4.7 (Expected Value). We compute the expected value of a random variable X by $E[X] = \int_{-\infty}^{\infty} xf(x) dx$.

Application of Definite Integral - Disk Method and Shell Method

Definition 2.4.8 (Disk Method). If the graph of $y = f(x)$ in the interval $[a, b]$ is revolved about the **x-axis**, the volume of the solid of revolution obtained is given by

$$\int_a^b \pi[f(x)]^2 dx.$$

Definition 2.4.9 (Shell Method). If the graph of $y = f(x)$ in the interval $[a, b]$ is revolved about the **y-axis**, the volume of the solid of revolution obtained is given by

$$\int_a^b 2\pi x[f(x)] dx.$$

Definition 2.4.10 (Arc Length). If f' is continuous on $[a, b]$, the arc length of the curve $y = f(x)$ between $x = a$ and $x = b$ is given by

$$\int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

Definition 2.4.11 (Surface Area). Suppose f' is continuous on $[a, b]$. When the graph of $y = f(x)$ between $x = a$ and $x = b$ is revolved about the x -axis, the area of the surface generated is given by

$$\int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx.$$

Integration of Odd and Even Functions

For an odd function f and $a > 0$,

$$\int_{-a}^a f(x) dx = 0.$$

For an even function f and $a > 0$,

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx.$$

Fundamental Theorem of Calculus

- If the function f and F are continuous on $[a, b]$ such that $F'(x) = f(x)$ for all $x \in (a, b)$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

- Let f be a continuous function on $[a, b]$ and $c \in [a, b]$. Define the function $F : [a, b] \rightarrow \mathbb{R}$ by

$$F(x) = \int_c^x f(t) dt.$$

Then F is continuous on $[a, b]$ and $F'(x) = f(x)$ for all $x \in (a, b)$.

2.5 Chapter 5: Numerical Methods

2.5.1 Method of Bisection

Steps of the method of bisection

Suppose we want to solve the equation $f(x) = 0$ where f is continuous. We do the following steps until we have obtained an interval containing the root that is within a desired error tolerance.

- Identify an interval $[a, b]$ for which $f(a)f(b) < 0$.
- Split the interval into two equal subintervals $[a, c]$ or $[c, b]$.
- Determine whether the root lies in (a, c) or (c, b) .
- Repeat the above until the length of the interval that contains the root is within the error tolerance.

Remark.

- The rationale of the method of bisection is the **Intermediate Value Theorem**. Recall the theorem here: Suppose that the function f is continuous on the interval $[a, b]$. Suppose $f(a) \neq f(b)$ and y is a real number lying strictly between $f(a)$ and $f(b)$. Then there exists $c \in (a, b)$ such that $f(c) = y$.
- Method of bisection is a way of **finding an interval** containing a root. When applying this method, a higher error tolerance means less cost in calculation, indicating that only a few repetitions can achieve the desired interval. If we are pursuing a more accurate interval (a lower error tolerance), then we need to repeat more times with a higher calculation cost. This is commonly seen in approximations and other numerical methods.
- The bisection method can only be applied on a **continuous** function or **on an interval that the function is continuous**. We can consider the function $f(x) = \frac{1}{x}$, the function is not continuous on the interval $[-1, 2]$. If we try to use the method of bisection to find a root in this interval, no matter how many times we attempt, we will eventually fail since there is no root within this interval but every step we can find the desired subintervals. On the other hand, by the first remark, we know that the rationale of the bisection method is the I.V.T. and continuity is also a condition of I.V.T..

2.5.2 Taylor Approximation

Series expansion of functions Let f be a function that has continuous derivatives of all orders on an interval I containing a . The n -th Taylor polynomial of f at a is the polynomial

$$f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

whereas the power series

$$f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots,$$

is called the Taylor series of f at a .

A Taylor series at 0 is sometimes called a Maclaurin series.

Taylor series	$f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots$
n-th Taylor polynomial	$f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$
Maclaurin series	$f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots$

Theorem 2.5.1 (Error analysis). Let f be a function that has continuous derivatives of all orders on an interval I containing a , and let $P_n(x)$ be the n -th Taylor polynomial of f at a . Then

$$f(x) - P_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1},$$

where c is a real number between x and a .

How to find the c to determine the upper bound of the error?

Look at the error term $\frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$, c only appears in the $(n+1)$ -th derivative. We can pay attention to the monotonic trends of $f^{(n+1)}(x)$ between x and a .

2.5.3 Newton's Method

Procedure

1. We begin with an initial guess x_0 as the root of function $f(x)$. Set $n = 0$.
2. Replace x_n by $x_n - \frac{f(x_n)}{f'(x_n)}$ and increase n by 1.
3. Repeat the above and stop when the stopping condition met.

Stopping condition

1. The difference between x_n and x_{n+1} is small enough, i.e. $|x_{n+1} - x_n|$ is small enough.
2. The value of $f(x_n)$ is close enough to 0, i.e. $|f(x_n)|$ is small enough.

Newton's method does not always work.

Refer to the Assignment 5 Question 3.

Possible obstacles:

- The derivative may be difficult to compute.
- The formula fails if $f'(x_n)$ is undefined for some n .
- The formula fails if $f'(x_n)$ is equal to 0 for some n .
- The iterates may not be convergent.

2.5.4 Trapezoidal Rule

Suppose f is continuous on $[a, b]$ and we divide $[a, b]$ into n equal subintervals with endpoints $x_0, x_1, \dots, x_n$ (with $x_0 = a$ and $x_n = b$). Then

$$\int_a^b f(x) dx \approx \frac{h}{2} \{f(x_0) + 2[f(x_1) + f(x_2) + \dots + f(x_{n-1})] + f(x_n)\}$$

where $h = \frac{1}{n}(b - a)$ is the width of each subinterval.

In general, $\int_a^b f(x) dx$ is **overestimated/underestimated** if f is **convex/concave** on the interval $[a, b]$.

Let f be twice differentiable on $[a, b]$ and K be a real number such that

$$|f''(x)| \leq K \text{ for all } x \in [a, b].$$

If we approximate $\int_a^b f(x) dx$ by the trapezoidal rule with n subintervals, the error is at most

$$\frac{(b - a)^3 K}{12n^2}.$$

2.5.5 Simpson's Rule

Suppose f is continuous on $[a, b]$ and we divide $[a, b]$ into $2m$ equal subintervals with endpoints $x_0, x_1, \dots, x_{2m}$ (with $x_0 = a$ and $x_{2m} = b$). Then

$$\begin{aligned} \int_a^b f(x) dx \approx & \frac{h}{3} \{f(x_0) + 4[f(x_1) + f(x_3) + \dots + f(x_{2m-1})] \\ & + 2[f(x_2) + f(x_4) + \dots + f(x_{2m-2})] + f(x_{2m})\} \end{aligned}$$

where $h = \frac{1}{2m}(b - a)$ is the width of each subinterval.

Let f be a function for which $f^{(4)}$ exists and is continuous on $[a, b]$, and K be a real number such that

$$|f^{(4)}(x)| \leq K \text{ for all } x \in [a, b].$$

If we approximate $\int_a^b f(x) dx$ by the Simpson's rule with $2m$ subintervals, the error is at most

$$\frac{(b - a)^5 K}{180(2m)^4}.$$

2.5.6 Summary of Numerical Methods

Method of Bisection	Find an interval that contains a root of a continuous function
Taylor Approximation	Find the approximated value of the function at some point
Newton's Method	Find an approximated root given some initial guess
Trapezoidal Rule	Find the approximation of an integration using the trapezoids
Simpson's Rule	Find the approximation of an integration using the interpolation in every subintervals

2.6 Chapter 6: Differential Equations

2.6.1 Basics

If the unknown function is a function of one variable, the resulting DE is called an **ordinary differential equation (ODE)**. If the unknown function is a function of two or more variables, the resulting DE is called a **partial differential equation (PDE)**. More details will be elaborated in course MATH3405 and MATH4406 respectively. **Order of DE** The **order** of a differential equation is the highest order derivative that appears in the equation.

Classifications of ODEs An ODE is said to be **linear** if it is of the form

$$F_n(x)y^{(n)} + \dots + F_2(x)y'' + F_1(x)y' + F_0(x)y = G(x).$$

And the ODE is said to be

- **homogeneous** if $G(x) = 0$;
- **with constant coefficients** if $F_k(x)$ is a constant for all k .

Initial Value Problem (IVP) The problem in solving ODE with appropriate initial condition(s) provided. Usually, the unknown function can be found without any constant and is uniquely determined.

2.6.2 First Order ODEs

Definition 2.6.1 (Separable First Order ODEs). A first order ODE is said to be **seperable** if it can be written in the form

$$M(x) = N(y) \frac{dy}{dx}$$

General solution to the separable first order ODE

The solution is given by

$$\int M(x) dx = \int N(y) dy$$

Definition 2.6.2 (Homogeneous of degree n). Let $f(x, y)$ be a function of two variables. We say that f is **homogeneous of degree n** if

$$f(\lambda x, \lambda y) = \lambda^n f(x, y) \text{ for all } x, y.$$

Intuitively, this means each term of $f(x, y)$ is of degree n .

Transforming into separable first order ODEs

If a first order ODE is of the form

$$M(x, y) = N(x, y) \frac{dy}{dx}$$

where both $M(x, y)$ and $N(x, y)$ are homogeneous of the same degree, then we may apply the substitution $y = vx$ to transform the ODE to a separable one.

Here, $v := v(x)$ is a function of x .

Integrating factor

We can solve a general linear first order ODE of the form

$$y' + P(x)y = Q(x)$$

by multiplying both sides by the **integrating factor** $e^{\int P(x) dx}$.

2.6.3 Second Order ODEs

We mainly discuss the solution of **linear second order ODEs with constant coefficients**, i.e. an ODE of the form

$$y'' + py' + qy = r(x),$$

where p and q are real numbers, and $r(x)$ is a function of x .

Initial Conditions and Uniqueness of the Solution

For a second order ODE, we usually need two initial conditions to determine a unique solution, and the initial conditions are usually in the form of

$$y(x_0) = y_1 \text{ and } y'(x_0) = y_2$$

where x_0, y_1, y_2 are real numbers.

Homogeneous second order ODEs

For a second order ODE in the form of

$$y'' + py' + qy = r(x),$$

it is **homogeneous** when $r(x) = 0$.

Definition 2.6.3 (Characteristic Equation). The function $y = e^{kx}$ satisfies the ODE

$$y'' + py' + q = 0$$

if and only if k is a root of the equation

$$\lambda^2 + p\lambda + q = 0,$$

which is called the **characteristic equation** associated with the ODE.

Distinct real roots

Consider the ODE

$$y'' + py' + qy = 0.$$

If the characteristic equation has two distinct real roots m and n , then the ODE has general solution

$$y = Ae^{mx} + Be^{nx}.$$

Double root (repeated root)

Consider the ODE

$$y'' + py' + qy = 0.$$

If the characteristic equation has a double real roots m then the ODE has general solution

$$y = Ae^{mx} + Bxe^{mx}.$$

Complex roots

Consider the ODE

$$y'' + py' + qy = 0.$$

If the characteristic equation has two complex roots $m \pm ni$, then the ODE has general solution

$$y = e^{mx}(A \cos nx + B \sin nx).$$

Non-homogeneous second order ODEs

Every solution to the ODE

$$y'' + py' + qy = r(x)$$

is of the form $y = y_c(x) + y_p(x)$, where

- $y_c(x)$ satisfies the associated homogeneous part $y'' + py' + qy = 0$, called the homogeneous solution of complementary solution;
- $y_p(x)$ is one solution to the original ODE, called a particular solution.

Form of $r(x)$	Try for particular solutions of the form
$p(x)$	$a_n x^n + \cdots + a_1 x + a_0$
e^{kx}	$a_0 e^{kx}$
$\sinh x$	$a_0 \sinh x + b_0 \cosh x$
$e^{kx} p(x)$	$e^{kx}(a_n x^n + \cdots + a_1 x + a_0)$

Remark.

- If $r(x)$ is a linear combination of the forms listed above, we may try a linear combination of the corresponding suggested forms of the particular solutions.
- Other forms like $\cosh x$ or $p(x) \sinh x$ can be handled in similar ways.

Steps for solving a non-homogeneous ODE with two initial conditions

1. Solve the homogeneous solution with two different constants.
2. Guess the particular solution based on the form of $r(x)$ with some constants.
3. Evaluate whether the particular solution is in the same form of the homogeneous solution. If yes, multiply x on the particular solution; if no, keep the particular solution for the next step.

4. Plug the particular solution into the non-homogeneous ODE and evaluate every unknown constant in the particular solution. (After plugging in, we should be able to evaluate all the constants in the particular solution.)
5. Add the homogeneous solution and the particular solution together to get the general solution. Use the two initial conditions to evaluate the two constants in the general solution.

2.7 Chapter 7: Matrices

2.7.1 Basic Terminology and Concepts

- row
- column
- size (dimension)
- (i, j) -entry
- submatrix
- equal matrices (size and every entry should be equal)
- square matrix
- main diagonal (collection of all diagonal entries)
- diagonal matrix (all off-diagonal entries are 0)
- triangular matrix
 - upper triangular matrix ($p_{ij} = 0$ whenever $i > j$)
 - lower triangular matrix ($p_{ij} = 0$ whenever $i < j$)

Remark.

- The *diagonal entries* of diagonal matrix are **not necessary to be non-zero**. The zero matrix is also a diagonal matrix.
- A diagonal matrix is default to be a square matrix, but triangular matrices are **not necessary to be square matrices**.

2.7.2 Matrix Operation

Addition

- defined entrywise and only defined on same size matrices;
- commutative;
- associative.

Definition 2.7.1. The **additive identity** O is the matrix satisfying $A + O = A$ for any matrix A , and O should be in the same size of A .

By the definition we know that the additive identity is indeed the zero matrix with same size.

The **additive inverse** of matrix $A = [a_{ij}]$ is defined by $-A = [-a_{ij}]$ such that $A + (-A) = O$.

Remark. We can define the matrix subtraction by considering $A - B$ as $A + (-B)$.

Scalar Multiplication

Definition 2.7.2. Let A be a matrix and k be a **scalar** (which can be taken as a real number or a constant number in this course). Then kA is the matrix obtained by multiplying every entry of A by k .

Distributive laws between addition and scalar multiplication

- $s(A + B) = sA + sB$
- $(s + t)A = sA + tA$
- $(st)A = s(tA)$

Transpose

Definition 2.7.3. Let $A = [a_{ij}]$ be a matrix. The **transpose** of A , denoted by A^T , is a matrix whose (i, j) -entry is the (j, i) -entry of A .

Remark: If A is a $m \times n$ matrix, then A^T is a $n \times m$ matrix.

Definition 2.7.4. A matrix A is said to be

- **symmetric** if $A^T = A$;
- **skew-symmetric** if $A^T = -A$.

If A, B are matrices with the same size and s is a scalar, then we have

- $(A + B)^T = A^T + B^T$;
- $(sA)^T = s(A^T)$;
- $(A^T)^T = A$.

Matrix Multiplication

Definition 2.7.5 (Dot Product). Given two ordered n -tuples $(a_1, a_2, \dots, a_n)$ and $(b_1, b_2, \dots, b_n)$, we define their **dot product** to be

$$(a_1, a_2, \dots, a_n) \cdot (b_1, b_2, \dots, b_n) = a_1b_1 + a_2b_2 + \dots + a_nb_n.$$

Definition 2.7.6. Let A be an $m \times n$ matrix and B be an $n \times k$ matrix. We define the product AB to be the $m \times k$ matrix $C = [c_{ij}]$ whose (i, j) -entry is the dot product of the i -th row of A and the j -th column of B , i.e.

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}.$$

Remark. The matrix multiplication is **not** commutative. BA may not be well defined as AB , and may have different size or unequal entry with AB .

Properties of matrix products

Let A, B, C be matrices and s be a scalar. We have the following properties whenever the operations are well-defined:

- $A(BC) = A(BC)$;
- $A(B + C) = AB + AC$;
- $(A + B)C = AC + BC$;
- $s(AB) = (sA)B = A(sB)$;
- $(AB)^T = B^T A^T$.

Definition 2.7.7. Let A be a **square** matrix and n be a positive integer. Naturally, we define

$$A^n = \underbrace{AA \cdots A}_{n \text{ times}}.$$

Alternative views of matrix multiplication

Suppose the matrix product AB is well-defined.

- The i -th row of AB is a **linear combination** of the rows of B , with the coefficients coming from the i -th row of A .
- The j -th column of AB is a **linear combination** of the columns of A , with the coefficients coming from the j -th row of B .

Matrix Inverse

Definition 2.7.8. The **identity matrix** I is a square matrix with all diagonal entries 1 and all other entries 0. We may write I_n for identity matrix with size $n \times n$. The product of identity matrix with any other matrix returns to the matrix itself (remember to check the size to well-define the matrix multiplication).

Definition 2.7.9. Let A be a square matrix. If $f(x)$ is a general polynomial, then we can naturally define $f(A)$ to be the matrix polynomial with same polynomial terms.

Definition 2.7.10. Let A be a square matrix. We say that B is an **inverse** of A if $AB = BA = I$. Notice that the inverse of a matrix is unique.

Remark.

- The inverse of A is usually denoted by A^{-1} .
- We can also define the left inverse and right inverse for a non-square matrix.
- B should be called a multiplicative inverse of A , but we usually call it the inverse of A .

Definition 2.7.11. A matrix with inverse is said to be **invertible (non-singular)**. A matrix without inverse is said to be **singular**.

Theorem 2.7.1. Let A and B be square matrices with the same size. If $AB = I$, then $BA = I$.

Properties of matrix inverse

Let A, B, C be invertible matrices of the same size, n be a positive integer and s be a scalar. Then we have

- $(A^{-1})^{-1} = A$;
- $(AB)^{-1} = B^{-1}A^{-1}$;
- $(A^n)^{-1} = (A^{-1})^n$;
- $(A^T)^{-1} = (A^{-1})^T$;
- $(sA)^{-1} = \frac{1}{s}A^{-1}$

2.7.3 Determinants

Definition 2.7.12. For a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, we define the **determinant** of A , denoted by $\det A$, by

$$\det A = ad - bc.$$

Remark.

- Alternative notations for the determinant include $|A|$ and $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$.
- The determinant is a scalar.
- It is easy to check that if $\det A = 0$, then A is not invertible.

Definition 2.7.13. For a 3×3 matrix $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$, we define

$$\det A = aei + bfg + cdh - ceg - afh - bdi.$$

Remark.

- Beware that 4×4 and higher-order determinants are defined in a different way.
- It is still true that if $\det A = 0$, then A is not invertible.

Definition 2.7.14. Let $A = [a_{ij}]$ be a square matrix.

- A_{ij} denotes the submatrix of A obtained from A by deleting row i and column j , and is still a square matrix with smaller size.
- The **(i, j) -cofactor of A** , denoted by $c_{ij}(A)$ or simply c_{ij} , is defined by $(-1)^{i+j} \det A_{ij}$.

Definition 2.7.15. The **cofactor matrix** of A is the matrix whose (i, j) -entry is the (i, j) -cofactor of A . The **transpose** of the cofactor matrix of A is called the **adjoint** of A , denoted by $\text{adj}A$.

Theorem 2.7.2. Let A be a square matrix. If $\det A \neq 0$, then

$$A^{-1} = \frac{1}{\det A} (\text{adj}A).$$

Properties of determinants Let A and B be square matrices with the same size. Then

1. A is invertible if and only if $\det A \neq 0$;
2. $\det AB = \det A \det B$;
3. $\det A^T = \det A$;
4. $\det A^{-1} = \frac{1}{\det A}$ if A is invertible.

2.8 Chapter 8: Vectors

2.8.1 Basics

Definition 2.8.1. A vector is a quantity with both **magnitude** and **direction**.

Notation: We usually draw an arrowed lined segment from an initial point A to a terminal point B . So we can denote the vector by $\overrightarrow{AB}$. Or we can use $\mathbf{v}$ or simply $\vec{v}$ in handwriting.

Definition 2.8.2. The length of a vector $\mathbf{v}$ is called the **norm** of $\mathbf{v}$ and is denoted by $\|\mathbf{v}\|$.

- A vector with norm 1 is called a **unit vector**;
- A vector with norm 0 is called **the zero vector** and is denoted by $\mathbf{0}$.
- Two vectors are **equal** if they are same in both magnitude and direction, **parallel** if they have the same or opposite direction(s).

Vector Operation

Scalar Multiplication

Let $\mathbf{v}$ be a non-zero vector, for a positive constant k , the vector $k\mathbf{v}$ is the vector in the same direction and k times as long as $\mathbf{v}$. If k is negative, then $k\mathbf{v}$ is the vector in the opposite direction and k times as long as $\mathbf{v}$. If k is 0, then $k\mathbf{v} = \mathbf{0}$.

Addition

To add two vectors $\mathbf{u}$ and $\mathbf{v}$, we move the initial point of $\mathbf{v}$ to the terminal point of $\mathbf{u}$. The sum $\mathbf{u} + \mathbf{v}$ is the vector whose initial point is the initial point of $\mathbf{u}$ and whose terminal point is the terminal point of $\mathbf{v}$. (tip-to-tail method) It is easy to see that $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$, known as the **parallelogram rule**.

Naturally, we define the subtraction by $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$.

Properties of vector operations

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors and s, t be scalars. We have:

- $(s + t)\mathbf{u} = s\mathbf{u} + t\mathbf{u}$;
- $(st)\mathbf{u} = s(t\mathbf{u})$;
- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$;
- $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$;
- $s(\mathbf{u} + \mathbf{v}) = s\mathbf{u} + s\mathbf{v}$.

Algebraically, a vector may be defined as

- a matrix with only one row (row vector); or
- a matrix with only one column (column vector).

We usually regard a vector as a column vector, use $\mathbb{R}^n$ to denote the set of all such vectors with n entries.

Vectors in Coordinate System

Two-Dimensional

Every vector $\begin{bmatrix} a \\ b \end{bmatrix}$ in $\mathbb{R}^2$ can be associated with the vector on the rectangular coordinate plane with initial point at the origin and terminal point at the point $P(a, b)$, and hence can be identified with the point P . Such a vector is also known as the **position vector** of P .

Three-Dimensional

Every vector $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ in $\mathbb{R}^3$ can be associated with the vector in the three-dimensional rectangular coordinate system with initial point at the origin and terminal point at the point $P(a, b, c)$, and hence can be identified with the point P . This is known as the position vector of P . (the labelling obeys the right hand rule)

In $\mathbb{R}^2$, we usually use $\mathbf{i}$ and $\mathbf{j}$ to denote the unit vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ respectively.

In $\mathbb{R}^3$, we usually use $\mathbf{i}, \mathbf{j}, \mathbf{k}$ to denote the unit vectors $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ respectively.

Every vector in $\mathbb{R}^2$ and $\mathbb{R}^3$ can be expressed as a linear combination of the standard unit vectors (also called coordinate vectors) defined above.

2.8.2 Dot Product

Definition 2.8.3. Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors. We define

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$. This is known as the **dot product** or **scalar product** of $\mathbf{u}$ and $\mathbf{v}$.

Remark.

- Two vectors are orthogonal (perpendicular) if and only if their dot product is zero.
- The zero vector is by definition orthogonal to any other vector.
- The dot product of $\mathbf{u}$ and $\mathbf{v}$ can be interpreted as the product of the lengths of $\mathbf{u}$ and $\mathbf{v}$ after one is projected onto the other.

Properties of dot product For any vector $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalar k , we have:

- $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$;
- $\mathbf{u} \cdot (k\mathbf{v}) = (k\mathbf{u}) \cdot \mathbf{v}$;
- $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$;
- $\mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2$.

For two vectors $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j}$ in $\mathbb{R}^2$, we have

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2.$$

For two vectors $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ in $\mathbb{R}^3$, we have

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3.$$

Angle between two vectors

For any two vectors $\mathbf{u} = (u_1, u_2, \dots, u_n)^T$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)^T$, we can define their dot product to be

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \dots + u_nv_n.$$

We can define the angle θ between $\mathbf{u}$ and $\mathbf{v}$ by

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

where $\theta \in [0, \pi]$.

2.8.3 Cross Product

Definition 2.8.4. Given two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$, we define the **cross product (vector product)** $\mathbf{u} \times \mathbf{v}$ to be the vector $\mathbf{w}$ such that

- $\|\mathbf{w}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta$, where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$;
- $\mathbf{w}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$ given by the right-hand rule.

Properties 1

- The cross product is a vector.
- We have $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ if and only if $\mathbf{u}$ and $\mathbf{v}$ are parallel, in which case the orientation does not matter.
- The cross product is skew-commutative, i.e.

$$\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u}) \quad \text{for any } \mathbf{u}, \mathbf{v} \in \mathbb{R}^3.$$

Properties 2

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors in $\mathbb{R}^3$ and s be a scalar. Then we have:

- $\mathbf{u} \times \mathbf{0} = \mathbf{0} \times \mathbf{u} = \mathbf{0}$;
- $\mathbf{u} \times \mathbf{u} = \mathbf{0}$;
- $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$;
- $(\mathbf{u} + \mathbf{v}) \times \mathbf{w} = \mathbf{u} \times \mathbf{w} + \mathbf{v} \times \mathbf{w}$;
- $s(\mathbf{u} + \mathbf{v}) = (s\mathbf{u}) \times \mathbf{v} = \mathbf{u} \times (s\mathbf{v})$.

Recall the unit vectors in the three-dimensional coordinate system. We can see that

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}, \mathbf{i} \times \mathbf{k} = -\mathbf{j}, \mathbf{j} \times \mathbf{i} = -\mathbf{k}, \mathbf{j} \times \mathbf{k} = \mathbf{i}, \mathbf{k} \times \mathbf{i} = \mathbf{j}, \mathbf{k} \times \mathbf{j} = -\mathbf{i}.$$

Given two vectors $\mathbf{u} = (u_1, u_2, u_3)^T$ and $\mathbf{v} = (v_1, v_2, v_3)^T$ in $\mathbb{R}^3$, we have

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

Geometric meaning of cross product

- Area of parallelogram ABCD = $\|\overrightarrow{AB} \times \overrightarrow{AC}\|$

- Area of $\triangle ABC = \frac{1}{2} \|\vec{AB} \times \vec{AC}\|$

Scalar Triple Product

Definition 2.8.5. Given three vectors $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^3$, the quantity

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$$

is called the **scalar triple product** of these vectors.

Given three vectors $\mathbf{u} = (u_1, u_2, u_3)^T$, $\mathbf{v} = (v_1, v_2, v_3)^T$ and $\mathbf{w} = (w_1, w_2, w_3)^T$ in $\mathbb{R}^3$, their scalar triple product is given by

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = \det [\mathbf{u} \quad \mathbf{v} \quad \mathbf{w}].$$

The absolute value of $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ is the volume of the parallelepiped formed by these three vectors. The scalar triple product $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ is equal to 0 if and only if these three vectors are coplanar.

Properties

Notice that $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$, the scalar triple product is sometimes written as

$$(\mathbf{u} \mathbf{v} \mathbf{w}).$$

Definition 2.8.6. We also define the following vector triple product

$$\mathbf{u} \times (\mathbf{v} \times \mathbf{w}).$$

- $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$
- $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) + \mathbf{v} \times (\mathbf{w} \times \mathbf{u}) + \mathbf{w} \times (\mathbf{u} \times \mathbf{v}) = \mathbf{0}$

Chapter 3

Revision Notes of MATH2822

3.1 Chapter 1: Coordinate Geometry

3.1.1 Lines in Three Dimensions

A straight line in the two-dimensional rectangular coordinate system can be expressed by

$$y = mx + c$$

where m is the [slope \(or gradient\)](#) of the straight line and c is the [y-intercept](#).

A straight line can be represented in the **vector form**:

$$\mathbf{r} = \mathbf{v} + t\mathbf{u}$$

Here $\mathbf{v}$ represents a specific point on the line and $\mathbf{u}$ represents the direction, while $\mathbf{r}$ represents the position vector of a general point on the line.

Definition 3.1.1. Two lines with same direction ($\mathbf{u}$) are parallel.

Definition 3.1.2. In three dimensions, it is possible that two lines are not parallel and do not have intersections. Such straight lines are said to be [skew lines](#).

Method to find the intersection of two lines given the vector forms:

1. If two lines have parallel direction (their direction vectors are parallel, maybe in opposite directions but still parallel), then two lines do not have any intersection except they overlap.
2. If two lines are not parallel, equal their coordinates respectively and try to solve the parameter t .
3. If there exist such t , plug in t to solve the coordinates of the intersection. If we cannot solve the t , two lines are skew lines.

Proposition 3.1.1. *In general, let L be a line passing through P and with direction $\mathbf{v}$. We can show that the distance from a point S to L is given by*

$$\frac{\|\overrightarrow{PS} \times \mathbf{v}\|}{\|\mathbf{v}\|}.$$

3.1.2 Planes in Three Dimensions

A plane can be uniquely determined by a point and the direction of a [normal \$\mathbf{n}\$](#) , which is a vector that is **orthogonal** to every vector on the plane.

In general, if a plane passes through the point (p, q, r) and has normal vector (A, B, C) , its equation is given by

$$A(x - p) + B(y - q) + C(z - r) = 0.$$

In vector form, this can be expressed as $\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$.

Conversely, given the equation of a plane in the form $Ax + By + Cz + D = 0$, we know that the vector (A, B, C) is normal to the plane.

Definition 3.1.3. If two planes have parallel normal vectors, they are parallel.

If two planes are not parallel, in three dimensions, they must intersect at a straight line.

Method to find the intersection of planes:

1. Since the intersection line lies in the two planes simultaneously, the normal vectors of two planes are orthogonal with the intersection. In other words, the direction of intersection line is same with the direction of cross product of two normal vectors. We compute the cross product to have the direction of intersection line.
2. To find a specific point on the plane, we can assign a value to one of the coordinates. Plug in the assigned value to compute the rest of coordinates using the equation of two planes.

Method to find the intersection of a line and a plane:

1. Transfer the expression of the line into the vector form with parameter t .
2. Plug x, y, z in forms of t in the equation of the plane to solve t . If such t exists, plug in to find the value of coordinates. If such t does not exist, the line is parallel with the plane.

Remark. We can firstly determine whether the line is parallel with the plane by compute the dot product of the direction vector of the line and the normal vector of the plane. If the dot product is 0, then they may be parallel or the line is on the plane.

Proposition 3.1.2. Generally, we can show that the distance from the point (p, q, r) to the plane $Ax + By + Cz + D = 0$ is given by

$$\frac{|Ap + Bq + Cr + D|}{\sqrt{A^2 + B^2 + C^2}}.$$

3.1.3 Graph of Functions

Definition 3.1.4. We define curve and surface as follows:

- Let f be a function of one variable x . The collection of points $(x, f(x))$ is the graph of $y = f(x)$, which is a **curve** in general.
- Let f be a function of two variables x and y . The collection of points $(x, y, f(x, y))$ is the graph of $z = f(x, y)$, which is a **surface** in general.

Definition 3.1.5. Implicit functions are in the form of $f(x, y) = 0$, while explicit functions are in the form of expressing a variable only by other variables.

Definition 3.1.6. If we take a collection of points with the same height (fix z and collect all pairs of corresponding (x, y)), and that gives us a **contour line** (also called a level curve). If take a top view, we get contour map.

Some special curves and surfaces:

Conic sections in $\mathbb{R}^2$

Conic sections can be expressed generally as

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

- **Parabola:** locus of points in that plane that are equidistant from the directrix and the focus
- **Ellipse:** locus of points that the sum of distances to the focal points is a constant
- **Hyperbola:** locus of points that the difference of distances to the focal points is a constant

Cylindrical surfaces and quadratic surfaces

- A curve $f(x, y) = 0$ in $\mathbb{R}^2$ can be used to generate a surface $g(x, y, z) = 0$ in $\mathbb{R}^3$ by defining $g(x, y, z) = f(x, y)$. The resulting surface is called a **cylindrical surface**.
- A **quadratic surface** is a surface in $\mathbb{R}^3$ that can be represented by an equation of degree 2 in x, y and z . Its general form is

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fzx + Gx + Hy + Iz + J = 0$$

and it is a natural generalization of the notion of conic sections in $\mathbb{R}^2$.

- A **sphere** centered at the origin with radius r has equation

$$x^2 + y^2 + z^2 = r^2.$$

- An **ellipsoid** is a surface of the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

horizontal cross section: ellipses, vertical cross section: ellipses

- An **elliptic paraboloid** is a surface of the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}.$$

horizontal cross section: ellipses, vertical cross section: parabolas

- A **hyperbolic paraboloid** is a surface of the form

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{z}{c}.$$

horizontal cross section: hyperbolas, vertical cross section: parabolas

- An **elliptic cone** is a surface of the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}.$$

horizontal cross section: ellipses, vertical cross section: hyperbolas

- A **hyperboloid of one sheet** is a surface of the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2} + 1.$$

horizontal cross section: ellipses, vertical cross section: hyperbolas

- An **hyperboloid of two sheets** is a surface of the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2} - 1.$$

horizontal cross section: ellipses, vertical cross section: hyperbolas

3.1.4 More coordinates systems

Definition 3.1.7 (Polar coordinates). We can use (r, θ) , $\theta \in [0, 2\pi)$ to represent a point in two dimensions with $x = r \cos \theta$ and $y = r \sin \theta$ while (x, y) in rectangular coordinates.

Definition 3.1.8 (Cylindrical coordinates). by adding the z -coordinate to the polar coordinate system, we get the **cylindrical coordinate system** which describes positions in three dimensions. The coordinates are denoted by (r, θ, z) .

Definition 3.1.9 (Latitudes and longitudes for positions on earth). We define the system as follows:

- The latitude measures the angle relative to the **equatorial plane** with north being positive and ranges from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$.
- The longitude measures the angle relative to the **prime meridian** with east being positive and ranges from $-\pi$ to π .

Definition 3.1.10 (Spherical coordinates). The **spherical coordinate system** uses a triple (ρ, ϕ, θ) to denote the position of a point P in three dimensions.

- ρ is the distance of P from the origin O
- ϕ is the angle with the positive z -axis to $\overrightarrow{OP}$, ranges from 0 to π
- θ is the same angle when P is expressed in cylindrical coordinates, ranges from 0 to 2π (Note that ϕ and θ play similar roles as the latitude and longitude respectively)

3.2 Chapter 2: Partial Differentiation

3.2.1 Partial derivative

Definition 3.2.1. Let $f(x, y)$ be a function. The limit

$$\lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h},$$

if exists, is known as the **partial derivatives** of f at (a, b) **with respect to x** . Similarly, we can define the partial derivative of f at (a, b) with respect to y as

$$\lim_{h \rightarrow 0} \frac{f(a, b+h) - f(a, b)}{h}.$$

The partial derivative of f at (a, b) with respect to x can be denoted by

$$f_x(a, b) \quad \text{or} \quad \frac{\partial f}{\partial x}(a, b).$$

To find the partial derivative of $f(x, y)$ with respect to a certain variable, just treat the other variable as constant and then differentiate as in the one-variable case. Rules and formulas in one-variable case can also be used.

Definition 3.2.2 (Second order partial derivative). We can carry out partial differentiation twice on $f(x, y)$ to obtain second order partial derivatives of f .

Notations: $f_{xx}, f_{yy}, f_{xy}, f_{yx}$ or $\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}$.

Theorem 3.2.1 (Mixed derivative theorem). Let $f(x, y)$ be a function. If all second order partial derivatives of f at (a, b) exist and are continuous, then $f_{xy}(a, b) = f_{yx}(a, b)$.

Theorem 3.2.2 (Chain rule). Let $w = f(x, y)$. If x and y are functions of t , then subject to some technical conditions on differentiability, we have

$$\frac{\partial w}{\partial t} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

If $x_0 = x(t_0)$ and $y_0 = y(t_0)$, then the chain rule can also be written as

$$w'(t_0) = f_x(x_0, y_0)x'(t_0) + f_y(x_0, y_0)y'(t_0).$$

Remark. Sometimes, we can use the branch diagrams to assist us.

Proposition 3.2.1 (Implicit differentiation). Suppose y is a function of x , defined implicitly by the equation $F(x, y) = 0$. Then, subject to some technical conditions on differentiability, we have

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

3.2.2 Directional derivatives

Definition 3.2.3. Let $\mathbf{u} = s\mathbf{i} + t\mathbf{j}$ be a unit vector. The **directional derivative of f at (a, b) in the direction of $\mathbf{u}$** , denoted $D_{\mathbf{u}}f(a, b)$, is defined to be

$$D_{\mathbf{u}}f(a, b) = \lim_{h \rightarrow 0} \frac{f(a + hs, b + ht) - f(a, b)}{h}.$$

Remark. We remark the follows:

- Setting $\mathbf{u} = \mathbf{i}$ and $\mathbf{u} = \mathbf{j}$ to reduce $D_{\mathbf{u}}f$ to f_x and f_y respectively.
- If we define $g(h) = f(a + hs, b + ht)$, then $D_{\mathbf{u}}f(a, b) = g'(0)$.

Definition 3.2.4 (Gradient). Let $f(x, y)$ be a function. If $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist at (a, b) , then we define

$$\nabla f(a, b) = \frac{\partial f}{\partial x}(a, b)\mathbf{i} + \frac{\partial f}{\partial y}(a, b)\mathbf{j}$$

to be the **gradient** of f at (a, b) .

By writing $g(h) = f(a + hs, b + ht)$ and using the chain rule, we readily see that

$$D_{\mathbf{u}}f(a, b) = (\nabla f(a, b)) \cdot \mathbf{u}.$$

3.2.3 Optimization

(Review of one-variable case) If $f(x)$ has a **stationary point** at $x = a$, i.e. $f'(a) = 0$, we can use second derivative test to determine the nature of the stationary point.

Proposition 3.2.2. For a function $f(x, y)$ of two variables, if $f(x, y)$ attains local extremum at an interior point (a, b) and if $\nabla f(a, b)$ exists, then $\nabla f(a, b) = \mathbf{0}$.

Definition 3.2.5 (Stationary point and critical point). We call (a, b) a **stationary point** if $\nabla f(a, b) = \mathbf{0}$. We call (a, b) a **critical point** if ∇f either does not exist or is equal to zero. A stationary point is a point where the tangent plane to the graph is horizontal. A stationary point may be a local maximum, a local minimum, or neither. A stationary point which is neither a local maximum nor a local minimum is called a **saddle point**.

Definition 3.2.6. For a two-variable function $f(x, y)$, we define $H_f = f_{xx}f_{yy} - f_{xy}f_{yx}$. This is called the **discriminant** or **Hessian** of f . Note that this can be expressed as a **determinant**:

$$H_f = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$$

Proposition 3.2.3. Hessian and nature of stationary points

- If Hessian is positive, the stationary point is a local minimum if $f_{xx}, f_{yy} > 0$, is a local maximum if $f_{xx}, f_{yy} < 0$
- If Hessian is negative, the stationary point is a saddle point
- If Hessian is 0, we cannot conclude anything, may consider alternative ways

Constrained optimization:

- If the variables are constrained by a surface with boundary, we need to discuss on boundary and inner part respectively.
- If the variables are constrained by a curve (an equation), consider reduction of variable or method of Lagrange multipliers.

Method of Lagrange multiplier 1:

Consider the problem

optimizing $f(x, y)$ subject to $g(x, y) = c$ where c is a constant.

Then, subject to technical conditions on differentiability, at a point (a, b) where extremum occurs, we have

$$\nabla f(a, b) = \lambda \nabla g(a, b)$$

for some real number λ known as the **Lagrange multiplier**.

Method of Lagrange multiplier 2:

Consider the problem

optimizing $f(x, y)$

subject to $g_1(x, y, z) = c_1$ and $g_2(x, y, z) = c_2$

where c_1 and c_2 are constants.

Then, subject to technical conditions on differentiability, at a point (a, b, c) where extremum occurs, we have

$$\nabla f(a, b, c) = \lambda_1 \nabla g_1(a, b, c) + \lambda_2 \nabla g_2(a, b, c)$$

for some real numbers λ_1 and λ_2 .

3.3 Chapter 3: More on Differentiation

3.3.1 Limits and Continuity

(Review of one-variable case) Let $f(x)$ be a function. We say that

- $\lim_{x \rightarrow c} f(x) = L$ if the value of $f(x)$ gets closer and closer to L as x gets closer and closer to c ;
- f is continuous at c if $\lim_{x \rightarrow c} f(x)$ exists and is equal to $f(c)$.

(Two-variable case) Let $f(x, y)$ be a function of two variables. We say that

- $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$ if the value of $f(x, y)$ gets closer and closer to L as (x, y) gets closer and closer to (a, b) ;
- f is continuous at (a, b) if $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ exists and is equal to $f(a, b)$.

Method to evaluate limits:

- Direct substitution
- Sandwich theorem (transform to polar coordinates)

Theorem 3.3.1 (Sandwich theorem). Suppose that D is a subset of $\mathbb{R}^2$ with $(a, b) \in D$, and that

$$g(x, y) \leq f(x, y) \leq h(x, y) \quad \text{for all } (x, y) \in D \setminus \{(a, b)\}.$$

If $\lim_{(x,y) \rightarrow (a,b)} g(x, y) = \lim_{(x,y) \rightarrow (a,b)} h(x, y) = L$, then $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$.

Remark. In two-variable case, if the limit does not exist, one of the following will suffice:

- The limit is different when restricted to two different directions (usually limit is different along two different curves);
- The limit does not exist along a certain direction.

3.3.2 Differentiability

(Review of one-variable case) Let $f(x)$ be a function. When we say that f is differentiable at $x = c$, it is in a sense characterized by the existence of a [tangent line](#), which provides a good approximation to the function value at [any](#) point that is close to c .

Note that if f is differentiable at $x = c$, then

$$\lim_{x \rightarrow c} \frac{f(x) - f(c) - f'(c)(x - c)}{x - c} = 0.$$

(Two-variable case) In a similar manner, given a two-variable function $f(x, y)$, we would like to define differentiability at a point $(x, y) = (a, b)$ that involves the existence of a [tangent plane](#) which provides a good approximation to the function value at [any](#) point that is close to (a, b) .

Definition 3.3.1. Let $f(x, y)$ be a function and (a, b) be an interior point in the domain. We say that f is [differentiable](#) at (a, b) if

$$\lim_{(x,y) \rightarrow (a,b)} \frac{f(x, y) - f(a, b) - f_x(a, b)(x - a) - f_y(a, b)(y - b)}{\|(x, y) - (a, b)\|} = 0.$$

The [derivative](#) of f is the 1×2 matrix $Df = [f_x \quad f_y]$.

Some further properties about differentiability:

- If $f(x, y)$ is differentiable at (a, b) , then the partial derivatives f_x and f_y at (a, b) must exist. But the converse may not be true.
- If f_x and f_y [exist and continuous](#) at (a, b) , then f is differentiable at (a, b) .
- If f is differentiable at (a, b) , then f is continuous at (a, b) .

3.3.3 Linear Approximations

Let $f(x, y)$ be a function of two variables. Subject to some technical conditions on differentiability, we have

$$\Delta f \approx f_x \Delta x + f_y \Delta y$$

for small changes Δx and Δy .

Remark. We have the following remarks:

- This is basically equivalent to using the tangent plane of the graph of $f(x, y)$ at a point to approximate f at nearby points.
- The right hand side is called the **differential** of f , denoted by df .

Definition 3.3.2. Linearisation Let $f(x, y)$ be a function and (a, b) be a point in the domain of f at which the partial derivatives f_x and f_y exist. We define

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

to be the **linearisation** of $f(x, y)$ at (a, b) .

Remark. Subject to some conditions on differentiability, we have

$$\lim_{(x, y) \rightarrow (a, b)} \frac{|f(x, y) - L(x, y)|}{\|(x, y) - (a, b)\|} = 0.$$

So the linearisation of f provides a good approximation to f in the above sense.

Newton's method - one-variable case:

Recall that in solving the one-variable equation $f(x) = 0$, we may start with an initial guess x_0 and apply the formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

to work out the iterates, which, in many cases, will converge to a root of the equation. Newton's method makes use of the tangent line to the graph $y = f(x)$ at the point $x = x_n$ to work out the next iterate, so it is a linear approximation in some sense.

Newton's method - two-variable case:

Suppose we want to solve the system of equations

$$\begin{cases} f(x, y) = 0 \\ g(x, y) = 0. \end{cases}$$

With an initial guess $\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$, we may use the formula

$$\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = \begin{bmatrix} x_n \\ y_n \end{bmatrix} - \begin{bmatrix} Df(x_n, y_n) \\ Dg(x_n, y_n) \end{bmatrix}^{-1} \begin{bmatrix} f(x_n, y_n) \\ g(x_n, y_n) \end{bmatrix}$$

to obtain iterates which, in many cases, will converge to roots of the system.

3.3.4 Higher-Order Approximations

(Review of one-variable case) Let f be a function that has continuous derivatives of all orders on an interval I containing a . The n -th Taylor polynomial of f at a is

$$P_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n.$$

We have

$$f(x) - P_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - a)^{n+1},$$

where c is a real number between x and a .

The following is an equivalent formulation. The n -th Taylor polynomial of f at a is

$$P_n(a+h) = f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \cdots + \frac{f^{(n)}(a)}{n!}h^n.$$

We have

$$f(a+h) - P_n(a+h) = \frac{f^{(n+1)}(a+th)}{(n+1)!}h^{n+1},$$

where $t \in (0, 1)$.

Two-variable case

- When $n = 1$, the Taylor polynomial is

$$P_1(a+h, b+k) = f(a, b) + f_x(a, b)h + f_y(a, b)k.$$

The error term is

$$\frac{1}{2!}(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2) \Big|_{(a+th, b+tk)}$$

for some $t \in (0, 1)$. This can also be expressed using the [differential operator](#) notation

$$\frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^2 f(a+th, b+tk).$$

- When $n = 2$, the Taylor polynomial is

$$\begin{aligned} P_2(a+h, b+k) &= f(a, b) + f_x(a, b)h + f_y(a, b)k \\ &\quad + \frac{1}{2!}[f_{xx}(a, b)h^2 + 2f_{xy}(a, b)hk + f_{yy}(a, b)k^2] \end{aligned}$$

while the error term is

$$\frac{1}{3!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^3 f(a+th, b+tk)$$

for some $t \in (0, 1)$. Note that the second Taylor polynomial can be abbreviated as

$$f(a, b) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right) f(a, b) + \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^2 f(a, b).$$

Definition 3.3.3 (General form of Taylor's theorem). Let $f(x, y)$ be a function with continuous partial derivatives up to order $n+1$. For any interior point (a, b) of the domain of f , we have

$$f(a+h, b+k) = \sum_{i=0}^n \frac{1}{i!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^i f(a, b) + \frac{1}{(n+1)!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^{n+1} f(a+th, b+tk)$$

for some $t \in (0, 1)$.

3.4 Chapter 4: Multiple Integrals

3.4.1 Double Integrals

(Review of one-variable case) Let $f(x)$ be a nice function and $a, b \in \mathbb{R}$ with $a < b$. The [definite integral](#)

$$\int_a^b f(x) dx$$

represents the [signed area](#) under the graph of $y = f(x)$ between $x = a$ and $x = b$.

This can be seen as the limit of the sum

$$\sum_{x=a}^{x=b} f(x) \Delta x$$

as $\Delta x \rightarrow 0$.

(Two-variable case) Let $f(x, y)$ be a function of two variables. We can likewise consider the [signed volume](#) under the graph of $z = f(x, y)$ between $x = a$ and $x = b$, and between $y = c$ and $y = d$.

Again, this can be seen as the limit of the sum

$$\sum_{x \in [a, b], y \in [c, d]} f(x, y) \Delta x \Delta y$$

as $\Delta x, \Delta y \rightarrow 0$.

Definition 3.4.1. The region bounded by $x = a, x = b, y = c$ and $y = d$ form a rectangle, which we denote by $D = [a, b] \times [c, d]$ (it means $D = \{(a, y) : a \leq x \leq b, c \leq y \leq d\}$). Writing $\Delta A = \Delta x \Delta y$, the summation becomes

$$\sum_D f(x, y) \Delta A.$$

As $\Delta A \rightarrow 0$ we get the [double integral](#)

$$\iint_D f(x, y) dA$$

or

$$\iint_D f(x, y) dx dy.$$

Definition 3.4.2 (Iterated integrals). We noted that

$$\int_0^1 \int_0^2 (10 - 4x - 2y) dy dx = \int_0^2 \int_0^1 (10 - 4x - 2y) dx dy$$

and they are called [iterated integrals](#). Both of them are equal to

$$\iint_D (10 - 4x - 2y) dA$$

where $D = [0, 1] \times [0, 2]$.

Theorem 3.4.1 (Fubini's theorem). Let $D = [a, b] \times [c, d]$ and $f(x, y)$ be continuous on D . Then

$$\iint_D f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx.$$

Definition 3.4.3 (Fubini's theorem (strong form)). Let $D = \{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$ where $f(x, y)$ is continuous on D and $g_1(x), g_2(x)$ are continuous on $[a, b]$. Then

$$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

Let $D = \{(x, y) : c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$ where $f(x, y)$ is continuous on D and $h_1(y), h_2(y)$ are continuous on $[c, d]$. Then

$$\iint_D f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$

3.4.2 Change of Variables

Let $x = g(u, v)$ and $y = h(u, v)$, where g and h have continuous partial derivatives on D' . If $(g, h) : D' \rightarrow D$ is bijective, then

$$\iint_D f(x, y) dx dy = \iint_{D'} f(g(u, v), h(u, v)) |J(u, v)| du dv.$$

Here, $J(u, v)$ is called the [Jacobian](#) of the transformation defined by

$$J(u, v) = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}.$$

It is also sometimes denoted by $\frac{\partial(x, y)}{\partial(u, v)}$.

Integration in Polar Coordinates:

Generally, subject to some technical conditions, we have

$$\text{Area of } D = \iint_{D'} r \, dr \, d\theta$$

where D' is the same region D but considered in polar coordinates.

The term $r \, dr \, d\theta$ is sometimes called the **area element** (or **area differential**).

If we replace the integrand 1 by an arbitrary function $f(x, y)$, then we have

$$\iint_D f(x, y) \, dx \, dy = \iint_{D'} f(r \cos \theta, r \sin \theta) \cdot r \, dr \, d\theta.$$

Definition 3.4.4 (Triple integrals). Let D be a region in three-dimensional space and $f(x, y, z)$ be a function of three variables x, y and z . Naturally, the integral

$$\iiint_D f(x, y, z) \, dV$$

represents the **four-dimensional volume** under the graph of f over D .

(Change of variables for triple integrals) Recall the following formula:

$$\iint_D f(x, y) \, dx \, dy = \iint_{D'} f(g(u, v), h(u, v)) |J(u, v)| \, du \, dv.$$

Subject to the same technical conditions, we also have

$$\iiint_D f(x, y, z) \, dx \, dy \, dz = \iiint_{D'} f(g_1(u, v, w), g_2(u, v, w), g_3(u, v, w)) |J(u, v, w)| \, du \, dv \, dw.$$

Triple integrals in cylindrical coordinates:

Suppose D is a region in $\mathbb{R}^3$ and D' is the same region considered in cylindrical coordinates. Then, subject to some technical conditions, we have

$$\text{Volume of } D = \iiint_{D'} r \, dr \, d\theta \, dz.$$

Triple integrals in spherical coordinates:

Suppose D is a region in $\mathbb{R}^3$ and D' is the same region considered in spherical coordinates. Then, subject to some technical conditions, we have

$$\text{Volume of } D = \iiint_{D'} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

3.5 Chapter 5: Systems of Linear Equations

3.5.1 Matrix Representations and Solutions of Linear Systems

Definition 3.5.1. Generally, a system of m linear equations in n unknowns can be written in the form

$$A\mathbf{x} = \mathbf{b}$$

where A is an $m \times n$ matrix (called the **coefficient matrix**), $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{b} \in \mathbb{R}^m$ (the vector $\mathbf{b}$ is sometimes called the **RHS vector**).

Moreover, we can denote the columns of A by $\mathbf{a}_1, \mathbf{a}_2, \dots$ and the entries of $\mathbf{x}$ by $x_1, x_2, \dots$. In this case, we can express $\mathbf{b}$ as a **linear combination** of the columns of A :

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \dots + x_n \mathbf{a}_n = \mathbf{b}.$$

Definition 3.5.2 (Augmented matrix). By appending the RHS vector to the coefficient matrix, we get the **augmented matrix** of the linear system. For example

$$\begin{cases} x_1 + 2x_2 + x_3 + x_4 = 3 \\ 3x_1 - x_2 - 3x_3 + 3x_4 = -1 \\ 2x_1 + 3x_2 + x_3 - 2x_4 = 4 \end{cases} \leftrightarrow \left[\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 3 \\ 3 & -1 & -3 & 3 & -1 \\ 2 & 3 & 1 & -2 & 4 \end{array} \right]$$

Remark. Note that

- each row of the augmented matrix corresponds to an equation;
- Each column except the last corresponds to an unknown.

Definition 3.5.3 (Consistency of linear system and existence of solutions). A linear system is said to be **consistent** if it has at least one solution. Hence, $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is a linear combination of the columns of A .

Not all linear systems are consistent.

Definition 3.5.4 (Homogeneous linear systems). When $\mathbf{b} = \mathbf{0}$, the linear system $A\mathbf{x} = \mathbf{b}$ becomes $A\mathbf{x} = \mathbf{0}$. Such system is called a **homogeneous** linear system and such system is always consistent as it has the **trivial solution** (all-zero solution).

If the homogeneous linear system is consistent and has a non-trivial solution, then the columns of A are linearly dependent.

Solving Linear Systems - Elementary Methods:

Definition 3.5.5 (Row echelon matrices). A matrix is said to be in **row echelon form** (and is called a **row echelon matrix**) if it satisfies:

1. Zero rows, if any, must be at the bottom.
2. The leading entry (first non-zero entry) of a non-zero row must be on the right of the leading entries in the rows above (which implies the entries below a leading entry must be 0).

Here are some examples and non-examples:

$$\begin{bmatrix} 1 & 0 & 7 \\ 0 & 2 & 7 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 3 & 0 & 6 \\ 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Definition 3.5.6 (Reduced row echelon matrices). A row echelon matrix is said to be in **reduced row echelon form** (and is called a **reduced row echelon matrix**) if it further satisfies:

1. Where there is a leading entry, the other entries in the same column must be 0.
2. The leading entry of a non-zero is 1 (called a leading 1).

Here are some examples and non-examples:

$$\begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 7 & 3 \\ 0 & 0 & 2 & 5 \\ 0 & 1 & 3 & 4 \end{bmatrix}, \begin{bmatrix} 4 & 4 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Use REF and RREF to solve a linear system:

If A is a row echelon matrix, we can easily solve the linear system $A\mathbf{x} = \mathbf{b}$ by means of **backward substitution**, which means solving the unknowns from the last row, and substituting the values of some unknowns for the rest in following calculations.

If A is a reduced row echelon matrix, we can read the solution to the linear system $A\mathbf{x} = \mathbf{b}$ directly from the augmented matrix $[A|\mathbf{b}]$.

Note that we have set variables corresponding to **non-leading columns** as **free variables**. The variables corresponding to **leading columns** become **basic variables**.

Definition 3.5.7 (Elementary row operations). Note that a linear system $A\mathbf{x} = \mathbf{b}$ will be easily solved if A is in REF or RREF. We can apply elementary row operations to the augmented matrix to transform A into a nicer form. Three elementary row operations (EROs) are:

1. Exchange two rows
2. Multiply a row by a nonzero constant
3. Add a multiple of a row to another row

Two matrices are said to be **row equivalent** if one can be obtained from the other by a sequence of EROs.

Remark. Two row equivalent matrices correspond to two equivalent linear systems.

Gaussian Elimination and steps:

Gaussian elimination can assist us to transform the matrix into an easily-solved form.

1. Identify the leftmost nonzero column.
2. Make the top entry of this column (this position is known as the **pivot position**) nonzero by a Type I ERO.
3. Make the entries below the pivot position zero by Type III EROs.
4. Ignore the row containing the pivot position and all rows above.
5. Repeat previous steps if there is a nonzero row in the submatrix that remains.
6. Make the entry in the last pivot position 1 by a Type II ERO.
7. Make all other entries in the same column 0 by Type III EROs.
8. Repeat step 6 and 7 with the row above, until all rows have been handled.

Example of Gaussian Elimination:

$$\begin{aligned}
 & \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 2 & 2 & 4 & 2 \\ 0 & 2 & -1 & 5 & 2 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{bmatrix} 0 & 2 & 2 & 4 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 2 & -1 & 5 & 2 \end{bmatrix} \xrightarrow{-r_1 + r_3 \rightarrow r_3} \begin{bmatrix} 0 & 2 & 2 & 4 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & -3 & 1 & 0 \end{bmatrix} \xrightarrow{3r_2 + r_3 \rightarrow r_3} \\
 & \begin{bmatrix} 0 & 2 & 2 & 4 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 4 & 3 \end{bmatrix} \xrightarrow{\frac{1}{4}r_3 \rightarrow r_3} \begin{bmatrix} 0 & 2 & 2 & 4 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & \frac{3}{4} \end{bmatrix} \xrightarrow{\begin{matrix} -r_3 + r_2 \rightarrow r_2 \\ -4r_3 + r_1 \rightarrow r_1 \end{matrix}} \begin{bmatrix} 0 & 2 & 2 & 0 & -\frac{1}{4} \\ 0 & 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 1 & \frac{3}{4} \end{bmatrix} \xrightarrow{-2r_2 + r_1 \rightarrow r_1} \\
 & \begin{bmatrix} 0 & 2 & 0 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 1 & \frac{3}{4} \end{bmatrix} \xrightarrow{\frac{1}{2}r_1 \rightarrow r_1} \begin{bmatrix} 0 & 1 & 0 & 0 & -\frac{3}{4} \\ 0 & 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}
 \end{aligned}$$

Proposition 3.5.1. A linear system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if the reduced row echelon form of the augmented matrix of the system has no row of the form $[0 \ 0 \ \dots \ 0 \ 1]$.

Example:

If the resulting augmented matrix after Gaussian elimination is like below, the linear system is not consistent:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right].$$

3.5.2 Rank and Nullity

Definition 3.5.8. Let A be an $m \times n$ matrix.

- The **rank** of A (denoted by $\text{rank } A$) is the number of nonzero rows in the reduced row echelon form of A .
- The **nullity** of A (denoted by $\text{nullity } A$) is defined to be $n - \text{rank } A$.

Remark. Note that

1. the rank of a matrix A is also the number of leading 1's in the reduced row echelon form of A ;
2. If the system $A\mathbf{x} = \mathbf{b}$ is consistent, the rank and nullity of A indicate the number of basic variables and free variables respectively.

Definition 3.5.9 (Span). Let S be a finite non-empty set of vectors in $\mathbb{R}^n$. The **span** of S , denoted by $\text{Span } S$, is defined to be the set of all linear combinations of the vectors in S .

That is, if $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$, then

$$\text{Span } S = \{a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + \dots + a_k\mathbf{u}_k : a_1, a_2, \dots, a_k \in \mathbb{R}\}.$$

Definition 3.5.10 (Generating set). Let $V \subseteq \mathbb{R}^n$. We say that a finite set S of vectors in $\mathbb{R}^n$ is a **generating set** for V (or simply S **generates** V) if $\text{Span } S = V$.

Definition 3.5.11 (Standard vector). A vector whose i -th entry is 1 and all other entries are 0 is called a **standard vector** (or **standard unit vector**) and is denoted by $\mathbf{e}_i$.

Proposition 3.5.2. Let A be an $m \times n$ matrix. The following statements are equivalent:

1. $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^m$.
2. The span of the columns of A is $\mathbb{R}^m$.
3. The reduced row echelon form of A has no zero row.
4. $\text{rank } A = m$.

Theorem 3.5.1. Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$. Then

$$\text{Span } S = \text{Span}(S \cup \{\mathbf{u}_{k+1}\}) \Leftrightarrow \mathbf{u}_{k+1} \in \text{Span } S.$$

Definition 3.5.12 (Linear dependence). Vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ in $\mathbb{R}^n$ are said to be **linearly dependent** if one of them can be expressed as a linear combination of the others. (We may also say that the set of such vectors is a linearly dependent set.)

In other words, there exist scalars $c_1, c_2, \dots, c_k$, not all 0, such that $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k = \mathbf{0}$.

Vectors which are not linearly dependent are said to be **linearly independent**. (We may call the set containing such vectors a linearly independent set.) That means none of the vectors can be expressed as a linear combination of the others.

Equivalently, the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ in $\mathbb{R}^n$ are linearly independent if $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k = \mathbf{0}$ implies $c_1 = c_2 = \dots = c_k = 0$.

Proposition 3.5.3. Let A be an $m \times n$ matrix. The following statements are equivalent:

1. The columns of A are linearly independent.
2. $A\mathbf{x} = \mathbf{b}$ has at most one solution for every $\mathbf{b} \in \mathbb{R}^m$.
3. $\text{nullity } A = 0$.
4. $\text{rank } A = n$.
5. The reduced row echelon form of A is $[\mathbf{e}_1 \ \mathbf{e}_2 \ \dots \ \mathbf{e}_n]$.
6. The system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

3.6 Chapter 6: More on Matrices and Determinants

3.6.1 Block Multiplication, EROs, and Matrix Multiplication

Block Multiplication In evaluating a matrix product AB , note that we need to guarantee that we partition A and B in the same pattern so that all the multiplications are well-defined. Example:

$$\begin{bmatrix} 1 & 2 & 1 & 0 & 5 & 6 \\ 3 & 4 & 0 & 1 & 7 & 8 \\ 7 & 7 & 2 & 0 & 0 & 0 \\ 7 & 7 & 0 & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 4 & 4 \\ 4 & 4 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} S & I & T \\ 7J & 2I & 0 \end{bmatrix} \begin{bmatrix} I \\ 4J \\ -I \end{bmatrix} = \begin{bmatrix} SI + I \cdot 4J + T \cdot (-I) \\ 7J \cdot I + 2I \cdot 4J + 0 \cdot (-I) \end{bmatrix} = \begin{bmatrix} S + 4J - T \\ 15J \end{bmatrix} =$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 15 & 15 \\ 15 & 15 \end{bmatrix}$$

EROs and Matrix Multiplication

Proposition 3.6.1. Applying an ERO on A is equivalent to multiplying on the left of A by a matrix E , where E is a matrix obtained by applying the same ERO on I .

Definition 3.6.1 (Elementary matrix). A matrix obtained from I by a single ERO is called an **elementary matrix**. An elementary matrix can be classified as Type I, Type II, Type III according to the type of the ERO from which the matrix is obtained from I .

Column Correspondence Property Let A be a matrix and R be its reduced row echelon form.

- If certain columns of R are linearly independent, so are the corresponding columns of A , and vice versa.
- If a certain column of R is a linear combination of other columns, then the same linear relation (with the same coefficients) holds for the corresponding columns of A , and vice versa.

3.6.2 Determinants

Definition 3.6.2. Determinant In general, we define the determinant of a square matrix using **cofactor expansion along the first row**:

Let A be an $n \times n$ matrix, where $n > 1$. We define

$$\det A = a_{11}c_{11} + a_{12}c_{12} + \cdots + a_{1n}c_{1n}.$$

Theorem 3.6.1 (Cofactor expansion theorem). Let A be an $n \times n$ matrix. If we multiply the elements in any row or column of A by their cofactors, then the sum of the resulting products (i.e. **cofactor expansion along any row or column of A**) is equal to $\det A$.

Remark. We can take advantages of some rows or columns with a lot of zero entries to simplify calculations.

Proposition 3.6.2. The determinants of a **triangular matrix** is the product of its diagonal entries.

$$\det A = \prod_{i=1}^n a_{ii} \quad \text{if } A \text{ is a triangular matrix.}$$

Proposition 3.6.3. The determinant of a **matrix in triangular block form** is the product of determinants of diagonal blocks.

$$\text{If } A = \begin{bmatrix} A_{11} & * & \cdots & * \\ O & A_{22} & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & A_{kk} \end{bmatrix} \text{ and all } A_{ii} \text{'s are square blocks, then } \det A = \prod_{i=1}^k \det A_{ii}.$$

Properties of Determinants: Let A and B be square matrices with the same size. Then

1. A is invertible if and only if $\det A \neq 0$
2. $\det AB = \det A \det B$
3. $\det A^T = \det A$
4. $\det A^{-1} = \frac{1}{\det A}$ if A is invertible

Proposition 3.6.4. If E is an elementary matrix, then E is invertible and E^{-1} is an elementary matrix of the same type.

Proposition 3.6.5. Let A be an $m \times n$ matrix and R be its reduced row echelon form. Then there exist elementary matrices $E_1, E_2, \dots, E_k$ of size $m \times m$ such that

$$E_k E_{k-1} \cdots E_2 E_1 A = R.$$

This can also be expressed as $PA = R$ where $P = E_k E_{k-1} \cdots E_2 E_1$ is a invertible matrix.

Effects of EROs on Determinants: Effects of different types of EROs on determinant:

- I: Switch the sign
- II: Multiplied by k
- III: Unchanged

Since $\det EA = \det E \det A$, we know that

- $\det E = 1$ if E is a Type I elementary matrix
- $\det E = k$ if E is a Type II elementary matrix obtained by multiplying a row by scalar k
- $\det E = 1$ if E is a Type III elementary matrix

Effects of EROs on determinants also apply to elementary **column operations**, so we can apply suitable column operations to transform a matrix into a nicer form to simplify calculations on determinants.

3.6.3 Invertible Matrix

(Review of Invertible Matrix) Recall that a **square** matrix A is said to be **invertible** if there exists a matrix B , called the **inverse** of A , such that $AB = BA = I$.

We have a formula for computing matrix inverse as follows

$$A^{-1} = \frac{1}{\det A}(\text{adj}A).$$

For a linear system $A\mathbf{x} = \mathbf{b}$ and A is invertible, we can solve the system by $\mathbf{x} = A^{-1}\mathbf{b}$.

Theorem 3.6.2 (Cramer's rule). *If A is an invertible square matrix, the solution to the linear system $A\mathbf{x} = \mathbf{b}$ is given by*

$$x_i = \frac{\det A_i}{\det A}$$

where A_i is the matrix obtained from A by replacing the i -th column by $\mathbf{b}$.

Theorem 3.6.3. *A square matrix A is invertible if and only if its reduced row echelon form is I . In other words,*

- if the reduced row echelon form of $[A|I]$ is $[R|B]$ where $R \neq I$, then A is not invertible.
- if the reduced row echelon form of $[A|I]$ is $[I|B]$, then B is the inverse of A .

Proposition 3.6.6. *The following statements are equivalent for an $n \times n$ matrix A :*

1. A is invertible.
2. The reduced row echelon form of A is I .
3. $\text{rank } A = n$ (i.e. $\text{nullity } A = 0$).
4. The span of the columns of A is $\mathbb{R}^n$.
5. $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^n$.
6. The columns of A are linearly independent.
7. $A\mathbf{x} = \mathbf{0}$ only has the trivial solution.
8. There exists a matrix B such that $BA = I$.
9. There exists a matrix C such that $AC = I$.
10. A is a product of elementary matrices.
11. $\det A \neq 0$.

3.6.4 Linear Transformation

Definition 3.6.3. A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be a **linear transformation** if

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \quad \text{and} \quad T(c\mathbf{u}) = cT(\mathbf{u})$$

for any $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and real number c .

Properties:

If T is a linear transformation, then

- $T(\mathbf{0}) = \mathbf{0}$ (notice two $\mathbf{0}$ may in different space)
- $T(-\mathbf{u}) = -T(\mathbf{u})$
- $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$
- $T(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n) = c_1T(\mathbf{u}_1) + c_2T(\mathbf{u}_2) + \cdots + c_nT(\mathbf{u}_n)$

Proposition 3.6.7. A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation if and only if there exists an $m \times n$ matrix A such that $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x}$. Such matrix A is called the **standard matrix** of T .

Definition 3.6.4. Null space Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation with standard matrix A . The **null space** of T (**kernel** of T) is the set of all $\mathbf{v} \in \mathbb{R}^n$ such that $T(\mathbf{v}) = \mathbf{0}$.

3.7 Chapter 7: Vector Spaces

3.7.1 Vector Space

Definition 3.7.1 (Vector space). A **vector space** is a set which allows **addition** and **scalar multiplication**, and satisfy:

- the addition is **commutative** and **associative**;
- every element has an additive inverse;
- addition and multiplication are **distributive**.

Remark. Note that

- $\mathbb{N}, \mathbb{Z}$, and the set of all matrices with real entries are not vector spaces.
- $\mathbb{R}[x], \mathbb{R}^4$, the set of all 3×4 matrices with real entries, the set of all solutions to the ODE $y'' - 7y' + 10y = 0$ are vector spaces.
- We accept that $\mathbb{R}^n$ is a vector space in this course.

Definition 3.7.2 (Subspace of $\mathbb{R}^n$). A subset $W \subseteq \mathbb{R}^n$ is said to be a **subspace** of $\mathbb{R}^n$ if it satisfies the following:

- $\mathbf{0} \in W$;
- If $\mathbf{x}, \mathbf{y} \in W$, then $\mathbf{x} + \mathbf{y} \in W$ (i.e. W is close under addition);
- If $\mathbf{x} \in W$ and $c \in \mathbb{R}$, then $c\mathbf{x} \in W$ (i.e. W is close under scalar multiplication).

Remark. Note that

- empty set is not a subspace;
- $\{\mathbf{0}\}$ is the zero subspace;
- $\{\mathbf{0}\}$ and $\mathbb{R}^n$ are trivial subspaces;
- The span of a finite non-empty set of vectors in $\mathbb{R}^n$.

3.7.2 Basis and Dimension

(Review of span and generating set) For $V \subseteq \mathbb{R}^n$, we say that S is a **generating set** for V if $\text{Span} S = V$.

Definition 3.7.3 (Basis). Let V be a subspace of $\mathbb{R}^n$. A **linearly independent** generating set for V is called a **basis** for V .

Remark. Note that

- a basis for V must be a subset of V .
- By convention, the only basis of the zero subspace is the **empty set**.

Definition 3.7.4 (Standard basis). The set of standard vectors $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ form a basis for $\mathbb{R}^n$. This is known as the **standard basis**.

Theorem 3.7.1. Every basis for $\mathbb{R}^n$ consists of exactly n vectors.

Proof. Consider a matrix with size $n \times k$ where k is the size of a set.

- A set of fewer than n vectors cannot generate $\mathbb{R}^n$.
- A set of more than n vectors must be linearly dependent.

□

Theorem 3.7.2 (Reduction and extension theorems). Let V be a non-zero subspace of $\mathbb{R}^n$. We have the following:

- (Reduction theorem) Every finite generating set of V contains a basis.
- (Extension theorem) Every linearly independent subset of V can be extended to a basis.

Definition 3.7.5 (Dimension). Any two bases for V contains the same number of vectors. This number is said to be the **dimension** of V and is denoted by $\dim V$.

Remark. Note that

- by convention the zero subspace is defined to have dimension 0.
- It follows from the definition that $\mathbb{R}^n$ has dimension n .

Theorem 3.7.3. Any two bases for V contain the same number of vectors.

Proposition 3.7.1. If V and W are subspaces of $\mathbb{R}^n$ such that $V \subseteq W$, then $\dim V \leq \dim W$. Equality holds if and only if $V = W$.

3.7.3 Subspaces Arising from Matrices

For each $m \times n$ matrix A , we have the following three subspaces:

Name	Notation	Subspace of	Definition
Row space	Row A	$\mathbb{R}^{1 \times n}$	The span of the rows of A
Column space	Col A	$\mathbb{R}^m$	The span of the columns of A
Null space	Null A	$\mathbb{R}^n$	The set of solutions to $A\mathbf{x} = \mathbf{0}$

An ERO will always produce a matrix with the same row space and null space, but not the same column space.

Three subspaces of A and those of its reduced row echelon form:

- For any matrix A , the dimension of **Row A** is equal to **rank A** . If R is the reduced row echelon form of A , then the set of nonzero rows in R will form a basis for Row A .
- For any matrix A , the dimension of **Col A** is equal to **rank A** . If R is the reduced row echelon form of A , then the set of columns in A corresponding to the leading columns of R will form a basis for Col A .
- For any matrix A , the dimension of **Null A** is equal to **nullity A** . If R is the reduced row echelon form of A , then the set of basic solutions to $R\mathbf{x} = \mathbf{0}$ will form a basis for Null A .

3.7.4 Coordinate Vectors and Matrix Representation of Linear Transformations

Definition 3.7.6 (Ordered bases and $\mathcal{B}$ -coordinate vector). Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k\}$ be an **ordered basis** for a subspace V . Then each $\mathbf{v} \in V$ can be written as a **unique** linear combination of the vectors in $\mathcal{B}$. Then each $\mathbf{v} \in V$ can be uniquely expressed in the form

$$\mathbf{v} = c_1 \mathbf{b}_1 + c_2 \mathbf{b}_2 + \dots + c_k \mathbf{b}_k.$$

Hence each $\mathbf{v} \in V$ corresponds to a unique choice of $c_1, c_2, \dots, c_k$. We define

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{bmatrix}$$

to be the **coordinate vector of $\mathbf{v}$ relative to $\mathcal{B}$** (or **$\mathcal{B}$ -coordinate vector of $\mathbf{v}$**).

Theorem 3.7.4. Let $\mathcal{B}$ be an ordered basis for $\mathbb{R}^n$. Then for any $\mathbf{v} \in \mathbb{R}^n$, we have

$$[\mathbf{v}]_{\mathcal{B}} = B^{-1}\mathbf{v},$$

where $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$, and B is the matrix that column vectors are $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n$.

Remark. Note that

- B is always invertible since its column vectors are always linearly independent.
- We do not have this formula if V is a non-trivial subspace of $\mathbb{R}^n$.

(Review on standard matrix for linear transformations) For any linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we can find an $m \times n$ matrix A such that $T(\mathbf{x}) = A\mathbf{x}$ for any $\mathbf{x} \in \mathbb{R}^n$. Such matrix A is called the **standard matrix** of T .

We can find A explicitly, namely, $A = [T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ \dots \ T(\mathbf{e}_n)]$.

(Matrix representation with respect to a particular basis) The $\mathcal{B}$ -matrix of a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is one satisfying

$$[T(\mathbf{v})]_{\mathcal{B}} = [T]_{\mathcal{B}}[\mathbf{v}]_{\mathcal{B}}$$

for all $\mathbf{v} \in \mathbb{R}^n$. Like the standard matrix, the $\mathcal{B}$ -matrix can be found column by column as well. For $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$, we have

$$[T]_{\mathcal{B}} = [[T(\mathbf{b}_1)]_{\mathcal{B}} \ [T(\mathbf{b}_2)]_{\mathcal{B}} \ \dots \ [T(\mathbf{b}_n)]_{\mathcal{B}}].$$

Definition 3.7.7 (Similarity of matrices). Let A and B be square matrices. We say A is **similar** to B if $B = PAP^{-1}$ for some invertible matrix P .

Remark. In some sense, similar matrices can be seen as matrices representing the same linear transformation with respect to different bases. Let A be the standard matrix of a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, and $\mathcal{B}$ be an ordered basis of $\mathbb{R}^n$. Then A and $[T]_{\mathcal{B}}$ are similar and

$$[T]_{\mathcal{B}} = B^{-1}AB$$

where B is the matrix with column vectors are elements in $\mathcal{B}$.

3.8 Chapter 8: Eigenvalues and Eigenvectors

(Motivation for diagonalisation) When we evaluate matrix powers, if the matrix can be decomposed to the form of $A = PDP^{-1}$, where D is a diagonal matrix, then it is easier for calculations.

Definition 3.8.1 (Definition of eigenvalues and eigenvectors). Let A be a square matrix. If $A\mathbf{x} = \lambda\mathbf{x}$ for some **non-zero** vector $\mathbf{x}$ and scalar λ ,

- λ is said to be an **eigenvalue** of A ;
- $\mathbf{x}$ is called an **eigenvector of A corresponding to the eigenvalue λ** , or in short, a **λ -eigenvector** of A .

Remark. While we assume the eigenvector $\mathbf{x}$ is non-zero, the eigenvalue λ can be zero.

Find λ -eigenvectors given the eigenvalue λ :

To find the λ -eigenvectors of a square matrix A , we should find the non-trivial solutions to the system $(A - \lambda I)\mathbf{x} = \mathbf{0}$.

Definition 3.8.2. Let A be an $n \times n$ matrix.

- The equation $\det(A - tI) = 0$ is called the **characteristic equation** of A .
- The left hand side of the characteristic equation, i.e. $\det(A - tI)$, is said to be the **characteristic polynomial** of A .
- **Eigenvalues** of A are thus roots of its characteristic equation, or zeros of its characteristic polynomial.
- The characteristic polynomial of A is a polynomial with degree n .
- We can also use $\det(tI - A)$ instead of $\det(A - tI)$.

Definition 3.8.3 (Eigenspaces). The set of all λ -eigenvectors of A is **almost** the same as the null space of $(A - \lambda I)$. We call $\text{Null}(A - \lambda I)$ the **eigenspace of A corresponding to λ** , or in short, the **λ -eigenspace of A** , which is also a subspace of $\mathbb{R}^n$.

Theorem 3.8.1 (Fundamental Theorem of Algebra). *A degree n polynomial (where $n > 1$) with coefficients in $\mathbb{C}$ has exactly n zeros in $\mathbb{C}$ (counting multiplicities).*

Remark. It follows that an $n \times n$ real-entries matrix A has exactly n eigenvalues in $\mathbb{C}$ (counting multiplicities). Even if the entries of A are real, the eigenvalues and eigenvectors of A could involve complex numbers.

Definition 3.8.4 (Algebraic Multiplicity and Geometric Multiplicity). We define the follows

- the **algebraic multiplicity** (or simply **multiplicity**) of an eigenvalue is the number of times it appears as a zero of the characteristic polynomial.
- The **geometric multiplicity** of an eigenvalue is the dimension of its corresponding eigenspace.

Theorem 3.8.2. *The algebraic multiplicity of an eigenvalue is always greater than or equal to its geometric multiplicity.*

Remark. The geometric multiplicity of an eigenvalue should always be positive. If the geometric multiplicity is 0, then the eigenspace only contains the zero vector, which is impossible. In this case, if the algebraic multiplicity is 1, the geometric multiplicity equals the algebraic multiplicity.

Proposition 3.8.1. *The eigenvalues of diagonal matrices and triangular matrices are their diagonal entries.*

Eigenvalues of related matrices:

Let λ be an eigenvalue of A .

- $c\lambda$ is an eigenvalue of cA for some scalar c
- λ^t is an eigenvalue of A^t for some power t
- $f(\cdot)$ is a polynomial, then $f(\lambda)$ is an eigenvalue of $f(A)$
- λ is an eigenvalue of a matrix similar to A

Definition 3.8.5 (Diagonalisable). We say that A is **diagonalisable** if we can factorize A into the form of PDP^{-1} where D is a diagonal matrix and P is some invertible matrix.

Remark. Note that

- not all matrices are diagonalisable.
- It is possible that a matrix is diagonalisable over $\mathbb{C}$ but not over $\mathbb{R}$. For example, $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.
- If a matrix is diagonalisable, the columns of P should be eigenvectors of A and the diagonal entries of D are the **corresponding** eigenvalues.

Theorem 3.8.3 (Multiplicity condition). *A matrix A is diagonalisable if and only if for each of its eigenvalues, the algebraic and geometric multiplicities are equal.*

Remark. Note that since the algebraic and geometric multiplicities must be equal when the former is equal to 1, it suffices to check the eigenvalues with algebraic multiplicity greater than 1 when checking whether a matrix is diagonalisable.

Applications to ODEs

If we are given an ODE system with the form of

$$\mathbf{y}' = A\mathbf{y} \tag{3.1}$$

for some diagonalisable coefficient matrix A and vectors $\mathbf{y}'$ and $\mathbf{y}$, we can solve the ODE system using diagonalisation of A .

Suppose we can diagonalize A into the form of PDP^{-1} by computing its eigenvalues and eigenvectors. Suppose $\mathbf{y} = P\mathbf{z}$ for some vector $\mathbf{z}$. Plug in (3.1), then we have

$$\begin{aligned} (P\mathbf{z})' &= A(P\mathbf{z}) \\ P\mathbf{z}' &= AP\mathbf{z} \\ \mathbf{z}' &= P^{-1}AP\mathbf{z} \\ \mathbf{z}' &= D\mathbf{z}. \end{aligned}$$

Hence, we can solve $\mathbf{z}$ easily and then compute $\mathbf{y}$ by $\mathbf{y} = P\mathbf{z}$.

Chapter 4

Practice Questions for MATH1821

4.1 Chapter 1: Basic Concepts

1. Consider a function $f : X \rightarrow Y$, where $X = \{1, 2, 3\}$, $Y = \{4, 5, 6\}$, and $f(1) = f(2) = 4$, $f(3) = 5$. $A = \{1, 3\}$, $B = \{5, 6\}$.
 - (a) What are the domain, codomain, and range of f ?
 - (b) Is f injective, surjective, and bijective?
 - (c) Find $f(A)$ and $f^{-1}(f(A))$.
 - (d) Find $f^{-1}(B)$ and $f(f^{-1}(B))$.
 - (e) What is the relationship between A and $f^{-1}(f(A))$, B and $f(f^{-1}(B))$?
 - (f) Consider a general function $f : X \rightarrow Y$, for what condition should f satisfies so that for any $A \subseteq X$, we always have $A = f^{-1}(f(A))$?
 - (g) Consider a general function $f : X \rightarrow Y$, for what condition should f satisfies so that for any $B \subseteq Y$, we always have $B = f(f^{-1}(B))$?

Suggested solution:

- (a) Domain: $\{1, 2, 3\}$, codomain: $\{4, 5, 6\}$, range: $\{4, 5\}$.
 - (b) f is not injective, not surjective, and not bijective.
 - (c) $f(A) = \{4, 5\}$, $f^{-1}(f(A)) = \{1, 2, 3\}$.
 - (d) $f^{-1}(B) = \{3\}$, $f(f^{-1}(B)) = \{5\}$.
 - (e) $A \subseteq f^{-1}(f(A))$, $B \subseteq f(f^{-1}(B))$.
 - (f) $f : X \rightarrow Y$ is injective **if and only if** $A = f^{-1}(f(A))$ for any $A \subseteq X$.
 - (g) $f : X \rightarrow Y$ is surjective **if and only if** $B = f(f^{-1}(B))$ for any $B \subseteq Y$.
2. Express the following complex numbers in polar form:
 - (a) $z = 1 + i$
 - (b) $(i^{25})^3$
 - (c) $z = -2 - 2\sqrt{3}i$

Suggested solution:

- (a) $z = \sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$
 - (b) $z = \cos \frac{\pi}{2} - i \sin \frac{\pi}{2}$
 - (c) $z = 4(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3})$

3. Prove that $S(n) : \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} \leq 2\sqrt{n} - 1$ is true for every positive integer n .

Suggested solution:

When $n = 1$, we have $\frac{1}{\sqrt{1}} = 1 \leq 2\sqrt{1} - 1 = 1$, hence $S(n)$ is true when $n = 1$.

Let k be an arbitrary integer such that $k \geq 1$. Assume that $S(k)$ is true.

When $n = k + 1$,

$$\begin{aligned} \text{LHS} &= \left(\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{k}} \right) + \frac{1}{\sqrt{k+1}} \\ &\leq 2\sqrt{k} - 1 + \frac{1}{\sqrt{k+1}} \\ &= 2\sqrt{k} - 1 + \frac{2}{2\sqrt{k+1}} \\ &\leq 2\sqrt{k} - 1 + \frac{2}{\sqrt{k+1} + \sqrt{k}} \\ &= 2\sqrt{k} - 1 + \frac{2(\sqrt{k+1} - \sqrt{k})}{(\sqrt{k+1} + \sqrt{k})(\sqrt{k+1} - \sqrt{k})} \\ &= 2\sqrt{k} - 1 + \frac{2(\sqrt{k+1} - \sqrt{k})}{(k+1) - k} \\ &= 2\sqrt{k} - 1 + 2(\sqrt{k+1} - \sqrt{k}) \\ &= 2\sqrt{k+1} - 1. \end{aligned}$$

Hence, $S(k+1)$ is true.

By the principle of mathematical induction, $S(n)$ holds for every positive integer n .

4. Let $-\pi < \theta < \pi$. Show that $\theta = 2 \arctan \left(\frac{\sin \theta}{1 + \cos \theta} \right)$.

Hint: try to use $\sin \theta$ and $\cos \theta$ to express $\tan \frac{\theta}{2}$.

Suggested Solutions:

Proof. Consider $\tan \frac{\theta}{2} = \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} = \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \cdot \frac{2 \cos \frac{\theta}{2}}{2 \cos \frac{\theta}{2}} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}}.$

Apply the double angle formulas, we get $\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$

Since $-\pi < \theta < \pi$, $-\frac{\pi}{2} < \frac{\theta}{2} < \frac{\pi}{2}$, $\theta = 2 \arctan \left(\frac{\sin \theta}{1 + \cos \theta} \right).$

□

4.2 Chapter 2: Limits and Continuity

1. Let $f(x) = \frac{x^2 - 1}{x - 1}$, find $\lim_{x \rightarrow 2} f(x)$, $\lim_{x \rightarrow 1} f(x)$.

Suggested solution:

Since $\lim_{x \rightarrow 2} (x - 1) = 1 \neq 0$, by the algebraic properties of limit,

$$\lim_{x \rightarrow 2} f(x) = \frac{\lim_{x \rightarrow 2} (x^2 - 1)}{\lim_{x \rightarrow 2} (x - 1)} = 3.$$

Since $\lim_{x \rightarrow 1} (x - 1) = 0$, we cannot apply algebraic properties of limit immediately.

Note that for x close to 1, $f(x) = \frac{(x+1)(x-1)}{x-1} = x+1$ provided that $x \neq 1$. It follows that

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x+1) = 2.$$

2. Consider

$$f(x) = \begin{cases} \frac{x^2 - 1}{x - 1} & \text{if } x < 1, \\ 2x & \text{if } x \geq 1. \end{cases}$$

Evaluate its limit at 1.

Suggested solution:

Note that $f(x)$ is given by different formulas on different sides of 1.

For any x slightly greater than 1, $f(x)$ is given the formula $f(x) = 2x$.

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} 2x = 2.$$

For any x slightly less than 1, $f(x)$ is given the formula $f(x) = \frac{x^2 - 1}{x - 1}$.

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x + 1) = 2.$$

Since

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x) = 2$$

we arrive at

$$\lim_{x \rightarrow 1} f(x) = 2.$$

3. Let $f(x) = g(x) \ln(2 + \sin \ln x)$ where g is a function whose domain contains $(0, 1)$. If $\lim_{x \rightarrow 0^+} |g(x)| = 0$, then evaluate $\lim_{x \rightarrow 0^+} |f(x)|$.

Suggested solution:

Note that for any $x \in (0, 1)$,

$$-1 \leq \sin \ln x \leq 1$$

$$1 \leq 2 + \sin \ln x \leq 3$$

Since $\ln x$ is strictly increasing,

$$0 = \ln 1 \leq \ln(2 + \sin \ln x) \leq \ln 3.$$

As $|g(x)| \geq 0$, we have

$$0 \leq |f(x)| = |g(x)| \ln(2 + \sin \ln x) \leq |g(x)| \ln 3.$$

As $\lim_{x \rightarrow 0^+} |g(x)| = 0$, $\lim_{x \rightarrow 0^+} |f(x)| = 0$ by sandwich theorem.

4. Let

$$f(x) = \begin{cases} \sqrt[3]{3x-1} & \text{if } x > 3 \\ \sin \frac{\pi x}{2} & \text{if } 1 < x \leq 3 \\ e^{1-x} & \text{if } x < 1 \end{cases}$$

Show that f is discontinuous at $x = 3$.

Suggested solution:

We need to calculate the limit of f at $x = 3$.

Notice that f has different formulas on left and right hand of $x = 3$, we need to calculate the one-sided limit at $x = 3$ separately.

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} \sin \frac{\pi x}{2} = \sin \frac{3\pi}{2} = -1$$

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} \sqrt[3]{3x-1} = \sqrt[3]{8} = 2$$

Since $\lim_{x \rightarrow 3^-} f(x) \neq \lim_{x \rightarrow 3^+} f(x)$, f is discontinuous at $x = 3$.

4.3 Chapter 3: Differentiation

1. Find the slope of the tangent line to the graph of $x^2 + y^2 = 9$ at point $(1, -2\sqrt{2})$.

Suggested solution: $f(x, y) = x^2 + y^2 - 9 = 0$

We take the derivative respect to x (consider y as a constant), hence we have

$$\frac{df(x, y)}{dx} = \frac{d(x^2 + y^2 - 9)}{dx} = 2x + 2y \frac{dy}{dx} = 0$$

which means

$$\frac{dy}{dx} = -\frac{x}{y} \text{ if } y \neq 0$$

At point $(1, -2\sqrt{2})$, the slope is $\frac{\sqrt{2}}{4}$.

2. Find $f'(x)$ of $f(x) = x^x$.

Suggested solution:

Consider $y = f(x) = x^x$. take logarithms respect on the both sides

$$\ln y = \ln x^x = x \ln x$$

Then take the derivative respect to x on both sides, we will have

$$\begin{aligned} \frac{dy}{dx}(\ln y) &= \frac{dy}{dx}(x \ln x) \\ \frac{1}{y} \frac{dy}{dx} &= \ln x + 1 \\ \frac{dy}{dx} &= y(\ln x + 1) = x^x(\ln x + 1) \end{aligned}$$

3. Find the point(s) of inflection of function $f(x) = -\frac{1}{6}x^4 + 7x$.

Suggested solution:

Notice that $f''(x) \leq 0$ for any x , hence the sign of $f''(x)$ has no change over any intervals. $f(x)$ has no point of inflection,

Remark. Some may argue that $f(0) = 0$ so that $(0, 0)$ is a point of inflection, however, please remember to check whether the sign of $f''(x)$ has changed at that point.

4. By applying the mean value theorem to the function $f(x) = \sqrt{x}$ over $[3, 4]$, prove that $12/7 < \sqrt{3} < 7/4$.

Suggested solution:

Since f is continuous on $[3, 4]$ and is differentiable on $(3, 4)$, the mean value theorem asserts that there is $c \in (a, b)$ such that

$$2 - \sqrt{3} = \frac{f(4) - f(3)}{4 - 3} = f'(c) = \frac{1}{2\sqrt{c}}$$

As $\sqrt{x}$ is increasing, we have

$$\begin{aligned} \sqrt{3} &< \sqrt{c} < \sqrt{4} = 2 \\ \frac{1}{4} &< \frac{1}{2\sqrt{c}} < \frac{1}{2\sqrt{3}} = \frac{1}{6}\sqrt{3} \end{aligned}$$

As a result,

$$\begin{aligned} \frac{1}{4} &< 2 - \sqrt{3} < \frac{1}{6}\sqrt{3} \\ 2 &< (1 + \frac{1}{6})\sqrt{3} \text{ and } \sqrt{3} < 2 - \frac{1}{4} \\ \frac{12}{7} &< \sqrt{3} < \frac{7}{4} \end{aligned}$$

5. Given a curve with equation

$$x^2y + y^3 + x^2 = 8$$

- Find $\frac{dy}{dx}$ by implicit differentiation.
- Find all points on the curve where its tangent line is horizontal.
- Find $\frac{d^2y}{dx^2}$ at the point(s) found in previous part.

Suggested solution:

(a) Implicit differentiate the given equation

$$\begin{aligned}\frac{d}{dx}(x^2y + y^3 + x^2) &= \frac{d}{dx}8 \\ 2xy + x^2 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} + 2x &= 0 \\ (x^2 + 3y^2) \frac{dy}{dx} &= -2x(1 + y) \\ \frac{dy}{dx} &= -\frac{2x(1 + y)}{x^2 + 3y^2}\end{aligned}$$

(b) To find a horizontal tangent line, we proceed to solve

$$\frac{dy}{dx} = -\frac{2x(1 + y)}{x^2 + 3y^2} = 0$$

and its solutions are $x = 0$ or $y = -1$. Substitute $x = 0$ into the original equation and we have $y = 2$. Substitute $y = -1$ into the original equation and we have $-1 = 8$, which is impossible, indicating that there is no point with $y = -1$ on the curve. Therefore, $y = -1$ is rejected. Hence $(0, 2)$ is the unique point on the curve at which the tangent line is horizontal, and the tangent line is $y = 2$.

(c) We implicit differentiate the equation we obtained in the first part to get

$$\begin{aligned}\frac{d}{dx}[(x^2 + 3y^2) \frac{dy}{dx}] &= \frac{d}{dx}[-2x(1 + y)] \\ (x^2 + 3y^2) \frac{d^2y}{dx^2} + \frac{d}{dx}(x^2 + 3y^2) \frac{dy}{dx} &= -2 \left[(1 + y) + x \frac{dy}{dx} \right]\end{aligned}$$

Since $\frac{dy}{dx} \Big|_{(x,y)=(0,2)} = 0$, we have

$$(0 + 12) \frac{d^2y}{dx^2} \Big|_{(x,y)=(0,2)} + 0 = -2(1 + 2 + 0) = -6$$

hence the $\frac{d^2y}{dx^2}$ at $(0, 2)$ is $-\frac{1}{2}$.

6. If $y = \cos 2x$, show that $\frac{d^n y}{dx^n} = 2^n \cos\left(2x + \frac{n\pi}{2}\right)$, where n is a positive integer. (Hint: use mathematical induction on n .)

Suggested solution:

We start with the case when $n = 1$, we have

$$\frac{d}{dx} \cos 2x = -2 \sin 2x = 2^1 \cos\left(2x + \frac{\pi}{2}\right)$$

Next, we assume that the quality holds when $n = k$ for some positive integer k . That is

$$\frac{d^k y}{dx^k} = 2^k \cos\left(2x + \frac{k\pi}{2}\right)$$

For $n = k + 1$, we have

$$\begin{aligned}\frac{d^{k+1}y}{dx^{k+1}} &= \frac{d}{dx} 2^k \cos\left(2x + \frac{k\pi}{2}\right) \\ &= 2^k \left[-2 \sin\left(2x + \frac{k\pi}{2}\right) \right] \\ &= -2^{k+1} \sin\left(2x + \frac{k\pi}{2} + \frac{\pi}{2} - \frac{\pi}{2}\right) \\ &= -2^{k+1} \sin\left(2x + \frac{k+1}{2}\pi - \frac{\pi}{2}\right) \\ &= 2^{k+1} \cos\left(2x + \frac{k+1}{2}\pi\right)\end{aligned}$$

By the principle of mathematical induction, we can conclude as desired.

7. Let $f(x) = xe^{1/x^2}$, $x > 0$. Determine all the vertical, horizontal, and oblique asymptotes of the graph of f , if any.

Suggested solution:

Since f is continuous on $(0, \infty)$, the only possible vertical asymptote is the y -axis. Let's test whether the y -axis is a vertical asymptote or not. (In fact, the y -axis is a vertical asymptote.) By l'Hôpital's rule

$$\lim_{x \rightarrow 0^+} xe^{1/x^2} = \lim_{x \rightarrow 0^+} \frac{e^{1/x^2}}{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \frac{-2e^{1/x^2} \frac{1}{x^3}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} 2e^{1/x^2} \frac{1}{x} = \infty$$

Next let's find the oblique asymptote.

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} e^{\frac{1}{x^2}} = 1$$

Then we compute

$$\lim_{x \rightarrow \infty} (f(x) - x) = \lim_{x \rightarrow \infty} x(e^{1/x^2} - 1) = \lim_{x \rightarrow \infty} \frac{e^{1/x^2} - 1}{1/x} = \lim_{x \rightarrow \infty} \frac{-(2/x^3)e^{1/x^2}}{-1/x^2} = \lim_{x \rightarrow \infty} \frac{2e^{1/x^2}}{x} = 0$$

So $y = x$ is an oblique asymptote of f , therefore f does not have any horizontal asymptote.

8. Let f be a differentiable function and $g(x) = 1 + \sin f(x)$. Given that g has an inverse function $h = g^{-1}$, $f(0) = 0$ and $f'(0) = 3$. Explain why h is differentiable at 1 and compute $h'(1)$.

Suggested solution:

Since g is a composition of differentiable functions, g is differentiable and

$$g'(x) = f'(x) \cos f(x)$$

Note that $g(0) = 1 + \sin f(0) = 1$ and $g'(0) = f'(0) \cos f(0) = 3 \neq 0$. By the inverse function theorem, g^{-1} is differentiable at $g(0) = 1$ and $h'(1) = (g^{-1})'(1) = 1/g'(0) = 1/3$.

9. Let $g(x) = (x^2 + 1)^{\sin x}$, $x \in \mathbb{R}$. First show that g is differentiable and find the derivative. Determine whether the following claims are true or false.

- (a) g has absolute maximum.
(b) g has infinitely many critical points.

Suggested solution:

Notice that we cannot discuss the differentiability for a function as the composition of a power function with the power and the base are two functions. Hence, we try to transfer the function to some other forms. Note that $1 + x^2$ is always positive and hence g is the same as

$$g(x) = e^{\ln g(x)} = e^{(\sin x) \ln(1+x^2)}$$

As $1 + x^2 > 0$, $1 + x^2$ and $\ln u$ are differentiable, $\ln(1 + x^2)$ is well-defined and is differentiable by the chain rule. As $\sin x$ is differentiable, the product $(\sin x) \ln(1 + x^2)$ is differentiable by the product rule. As e^u is differentiable, g is differentiable by the chain rule.

Derivative of g can be obtained by logarithmic differentiation since directly differentiating is hard to deal with.

$$\begin{aligned} \frac{g'(x)}{g(x)} &= (\ln g(x))' = \cos x \ln(x^2 + 1) + \frac{\sin x}{x^2 + 1} 2x \\ g'(x) &= \left(\cos x \ln(x^2 + 1) + \frac{2x \sin x}{x^2 + 1} \right) (x^2 + 1)^{\sin x} \end{aligned}$$

- (a) False. For any $k \in \mathbb{N}$, consider $g(2k\pi + \pi/2)$, we have

$$g(2k\pi + \pi/2) = ((2k\pi + \pi/2)^2 + 1)^{\sin(2k\pi + \pi/2)} = (2k\pi + \pi/2)^2 + 1$$

$$\lim_{k \rightarrow \infty} g(2k\pi + \pi/2) = \infty$$

So g does not have any absolute maximum.

(b) True. Note that for any $k \in \mathbb{N}$,

$$g(k\pi) = (k^2\pi^2 + 1)^{\sin(k\pi)} = 1 = ((k+1)^2\pi^2 + 1)^{\sin((k+1)\pi)} = g((k+1)\pi)$$

By the Rolle's Theorem, $g'(c_k) = 0$ for some $c_k \in (k\pi, (k+1)\pi)$.

4.4 Chapter 4: Integration

1. Find $\int \frac{1}{1-2x} dx$.

Suggested solution:

We make the substitution $u = 1 - 2x$ and $\frac{du}{dx} = -2$. Then

$$du = \left(\frac{du}{dx}\right) dx = -2 dx.$$

So we have

$$\begin{aligned} \int \frac{1}{1-2x} dx &= \int \frac{1}{u} \left(-\frac{1}{2} du\right) = -\frac{1}{2} \int \frac{1}{u} du \\ &= -\frac{1}{2} \ln|u| + C = -\frac{1}{2} \ln|1-2x| + C. \end{aligned}$$

2. Find $\int 2x\sqrt{1+x^2} dx$.

Suggested solution:

We make the substitution $u = 1 + x^2$ and $\frac{du}{dx} = 2x$. Then

$$du = \left(\frac{du}{dx}\right) dx = 2x dx.$$

So we have

$$\int 2x\sqrt{1+x^2} dx = \int \sqrt{u} du = \frac{2}{3} u^{3/2} + C = \frac{2}{3} (1+x^2)^{3/2} + C.$$

3. Evaluate $\int \tan^2 x dx$.

Suggested solution:

We can consider that $\tan x = \frac{\sin x}{\cos x}$. Now we transfer the original integration to $\int \frac{\sin^2 x}{\cos^2 x} dx$. Here we have the power forms of $\sin x$ and $\cos x$, we can recall that $\sin^2 x + \cos^2 x = 1$, let's use this to substitute $\sin^2 x$ to see what will happen.

$$\int \frac{\sin^2 x}{\cos^2 x} dx = \int \frac{1 - \cos^2 x}{\cos^2 x} dx = \int \left(\frac{1}{\cos^2 x} - 1\right) dx = \int (\sec^2 x - 1) dx = \tan x - x + C.$$

4. Evaluate $\int \sin 2x dx$.

Suggested solution:

Since this integration is in power 1 and have no other special features, let's try to integrate it directly.

$$\int \sin 2x dx = -\frac{1}{2} \cos 2x + C.$$

Remark. We have following two formulas for this type of integration:

- $\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$

$$\bullet \int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

5. Evaluate $\int \cos^2 3x dx$.

Suggested solution:

We see some power term, let's consider the double/triple angle formula. We know that

$$\cos 2x = \cos^2 x + \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x.$$

Here we have the term we want and $\cos^2 x = \frac{\cos 2x + 1}{2}$. So

$$\int \cos^2 3x dx = \int \frac{\cos 6x + 1}{2} dx = \frac{1}{12} \sin 6x + \frac{1}{2} x + C.$$

6. Evaluate $\int \sin 3x \cos 5x dx$.

Suggested solution:

Noticed that the integrator is a product of two trigonometric functions, we can consider the product-to-sum formula.

$$\sin \alpha \sin \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)].$$

So we have

$$\int \sin 3x \cos 5x dx = -\frac{1}{16} \cos 8x + \frac{1}{4} \cos 2x + C.$$

7. Evaluate $\int \frac{dx}{x^2 + 4}$.

Suggested solution:

Motivation: We know that $\int \frac{1}{x^2 + 1} dx = \tan^{-1} x + C$. Then

$$\int \frac{dx}{x^2 + 4} = \frac{1}{4} \int \frac{dx}{\frac{1}{4}x^2 + 1} = \frac{1}{2} \int \frac{du}{u^2 + 1} \quad (\text{for } u = \frac{x}{2}) = \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + C.$$

Hence we have a generalized result for $a > 0$:

$$\int \frac{b}{x^2 + a} dx = \frac{b}{\sqrt{a}} \tan^{-1} \left(\frac{x}{\sqrt{a}} \right) + C.$$

We have two extra similar practices:

$$\begin{aligned} \bullet \int \frac{dx}{\sqrt{9-x^2}} &= \sin^{-1} \left(\frac{x}{3} \right) + C \\ \bullet \int \frac{dx}{x^2 + 4x + 13} &= \frac{1}{3} \tan^{-1} \left(\frac{x+2}{3} \right) + C \end{aligned}$$

8. Let $I_n = \int (\ln t)^n dt$ for any integer $n \geq 1$.

- (a) Use integration by parts to obtain a formula relating I_n and I_{n-1} for any $n \geq 2$.
 (b) Find I_1 . Use your formula in the previous part to compute I_2 and I_3 .

Suggested solution:

- (a) Integration by parts to get

$$\begin{aligned} I_n &= t(\ln t)^n - \int t d((\ln t)^n) \\ &= t(\ln t)^n - n \int (\ln t)^{n-1} dt \\ &= t(\ln t)^n - nI_{n-1} \end{aligned}$$

(b) It is known that $I_1 = t \ln t - t + C$. Using the formula in previous part,

$$\begin{aligned} I_2 &= t(\ln t)^2 - 2I_1 = t(\ln t)^2 - 2t \ln t + 2t + C \\ I_3 &= t(\ln t)^3 - 3I_2 = t(\ln t)^3 - 3[t(\ln t)^2 - 2t(\ln t - 1)] + C \\ &= t(\ln t)^3 - 3t(\ln t)^2 + 6t(\ln t) - 6t + C \end{aligned}$$

9. We have two functions $y = \ln x$ and $y = \frac{1}{x}$. They have an intersection in the first quadrant, suppose the x-coordinate of the intersection is a . Find the area of the region between these two functions and between the two straight lines $x = a$ and $x = e$. (final answer should only contains terms without any logarithm)

Suggested solution:

We should check the relationship between a and e to confirm with straight line is on the left.

We know that the function value of $y = \ln x$ is always smaller than the function value of $y = \frac{1}{x}$ when $x < a$, and greater when $x > a$. Test the function value when $x = e$. $\ln e = 1$ and $\frac{1}{e} < 1 < \ln e$. Hence we can conclude that the straight line $x = e$ is on the right hand side of the straight line $x = a$.

Now we can evaluate the area by

$$\int_a^e \ln x - \frac{1}{x} dx = [x(\ln x - 1) - \ln |x|]_a^e = -1 - a(\ln a - 1) + \ln a$$

In order to escape some terms of logarithm, we noticed that $\ln a = \frac{1}{a}$ since a is the x-coordinate of the intersection. So we can substitute $\ln a$ by $\frac{1}{a}$. We eventually have the area is $-2 + a + \frac{1}{a}$.

Remark. Is there any method that we can confirm this answer makes sense?

We know that the area of the region cannot be negative, so let's test the relationship between $-2 + a + \frac{1}{a}$ and 0. We know that a is positive, by the basic inequality, we always have

$$-2 + a + \frac{1}{a} \geq -2 + 2\sqrt{a \cdot \frac{1}{a}} = -2 + 2 = 0$$

So the area is always non-negative, indicating this answer makes sense.

10. Differentiate $\ln(x + \sqrt{3 + x^2})$ and $x\sqrt{3 + x^2}$. Hence find $\int \sqrt{3 + x^2} dx$.

Suggested solution:

$$\begin{aligned} (\ln(x + \sqrt{3 + x^2}))' &= \frac{1}{x + \sqrt{3 + x^2}} \left(1 + \frac{1}{2} \cdot 2x \frac{1}{\sqrt{3 + x^2}} \right) \\ &= \frac{1}{x + \sqrt{3 + x^2}} \frac{x + \sqrt{3 + x^2}}{\sqrt{3 + x^2}} \\ &= \frac{1}{\sqrt{3 + x^2}} \\ (x\sqrt{3 + x^2})' &= \sqrt{3 + x^2} + x \cdot \frac{1}{2} \cdot 2x \frac{1}{\sqrt{3 + x^2}} \\ &= \sqrt{3 + x^2} + \frac{x^2}{\sqrt{3 + x^2}} \\ &= \frac{3 + 2x^2}{\sqrt{3 + x^2}} \end{aligned}$$

We can write

$$\sqrt{3 + x^2} = \frac{1}{2} \left(\frac{3 + 2x^2}{\sqrt{3 + x^2}} + \frac{3}{\sqrt{3 + x^2}} \right)$$

Then by previous results, we have

$$\begin{aligned} \int \sqrt{3 + x^2} dx &= \frac{1}{2} \int \frac{3 + 2x^2}{\sqrt{3 + x^2}} dx + \frac{3}{2} \int \frac{1}{\sqrt{3 + x^2}} dx \\ &= \frac{1}{2} x\sqrt{3 + x^2} + \frac{3}{2} \ln(x + \sqrt{3 + x^2}) + C. \end{aligned}$$

11. Find a reduction formula of $I_n = \int \frac{1}{(x^2 + a^2)^n} dx$ where $n > 1$ and $a > 0$.

Recall that I_1 is

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctan \frac{x}{a} + C$$

Suggested solution:

$$\begin{aligned} I_n &= \frac{1}{a^2} \int \frac{(x^2 + a^2) - x^2}{(x^2 + a^2)^n} dx = \frac{1}{a^2} \left(I_{n-1} - \int \frac{x^2}{(x^2 + a^2)^n} dx \right) \\ a^2 I_n &= I_{n-1} + \frac{1}{2(n-1)} \left(\frac{x}{(x^2 + a^2)^{n-1}} - \int \frac{dx}{(x^2 + a^2)^{n-1}} \right) \\ I_n &= \frac{2n-3}{2a^2(n-1)} I_{n-1} + \frac{1}{2a^2(n-1)} \cdot \frac{x}{(x^2 + a^2)^{n-1}} \end{aligned}$$

12. Determine whether $\int_a^\infty \frac{1}{x} e^{2+\sin 1/x} dx$ exists or not.

Suggested solution:

Note that for any x ,

$$2 + \sin(1/x) \geq 2 + (-1) = 1$$

$$e^{2+\sin(1/x)} \geq e^1 = e > 0$$

It follows for any $x > 1$,

$$0 \leq \frac{e}{x} \leq \frac{1}{x} e^{2+\sin(1/x)}$$

For any $b > 1$, by the fundamental theorem of calculus,

$$\int_1^b \frac{e}{x} dx = e[\ln x]_1^b = e \ln b \rightarrow \infty \text{ as } b \rightarrow \infty$$

Hence the given improper integral does not exist and diverges to ∞ .

4.5 Chapter 5: Numerical Methods

1. Consider the graph of the equation

$$xy^2 - e^{-(x^2+y)} = 0.$$

Given that it passes through the point $P(1, -1)$ and $y = f(x)$ is the local solution of this equation at P .

- (a) Find the equation of the tangent line to the graph of the equation at P .
 (b) Let Q be a point of the graph near P with coordinates $(1.1, \beta)$. Use the 2nd Taylor polynomial of f at the reference point $P(1, -1)$ to approximate β .

Suggested solution:

- (a) Implicit differentiate the equation with respect to x to get

$$y^2 + 2xyy' + (2x + y')e^{-(x^2+y)} = 0.$$

Substitute $(x, y) = (1, -1)$ to get

$$1 - 2y' + 2 + y' = 0.$$

So that $y' = 3$. It follows that the equation of the tangent line at point P is

$$y = -1 + 3(x - 1) = 3x - 4.$$

(b) Implicit differentiate the equation

$$y^2 + 2xyy' + (2x + y')e^{-(x^2+y)} = 0$$

with respect to x to get

$$2yy' + 2(yy' + x(y')^2 + xy y'') + e^{-(x^2+y)}(-(2x + y')^2 + (2 + y'')) = 0.$$

Substitute $(x, y, y') = (1, -1, 3)$ and get

$$-6 + 2(-3 + 9 - y'') + (-(2 + 3)^2 + (2 + y'')) = 0$$

$$-6 + 12 - 2y'' - 25 + 2 + y'' = 0.$$

So that $y'' = -17$. It follows that the second Taylor polynomial at 1 is

$$P_2(x) = f(1) + f'(1)(x - 1) + \frac{1}{2}f''(1)(x - 1)^2 = -1 + 3(x - 1) - \frac{17}{2}(x - 1)^2$$

and thus $\beta \approx P_2(1.1) = -0.785$.

2. Let $f(x) = x \arctan x - \int_1^x \arctan t \, dt$.

(a) Without computing $\int \arctan t \, dt$, find f' and f'' .

(b) Find the second degree Taylor polynomial of f at the reference point $\alpha = 1$.

(c) Use the polynomial in part (b) to estimate $f(1.1)$ and find an upper bound for the approximation error.

Suggested solution:

(a) By the fundamental theorem of calculus,

$$f'(x) = x \cdot (\arctan x)' + \arctan x - \arctan x = \frac{x}{1 + x^2}.$$

Differentiate f' to get

$$f''(x) = \frac{1 - x^2}{(1 + x^2)^2}.$$

(b) Using the formula of f and part (a), $f(1) = 1 \cdot \arctan 1 - 0 = \pi/4$, $f'(1) = 1/2$ and $f''(1) = 0$. As a result, the second degree Taylor polynomial of f is

$$P_2(x) = \frac{\pi}{4} + \frac{1}{2} \cdot (x - 1).$$

(c) Firstly, $f(1.1) \approx P_2(1.1) = \frac{\pi}{4} + \frac{1}{20}$. We then proceed to find an upper bound for the error committed when $f(1.1)$ is approximated by $P_2(1.1)$. Further differentiation yields

$$f^{(3)}(x) = \frac{2x(x^2 - 3)}{(x^2 + 1)^3}$$

and hence the error committed is given by

$$|f(1.1) - P_2(1.1)| = \frac{1}{3!} |f^{(3)}(c)| 0.1^3 = \frac{2c(3 - c^2)}{6000(c^2 + 1)^3}$$

for some $c \in (1, 1.1)$. It follows that an upper bound is

$$\frac{2(1.1)(3 - 1^2)}{6000(1^2 + 1)^3} = \frac{11}{120000}$$

as $0 \leq 3 - c^2 \leq 3 - 1^2$ and $0 \leq (c^2 + 1)^{-3} \leq (1^2 + 1)^{-3}$ for all $c \in (1, 1.1)$.

3. Use Newton's method to find the root of the following equation

$$xe^{-x} = 0,$$

with initial points at 0.5 and 1.1 respectively. (Stopping condition: $|x_{n+1} - x_n| < 0.01$)

Suggested solution:

Here, we have $f(x) = xe^{-x}$ and $f'(x) = (1-x)e^{-x}$. With initial points 0.5 and 1.1, we can have the following tables:

n	x_n	$f(x_n)$	$f'(x_n)$	$ x_{n+1} - x_n $
0	0.5000	0.3033	0.3033	
1	-0.5000	-0.8244	2.4731	1.0000
2	-0.1667	-0.1969	1.3783	0.3333
3	-0.0238	-0.0244	1.0485	0.1429
4	-0.0006	-0.0006	1.0011	0.0233
5	-0.0000	-0.0000	1.0000	0.0006

n	x_n	$f(x_n)$	$f'(x_n)$	$ x_{n+1} - x_n $
0	1.1000	0.3662	-0.0333	
1	12.1000	0.0001	-0.0001	11.0000
2	13.1901	0.0000	-0.0000	1.0901
3	14.2721	0.0000	-0.0000	1.0820
4	15.3475	0.0000	-0.0000	1.0753
5	16.4172	0.0000	-0.0000	1.0697
...	...	...	...	...

Observing the above tables, when the initial point is 0.5, our final x_n is around 0, which is a good approximation of the root. But it seems not the case when the initial point is 1.1. In fact, when the initial point is 1.1, the sequence $\{x_n\}$ is divergent. So we cannot use the Newton's method with initial point 1.1 to approximate the root.

For $f(x) = xe^{-x}$, in course MATH3904 or MATH3601, we may learn that the x_n obtained by Newton's method will converge to the root $x^* = 0$ if we have an initial point $x_0 < 1$ and diverge to infinity if $x_0 > 1$.

4.6 Chapter 6: Differential Equations

1. $y' = \frac{xy \sin x}{y+1}$, $y(0) = 1$.
2. $xy' = y + xe^{y/x}$, $y(1) = 0$.
3. $x^2y' + 2xy = \ln x$, $y(1) = 2$.
4. $(x^2 + 1)y' + 3x(y - 1) = 0$, $y(0) = 2$.
5. $y' = \frac{y^2}{1+x^2}$, $y(0) = 1$.
6. $2y' - y = 4 \sin 3x$, $y(0) = 1$.
7. $y' \sin x + 2y \cos x = \cos x$.
8. $y' + 2y = xy^{-2}$.
9. $y'' - y' - 6y = 0$, $y(0) = 1$, $y'(0) = -7$.
10. $y'' + 4y' + 5y = 0$, $y(0) = -2$, $y'(0) = 5$.

11. $y'' + 6y' + 9y = 0, y(0) = 2, y'(0) = -7.$
12. $x'' - 4x = 3e^{-2t}.$
13. $x'' + 3x' = -2t.$
14. $x'' - 4x' = 0.$

Suggested solution:

1. The ODE is a first order separable ODE.

$$\begin{aligned}\int \frac{y+1}{y} dy &= \int x \sin x dx \\ \int \left(1 + \frac{1}{y}\right) dy &= \int x d(-\cos x) \\ y + \ln |y| &= -x \cos x + \int \cos x dx = -x \cos x + \sin x + C\end{aligned}$$

Since $y(0) = 1$, we have $1 + \ln 1 = 1 = -0 \cos 0 + \sin 0 + C$, so $C = 1$. The solution is $y + \ln |y| = -x \cos x + \sin x + 1$.

2. Consider substitute y as $y = vx$ for v is a function of x . Then $y' = v + v'x$. Substitute into the original ODE, we have

$$\begin{aligned}v + xv' &= \frac{y}{x} + e^{y/x} = v + e^v \\ xv' &= e^v \\ \int \frac{dv}{e^v} &= \int \frac{dx}{x} \\ -e^{-v} &= \ln |x| + C\end{aligned}$$

Since $y(1) = 0$, we have $-1 = -e^0 = \ln 1 + C$, so $C = -1$. The solution in terms of y is $e^{-y/x} = 1 - \ln |x|$.

3. Notice that the LHS of the ODE is the derivative of x^2y , so we have

$$x^2y = \int \ln x dx = x \ln x - \int x d(\ln x) = x \ln x - \int dx = x \ln x - x + C$$

Since $y(1) = 2$, we have $2 = 1 \ln 1 - 1 + C$, so $C = 3$. The solution is $x^2y = x \ln x - x + 3$.

4. We try to use the integrating factor to solve this ODE. Simplify the ODE, we have

$$\begin{aligned}(x^2 + 1)y' + 3xy &= 3x \\ y' + \frac{3x}{x^2 + 1}y &= \frac{3x}{x^2 + 1}\end{aligned}$$

Take the integrating factor as $e^{\int \frac{3x}{x^2+1} dx} = e^{\ln(x^2+1)^{3/2}} = (x^2 + 1)^{3/2}$. Thus we have

$$\begin{aligned}((x^2 + 1)^{\frac{3}{2}}y)' &= 3x(x^2 + 1)^{\frac{1}{2}} \\ (x^2 + 1)^{\frac{3}{2}}y &= \int 3x(x^2 + 1)^{\frac{1}{2}} dx = \int \frac{3}{2}u^{\frac{1}{2}} du = u^{\frac{3}{2}} + C = (x^2 + 1)^{\frac{3}{2}} + C \text{ with } u = x^2 + 1\end{aligned}$$

Since $y(0) = 2$, we have $2 = 1 + C$, so $C = 1$. The solution is $y = 1 + (x^2 + 1)^{-\frac{3}{2}}$.

5. This is a first order separable ODE.

$$\begin{aligned}\int \frac{dy}{y^2} &= \int \frac{dx}{1+x^2} \\ -\frac{1}{y} &= \arctan x + C\end{aligned}$$

Since $y(0) = 1$, we have $-1 = 0 + C$, so $C = 1$. The solution is $y = \frac{1}{1 - \arctan x}$.

6. Simplifying the ODE, we have

$$y' - \frac{1}{2}y = 2 \sin 3x.$$

The integrating factor is $e^{\int -\frac{1}{2} dx} = e^{-\frac{x}{2}}$. Then

$$(e^{-\frac{x}{2}}y)' = 2e^{-\frac{x}{2}} \sin 3x$$

$$e^{-\frac{x}{2}}y = \int 2e^{-\frac{x}{2}} \sin 3x dx = 2 \int e^{-\frac{x}{2}} \sin 3x dx = -\frac{24}{37}e^{-\frac{x}{2}} \cos 3x - \frac{4}{37}e^{-\frac{x}{2}} \sin 3x + C.$$

Then

$$y = \frac{24}{37} \cos 3x - \frac{4}{37} \sin 3x + Ce^{\frac{x}{2}}.$$

Since $y(0) = 1$, we have $-\frac{24}{37} + C = 1$, so $C = \frac{61}{37}$. The solution is $y = \frac{24}{37} \cos 3x - \frac{4}{37} \sin 3x + \frac{61}{37}e^{\frac{x}{2}}$.

7. Simplifying the ODE, we have

$$y' + 2y \cot x = \cot x.$$

The integrating factor is $e^{\int 2 \cot x dx} = e^{2 \ln \sin x} = \sin^2 x$. Then

$$y \sin^2 x = \int \sin^2 x \cot x dx = \int \sin x \cos x dx = -\frac{1}{4} \cos 2x + C.$$

The solution is $y = \frac{C - \cos 2x}{4 \sin^2 x}$.

8. This is a Bernoulli Equation. Consider $z = y^{1-(-2)} = y^3$, then $z' = 3y^2 y'$. Substitute into the ODE, we have

$$3y^2 y' + 2(3y^2)y = xy^{-2}(3y^2)$$

$$z' + 6z = 3x.$$

The integrating factor is $e^{\int 6 dx} = e^{6x}$. So we have

$$e^{6x}z = \int 3xe^{6x} dx = e^{6x}\left(\frac{1}{2}x - \frac{1}{12}\right) + C.$$

The solution is $y^3 = \frac{1}{2}x - \frac{1}{12} + Ce^{-6x}$.

9. Solve the characteristic equation

$$\lambda^2 - \lambda - 6 = (\lambda - 3)(\lambda + 2) = 0,$$

a general solution of the ODE is

$$y(x) = Ae^{-2x} + Be^{3x}.$$

Impose the initial condition to get

$$y(0) = A + B = 1$$

$$y'(0) = -2A + 3B = -7$$

Solve the system of equations to get $A = 2$ and $B = -1$. The solution is $y(x) = 2e^{-2x} - e^{3x}$.

10. Solve the characteristic equation

$$\lambda^2 + 4\lambda + 5 = (\lambda + 2 + \sqrt{-1})(\lambda + 2 - \sqrt{-1}) = 0,$$

a general solution of the ODE is

$$y(x) = e^{-2x}(A \cos x + B \sin x).$$

Impose the initial condition to get

$$y(0) = A = -2$$

$$y'(0) = -2A + B = 5$$

Solve the system of equations to get $A = -2$ and $B = 1$. The solution of the ODE is $y(x) = e^{-2x}(-2 \cos x + \sin x)$.

11. Solve the characteristic equation

$$\lambda^2 + 6\lambda + 9 = (\lambda + 3)^2 = 0,$$

a general solution of the ODE is

$$y(x) = Ae^{-3x} + Bxe^{-3x}.$$

Impose the initial condition to get

$$y(0) = A = 2$$

$$y'(0) = -3A + B = -7$$

Solve the system of equations to get $A = 2$ and $B = -1$. The solution is $y(x) = e^{-3x}(2 - x)$.

12. First, solve the homogeneous solution.

$$x_c(t) = Ae^{2t} + Be^{-2t}.$$

Next, we guess the particular solution as the form of

$$x_p(t) = Cte^{-2t}.$$

(We can try $x_p(t) = Ce^{-2t}$ first and find that it is in the form already seen in the homogeneous solution, so we need to adjust it to the above form.)

Plug the particular solution into the ODE and solve for C .

$$C(-4e^{-2t} + 4te^{-2t}) - 4Cte^{-2t} = -4Ce^{-2t} = 3e^{-2t}$$

So we have $C = -\frac{3}{4}$. The general solution is $x(t) = Ae^{2t} + Be^{-2t} - \frac{3}{4}te^{-2t}$.

13. First, we solve the homogeneous solution.

$$x'' + 3x' = (x' + 3x)' = 0$$

Then we have

$$x' + 3x = \int 0 dt = A.$$

Calculate the integrating factor $e^{\int 3 dt} = e^{3t}$. Hence

$$e^{3t}x = \int Ae^{3t} dt = A'e^{3t} + B.$$

So

$$x_c(t) = A' + Be^{-3t}.$$

Next, we guess the particular solution as the form of

$$x_p(t) = Ct^2 + Dt + E.$$

Plug the particular solution into the ODE and solve for C, D and E .

$$2C + 3(2Ct + D) = 2C + 6Ct + 3D = -2t$$

Then we have $C = -\frac{1}{3}$ and $D = \frac{2}{9}$. The general solution is $x(t) = A' + Be^{-3t} - \frac{1}{3}t^2 + \frac{2}{9}t + E = F + Be^{-3t} - \frac{1}{3}t^2 + \frac{2}{9}t$.

14. Notice that

$$x'' - 4x' = (x' - 4x)' = 0.$$

Then

$$x' - 4x = \int 0 dt = A.$$

The integrating factor $e^{\int -4 dt} = e^{-4t}$. So

$$e^{-4t}x = \int Ae^{-4t} dt = Ce^{-4t} + B.$$

The solution is $x(t) = C + Be^{4t}$.

4.7 Chapter 7: Matrices

1. Let $A = \begin{bmatrix} 2 & -1 & 4 \\ 3 & 3 & 2 \\ 1 & 1 & 1 \end{bmatrix}$. Find the $(3, 2)$ -entry of $\left(\frac{1}{3}A\right)^{-1}$.

Suggested Solutions:

$\det A = 2 \cdot 3 + 3 \cdot 4 + (-2) - 4 \cdot 3 - 2 \cdot 2 - (-3) = 3$. Next, note that

$$\left(\frac{1}{3}A\right)^{-1} = 3A^{-1}.$$

So we proceed to find the $(3, 2)$ entry of A^{-1} :

$$\frac{1}{\det A} C_{23} = \frac{1}{3}(-1)^{2+3} \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} = -1.$$

As a result, the $(3, 2)$ entry is $3 \cdot (-1) = -3$.

2. Compute the determinants of the following matrices:

$$A = \begin{bmatrix} 5 & 7 \\ -3 & -4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & -1 \\ 2 & 5 & 7 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 1 & 3 \\ -3 & 1 & -3 \\ 3 & -1 & 1 \end{bmatrix}.$$

Are they invertible? Find the inverse whenever the matrix is invertible.

Suggested Solutions:

Since $\det A = \begin{vmatrix} 5 & 7 \\ -3 & -4 \end{vmatrix} = 1 \neq 0$, A is invertible. In addition,

$$A^{-1} = \begin{bmatrix} -4 & -7 \\ 3 & 5 \end{bmatrix}.$$

Since $\det B = \begin{vmatrix} 1 & 4 & 2 \\ 0 & 1 & -1 \\ 2 & 5 & 7 \end{vmatrix} = 7 - 8 + 5 - 4 = 0$, B is singular.

Since $\det P = \begin{vmatrix} 1 & 1 & 3 \\ -3 & 1 & -3 \\ 3 & -1 & 1 \end{vmatrix} = 1 - 9 + 9 - 3 + 3 - 9 = -8 \neq 0$, P is invertible. Note that the cofactors of P are

$$C_{11} = \begin{vmatrix} 1 & -3 \\ -1 & 1 \end{vmatrix} = -2, C_{12} = -\begin{vmatrix} -3 & -3 \\ 3 & 1 \end{vmatrix} = -6, C_{13} = \begin{vmatrix} -3 & 1 \\ 3 & -1 \end{vmatrix} = 0, C_{21} = -\begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = -4, C_{22} = \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} = -8,$$

$$C_{23} = -\begin{vmatrix} 1 & 1 \\ 3 & -1 \end{vmatrix} = 4, C_{31} = \begin{vmatrix} 1 & 3 \\ 1 & -3 \end{vmatrix} = -6, C_{32} = -\begin{vmatrix} 1 & 3 \\ -3 & -3 \end{vmatrix} = -6, C_{33} = \begin{vmatrix} 1 & 1 \\ -3 & 1 \end{vmatrix} = 4.$$

It follows that

$$P^{-1} = \frac{1}{\det P} (C_{ij})^T = -\frac{1}{8} \begin{bmatrix} -2 & -4 & -6 \\ -6 & -8 & -6 \\ 0 & 4 & 4 \end{bmatrix} = \begin{bmatrix} 1/4 & 1/2 & 3/4 \\ 3/4 & 1 & 3/4 \\ 0 & -1/2 & -1/2 \end{bmatrix}.$$

3. Let A and B be square matrices in 3×3 . Suppose that $\det A = 2, \det B = -1$. Find

- $\det(3A)$,
- $\det(2B^{-1})$,
- $\det(B^T A^{-1})$.

Suggested Solutions:

- $\det(3A) = 3^3 \det A = 27 \cdot 2 = 54$.

$$(b) \det(2B^{-1}) = 2^3 \det(B^{-1}) = 8 \cdot \frac{1}{\det B} = 8 \cdot \frac{1}{-1} = -8.$$

$$(c) \det(B^T A^{-1}) = \det(B^T) \det(A^{-1}) = \det B \frac{1}{\det A} = (-1) \cdot \frac{1}{2} = -\frac{1}{2}.$$

Remark. If A is a $n \times n$ square matrix, $\det(kA) = k^n \det A$ for any scalar k and $n \in \mathbb{N}$. This proposition can be proved by the mathematical induction using the cofactor expansion of calculating the determinant of high-dimension matrix.

4. Let A be a 2×2 matrix which satisfies $2A^2 - 3A - 5I = O$. If $\det(I + A) \neq 0$, find $\det A$.

Suggested Solutions:

The equation is equivalent to $(2A - 5I)(A + I) = O$. As $\det(A + I) \neq 0$, $A + I$ is invertible and we can use $(A + I)^{-1}$ to represent the inverse of $A + I$. So we have

$$2A - 5I = (2A - 5I)(A + I)(A + I)^{-1} = O(A + I)^{-1} = O,$$

$$A = (5/2)I.$$

Thus $\det A = \det((5/2)I) = (5/2)^2 \det I = 25/4$.

5. In Assignment 7, we have the definition of nilpotent matrix given by $A^k = O$ for some positive integer k .

(a) Show that the determinant of a nilpotent matrix is 0.

(b) By part(a), if a matrix is singular (not invertible), can we conclude the matrix is nilpotent?

Suggested Solutions:

(a) Given a nilpotent matrix A and $A^k = O$, we have $\det(A^k) = \det O = 0$. Since $\det(A^k) = (\det A)^k$, we have $(\det A)^k = 0$, $\det A = 0$. So the determinant of a nilpotent matrix is 0.

(b) No. If a matrix is singular, then its determinant is 0, but we cannot conclude whether it is nilpotent or not. We can consider an example $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. $\det B = 0$ but $B^k = B$ for any $k \in \mathbb{N}$ so B is not nilpotent.

Remarks:

Some may argue that a singular matrix have determinant of 0, and the determinant of its power should be 0 as well, which is equal to the determinant of the zero matrix, so that we can conclude its power is the zero matrix, and the matrix is nilpotent. The problem of this argument is that, we cannot conclude that two matrices are equal if they have equal determinant, so we cannot conclude the power of singular matrix is just the zero matrix. So for singular matrices, some of them are nilpotent while some of them are not. But for nilpotent matrices, all of them are singular.

6. Let $A = \begin{bmatrix} 1 & 0 & 2 & 1 \\ -1 & x & 2 & -1 \\ 2 & 0 & 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & y & 0 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$ where x and y are real non-negative constants. If AB is

singular, find the value of x and y .

Suggested Solutions:

Compute that $AB = \begin{bmatrix} 0 & y & 1 \\ 4 & x - y - 4 & x - 1 \\ 3 & 2y - 7 & -2 \end{bmatrix}$. Then $\det(AB) = (3x + 16)(y - 1) = 0$ since AB is singular.

Since $x, y > 0$, the solution for the equation is $y = 1$ and x can be any non-negative numbers.

7. Let A and B are square matrices with $AB = O$. Prove or disprove the following statements.

(a) $BA = O$.

(b) Either $A = O$ or $B = O$.

(c) If A is invertible, then $B = O$.

Suggested Solutions:

- (a) *Disproof.* Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$. Then $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ but $BA = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$ which is not the zero matrix.
- (b) *Disproof.* By the same counterexample in part (a).
- (c) *Proof.* Since A is invertible, we have $A^{-1}AB = A^{-1}O$, that is $B = O$.

4.8 Chapter 8: Vectors

1. Suppose $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^3$. Determine whether the statement is true or false: if $\mathbf{u} \times \mathbf{x} = \mathbf{v} \times \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^3$, then $\mathbf{u} = \mathbf{v}$.

Suggested Solutions:

The statement is *true*. Since the equality holds for any $\mathbf{x} \in \mathbb{R}^3$, we can substitute $\mathbf{x}$ by $\mathbf{u}$, then we get $\mathbf{0} = \mathbf{v} \times \mathbf{u}$. This means $\mathbf{v}$ and $\mathbf{u}$ are parallel. Without loss of generality, let $\mathbf{v} = k\mathbf{u}$ for some $k \in \mathbb{R}$. Then $\mathbf{u} \times \mathbf{x} = (k\mathbf{u}) \times \mathbf{x} = k(\mathbf{u} \times \mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^3$.

- If $\mathbf{u} = \mathbf{0}$, then $\mathbf{v} = k\mathbf{u} = \mathbf{0} = \mathbf{u}$.
- If $\mathbf{u} \neq \mathbf{0}$, then $\mathbf{u} \times \mathbf{x}$ is nonzero for those $\mathbf{x}$ not parallel to $\mathbf{u}$. So we must have $k = 1$, and hence $\mathbf{v} = \mathbf{u}$.

This shows $\mathbf{u} = \mathbf{v}$ in any case.

2. Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and $\mathbf{x}$ be vectors in $\mathbb{R}^3$. Prove the following identities:

- (a) $\|\mathbf{u} \times \mathbf{v}\|^2 = \|\mathbf{u}\|^2\|\mathbf{v}\|^2 - (\mathbf{u} \cdot \mathbf{v})^2$.
- (b) $(\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{w} \times \mathbf{x}) = \begin{vmatrix} \mathbf{u} \cdot \mathbf{w} & \mathbf{u} \cdot \mathbf{x} \\ \mathbf{v} \cdot \mathbf{w} & \mathbf{v} \cdot \mathbf{x} \end{vmatrix}$.

Suggested Solutions:

- (a) Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$. Then we have

$$\begin{aligned} \|\mathbf{u} \times \mathbf{v}\|^2 &= \|\mathbf{u}\|^2\|\mathbf{v}\|^2 \sin^2 \theta \\ &= \|\mathbf{u}\|^2\|\mathbf{v}\|^2 (1 - \cos^2 \theta) \\ &= \|\mathbf{u}\|^2\|\mathbf{v}\|^2 - \|\mathbf{u}\|^2\|\mathbf{v}\|^2 \cos^2 \theta \\ &= \|\mathbf{u}\|^2\|\mathbf{v}\|^2 - (\mathbf{u} \cdot \mathbf{v})^2. \end{aligned}$$

- (b) Use the property of vector triple product: $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$ and consider $(\mathbf{w} \times \mathbf{x})$ as a whole, we have

$$\begin{aligned} (\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{w} \times \mathbf{x}) &= \mathbf{u} \cdot (\mathbf{v} \times (\mathbf{w} \times \mathbf{x})) \\ &= (\mathbf{v} \times (\mathbf{w} \times \mathbf{x})) \cdot \mathbf{u} \\ &= ((\mathbf{v} \cdot \mathbf{x})\mathbf{w} - (\mathbf{v} \cdot \mathbf{w})\mathbf{x}) \cdot \mathbf{u} \\ &= (\mathbf{v} \cdot \mathbf{x})(\mathbf{w} \cdot \mathbf{u}) - (\mathbf{v} \cdot \mathbf{w})(\mathbf{x} \cdot \mathbf{u}) \\ &= \begin{vmatrix} \mathbf{u} \cdot \mathbf{w} & \mathbf{u} \cdot \mathbf{x} \\ \mathbf{v} \cdot \mathbf{w} & \mathbf{v} \cdot \mathbf{x} \end{vmatrix}. \end{aligned}$$

Chapter 5

Practice Questions for MATH2822

5.1 Chapter 1: Coordinate Geometry

1. Find the vector expression of the line passing through the point $(-1, 1, 3)$ and $(1, 1, 2)$.

Suggested Solutions:

We need to find the direction of the line first.

$$\mathbf{u} = (-1, 1, 3) - (1, 1, 2) = (-2, 0, 1)$$

So the vector form expression of the desired line is

$$(-1, 1, 3) + t(-2, 0, 1).$$

2. Find the distance between the point $(1, -2, 3)$ and the line $l : x = 2t - 5, y = 3 - t, z = 4$.

Suggested solution:

First, we randomly find a point P on the line. By selecting $t = 0$, we have $P(-5, 3, 4)$. Given the point S as $(1, -2, 3)$, we have $\overrightarrow{PS} = (1, -2, 3) - (-5, 3, 4) = (6, -5, -1)$. Next, we need to find the direction of the line. By selecting $t = 1$, we have another point Q on the line is $(-3, 2, 4)$. The direction of the line is given by $\mathbf{v} = \overrightarrow{PQ} = (-3, 2, 4) - (-5, 3, 4) = (2, -1, 0)$. By the formula, we have the distance from the point S to the line is given by

$$\frac{\|\overrightarrow{PS} \times \mathbf{v}\|}{|\mathbf{v}|} = \frac{\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 6 & -5 & -1 \\ 2 & -1 & 0 \end{vmatrix}}{\sqrt{5}} = \frac{|-\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}|}{\sqrt{5}} = \frac{\sqrt{21}}{\sqrt{5}}.$$

3. Find the distance from the point $(3, 5, 0)$ to the plane $5x - 2y + 6z = 13$.

Suggested solution:

Given the formula of finding the distance from a point to a plane, we have

$$\frac{|5 \times 3 - 2 \times 5 - 13|}{\sqrt{5^2 + 2^2 + 6^2}} = \frac{8}{\sqrt{65}}.$$

5.2 Chapter 2: Partial Differentiation

1. Find $\frac{\partial z}{\partial x}$ if the equation

$$yz - \ln z = x + y$$

defines z as a function of the two independent variables x and y and the partial derivative exists.

Suggested solution:

We differentiate both sides of the equation with respect to x , holding y constant and treating z as a differentiable function of x :

$$\begin{aligned}\frac{\partial}{\partial x}(yz) - \frac{\partial}{\partial x} \ln z &= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial x} \\ y \frac{\partial z}{\partial x} - \frac{1}{z} \frac{\partial z}{\partial x} &= 1 + 0 \\ \left(y - \frac{1}{z}\right) \frac{\partial z}{\partial x} &= 1 \\ \frac{\partial z}{\partial x} &= \frac{z}{yz - 1}.\end{aligned}$$

2. Find f_x and f_y as functions if

$$f(x, y) = \frac{2y}{y + \cos x}.$$

Suggested solution:

We treat f as a quotient. With y held constant, we get

$$\begin{aligned}f_x &= \frac{\partial}{\partial x} \left(\frac{2y}{y + \cos x} \right) = \frac{(y + \cos x) \frac{\partial}{\partial x}(2y) - 2y \frac{\partial}{\partial x}(y + \cos x)}{(y + \cos x)^2} \\ &= \frac{(y + \cos x)(0) - 2y(-\sin x)}{(y + \cos x)^2} = \frac{2y \sin x}{(y + \cos x)^2}.\end{aligned}$$

With x held constant, we get

$$\begin{aligned}f_y &= \frac{\partial}{\partial y} \left(\frac{2y}{y + \cos x} \right) = \frac{(y + \cos x) \frac{\partial}{\partial y}(2y) - 2y \frac{\partial}{\partial y}(y + \cos x)}{(y + \cos x)^2} \\ &= \frac{(y + \cos x)(2) - 2y(1)}{(y + \cos x)^2} = \frac{2 \cos x}{(y + \cos x)^2}.\end{aligned}$$

3. Find the derivative of $f(x, y) = xe^y + \cos(xy)$ at the point $(2, 0)$ in the direction of $\mathbf{v} = 3\mathbf{i} - 4\mathbf{j}$.

Suggested solution:

The direction of $\mathbf{v}$ is the unit vector obtained by dividing $\mathbf{v}$ by its length:

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{\mathbf{v}}{5} = \frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}.$$

The partial derivatives of f are everywhere continuous and at $(2, 0)$ are given by

$$\begin{aligned}f_x(2, 0) &= (e^y - y \sin(xy))_{(2,0)} = e^0 - 0 = 1 \\ f_y(2, 0) &= (xe^y - x \sin(xy))_{(2,0)} = 2e^0 - 2 \cdot 0 = 2.\end{aligned}$$

The gradient of f at $(2, 0)$ is

$$\nabla f|_{(2,0)} = f_x(2, 0)\mathbf{i} + f_y(2, 0)\mathbf{j} = \mathbf{i} + 2\mathbf{j}.$$

The derivative of f at $(2, 0)$ in the direction of $\mathbf{v}$ is therefore

$$\begin{aligned}D_{\mathbf{u}}f|_{(2,0)} &= \nabla f|_{(2,0)} \cdot \mathbf{u} \\ &= (\mathbf{i} + 2\mathbf{j}) \cdot \left(\frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}\right) = \frac{3}{5} - \frac{8}{5} = -1.\end{aligned}$$

4. Consider the function $f(x, y) = (y - x^2)^2 - x^2$.

(a) Find all the local extremum points of f on $\mathbb{R}^2$.

(b) Consider the minimization of f subject to no constraint on x and the constraint $-1 \leq y \leq 1$ on y . Show that there exists at least one global minimum point and find all these points.

Suggested solution:

- (a) The partial derivatives of
- f
- are

$$\begin{aligned}f_x &= -2(2y+1)x + 4x^3 \\f_y &= 2y - 2x^2\end{aligned}$$

When $f_x = f_y = 0$, we have $x = y = 0$. The Hessian of f is given by

$$H_f = \begin{vmatrix} -2(2y+1) + 12x^2 & -4x \\ -4x & 2 \end{vmatrix}$$

$H_f(0,0) = \begin{vmatrix} -2 & 0 \\ 0 & 2 \end{vmatrix} = -4$. So $(0,0)$ is a saddle point, the function f does not have any local extremum point.

- (b) By part (a), we know that no interior point will be a local extremum point, further no interior point will be a global extremum point. Next we only need to consider boundary points.

When $y = 1$, $f(x,1) = (1-x^2)^2 - x^2$. By taking $x^2 = a, a \geq 0$, we have $f(x,1) = g(a) = (1-a)^2 - a = a^2 - 3a + 1 = (a - \frac{3}{2})^2 - \frac{5}{4} \geq -\frac{5}{4}$, equality holds when $a = x^2 = \frac{3}{2}$, then $x = \pm \frac{\sqrt{6}}{2}$.

Similarly, when $y = -1$, we have $f(x,-1) = h(a) = (1+a)^2 - a = a^2 + a + 1 \geq 1 > -\frac{5}{4}$. So the global minimum will be attained when $y = 1$. The global minimum points are $(\frac{\sqrt{6}}{2}, 1)$ and $(-\frac{\sqrt{6}}{2}, 1)$, the global minimum value is $-\frac{5}{4}$.

5. Consider the function
- $f(x,y) = 8xy + \frac{1}{4}(x-y)^4$
- . Find all the local minimum points of
- f
- on
- $\mathbb{R}^2$
- .

Suggested solution:

The partial derivatives of f are given by

$$\begin{aligned}f_x &= 8y + (x-y)^3 \\f_y &= 8x - (x-y)^3\end{aligned}$$

When $f_x = f_y = 0$, we have $(x,y) = (0,0)$ or $(1,-1)$ or $(-1,1)$.

The Hessian of the function is

$$H_f = \begin{vmatrix} 3(x-y)^2 & 8-3(x-y)^2 \\ 8-3(x-y)^2 & 3(x-y)^2 \end{vmatrix}.$$

$H_f(0,0) = \begin{vmatrix} 0 & 8 \\ 8 & 0 \end{vmatrix} = -64 < 0$, so $(0,0)$ is a saddle point.

$H_f(1,-1) = \begin{vmatrix} 12 & -4 \\ -4 & 12 \end{vmatrix} = 128 > 0$, so $(1,-1)$ is a local minimum point.

$H_f(-1,1) = \begin{vmatrix} 12 & -4 \\ -4 & 12 \end{vmatrix} = 128 > 0$, so $(-1,1)$ is a local minimum point.

In summary, $(1,-1)$ and $(-1,1)$ are local minimum points.

6. The plane
- $x + y + z = 1$
- cuts the cylinder
- $x^2 + y^2 = 1$
- in an ellipse. Find the points on the ellipse that lie closest to and farthest from the origin.

Suggested solution:

We find the extreme values of

$$f(x,y,z) = x^2 + y^2 + z^2$$

subject to the constraints

$$\begin{aligned}g_1(x,y,z) &= x^2 + y^2 - 1 = 0 \\g_2(x,y,z) &= x + y + z - 1 = 0.\end{aligned}$$

By method of Lagrange multiplier, we have

$$\begin{aligned}\nabla f &= \lambda \nabla g_1 + \mu \nabla g_2 \\ 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k} &= \lambda(2x\mathbf{i} + 2y\mathbf{j}) + \mu(\mathbf{i} + \mathbf{j} + \mathbf{k}) \\ 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k} &= (2\lambda x + \mu)\mathbf{i} + (2\lambda y + \mu)\mathbf{j} + \mu\mathbf{k}\end{aligned}$$

or

$$2x = 2\lambda x + \mu, \quad 2y = 2\lambda y + \mu, \quad 2z = \mu.$$

The scalar equations further yield

$$\begin{aligned}2x &= 2\lambda x + 2z \Rightarrow (1 - \lambda)x = z, \\ 2y &= 2\lambda y + 2z \Rightarrow (1 - \lambda)y = z.\end{aligned}$$

The above two equations are satisfied simultaneously if either $\lambda = 1$ and $z = 0$ or $\lambda \neq 1$ and $x = y = z/(1 - \lambda)$. If $z = 0$, then solving g_1 and g_2 simultaneously we have two points $(1, 0, 0)$ and $(0, 1, 0)$. If $x = y$, then g_1 and g_2 give

$$\begin{aligned}x^2 + x^2 - 1 &= 0 \\ 2x^2 &= 1 \\ x &= \pm \frac{\sqrt{2}}{2} \\ x + x + z - 1 &= 0 \\ z &= 1 - 2x = 1 \mp \sqrt{2}.\end{aligned}$$

The corresponding points on the ellipse are

$$P_1 = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 1 - \sqrt{2}\right) \quad \text{and} \quad P_2 = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 1 + \sqrt{2}\right).$$

By computing the distance of $(1, 0, 0)$, $(0, 1, 0)$, P_1, P_2 from the origin, we have the points on the ellipse closet to the origin are $(1, 0, 0)$ and $(0, 1, 0)$. The point on the ellipse farthest from the origin is P_2 .

5.3 Chapter 3: More on Differentiation

1. Evaluate whether the following limit exist or not, if yes, please find the limit:

$$\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{2x^2 + y^2 - 3z^2}{x^2 + y^2 + z^2}.$$

Suggested solution:

When $y = z = 0$, $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{2x^2 + y^2 - 3z^2}{x^2 + y^2 + z^2} = \lim_{x \rightarrow 0} \frac{2x^2}{x^2} = 2$; When $x = z = 0$, $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{2x^2 + y^2 - 3z^2}{x^2 + y^2 + z^2} = \lim_{y \rightarrow 0} \frac{y^2}{y^2} = 1$. In two cases, limits are not equal, so the limit does not exist.

2. The function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined by

$$f(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0). \end{cases}$$

Is the function continuous at $(0, 0)$?

Suggested solution:

The function is not continuous at $(0, 0)$.

$$\begin{aligned}\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} &= \lim_{r \rightarrow 0} \frac{r^2(\cos^2 \theta - \sin^2 \theta)}{r^2(\cos^2 \theta + \sin^2 \theta)} \quad (\text{change into polar coordinates}) \\ &= \lim_{r \rightarrow 0} (\cos^2 \theta - \sin^2 \theta) \quad (\text{using } \cos^2 \theta + \sin^2 \theta = 1) \\ &= \cos^2 \theta - \sin^2 \theta, \text{ where } \theta \text{ is arbitrary.}\end{aligned}$$

When $\theta = 0$, $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} = 1$; when $\theta = \frac{\pi}{2}$, $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} = -1$. Since two limits are not equal, the limit does not exist, further the function is not continuous at $(0, 0)$.

Remark: The limit does not exist can also be proved by choosing two different curves which both pass through $(0, 0)$ and finding the limits are different along these two curves.

3. Define function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x, y) = e^x \sin(x + y)$. Find the second Taylor polynomial of f at point $(0, 0)$.

Suggested solution:

We can find the derivative and Hessian matrix of f as follows:

$$\begin{aligned}Df &= (e^x \sin(x + y) + e^x \cos(x + y) \quad e^x \cos(x + y)) \\ Hf &= \begin{pmatrix} 2e^x \cos(x + y) & e^x(-\sin(x + y) + \cos(x + y)) \\ e^x(\cos(x + y) - \sin(x + y)) & -e^x \sin(x + y) \end{pmatrix}\end{aligned}$$

And

$$Df(0, 0) = (1 \quad 1),$$

$$Hf(0, 0) = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then the second Taylor polynomial of f at point $(0, 0)$ is

$$P_2(x, y) = f(0, 0) + Df(0, 0) \begin{pmatrix} x \\ y \end{pmatrix} + \frac{1}{2} \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = x + y + x^2 + xy.$$

5.4 Chapter 4: Multiple Integrals

1. Evaluate the volume under the graph of $z = \sin x \cos y$ over the rectangle $[0, \pi] \times [0, \frac{\pi}{2}]$.

Suggested solution:

$$\begin{aligned}\int_0^\pi \int_0^{\frac{\pi}{2}} \sin x \cos y \, dy \, dx &= \int_0^\pi \sin x \sin y \Big|_{y=0}^{y=\frac{\pi}{2}} dx \\ &= \int_0^\pi \sin x \, dx \\ &= -\cos x \Big|_{x=0}^{x=\pi} \\ &= 1 - (-1) \\ &= 2.\end{aligned}$$

2. Evaluate the following integrals:

(a) $\iint_D x^2 y \, dA$, where D is a triangle in $\mathbb{R}^2$ with vertices at $(0, 0)$, $(1, 0)$ and $(0, 1)$.

Suggested solution:

The triangle is in the first quadrant, and the line passing through points $(1, 0)$ and $(0, 1)$ is $y = 1 - x$. So for any fixed $x \in [0, 1]$, the range for corresponding y is from 0 to $1 - x$.

$$\begin{aligned}\iint_D x^2 y \, dA &= \int_0^1 \int_0^{1-x} x^2 y \, dy \, dx \\ &= \int_0^1 \frac{1}{2} x^2 y^2 \Big|_{y=0}^{y=1-x} dx \\ &= \int_0^1 \frac{1}{2} x^2 (1-x)^2 dx \\ &= \left(\frac{1}{10} x^5 - \frac{1}{4} x^4 + \frac{1}{6} x^3 \right) \Big|_{x=0}^{x=1} \\ &= \frac{1}{60}.\end{aligned}$$

- (b) $\iint_D x \, dA$, where D is the bounded region in $\mathbb{R}^2$ bounded by $y = x$ and parabola $y = -2x^2 - 2x + 2$.

Suggested solution:

Consider the following system:

$$\begin{cases} y = x \\ y = -2x^2 - 2x + 2 \end{cases}$$

The two solutions are $x = \frac{1}{2}$ and $x = -2$, which are the bounds for x .

$$\begin{aligned}\iint_D x \, dA &= \int_{-2}^{\frac{1}{2}} \int_x^{-2x^2-2x+2} x \, dy \, dx \\ &= \int_{-2}^{\frac{1}{2}} xy \Big|_{y=x}^{y=-2x^2-2x+2} dx \\ &= \int_{-2}^{\frac{1}{2}} -2x^3 - 3x^2 + 2x \, dx \\ &= \left(-\frac{1}{2} x^4 - x^3 + x^2 \right) \Big|_{-2}^{\frac{1}{2}} \\ &= -\frac{125}{32}.\end{aligned}$$

- (c) $\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2} \, dx \, dy$.

Suggested solution:

We can consider changing the order of integration as the anti-derivative of e^{x^2} is hard to find:

$$\begin{aligned}\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2} \, dx \, dy &= \int_0^1 \int_0^{2x} e^{x^2} \, dy \, dx \\ &= \int_0^1 e^{x^2} y \Big|_{y=0}^{y=2x} dx \\ &= \int_0^1 2xe^{x^2} \, dx \\ &= e^{x^2} \Big|_{x=0}^{x=1} \\ &= e - 1.\end{aligned}$$

- (d) $\iiint_D \frac{1}{\sqrt{1-z^2}} \, dV$, where D is the region in $\mathbb{R}^3$ bounded between the elliptic paraboloid $y = x^2 + 4z^2$ and the plane $y = 4$.

Suggested solution:

Notice that $\frac{1}{\sqrt{1-z^2}}$ is not bounded on D when z tends to ± 1 , we are only able to find the approximation of integration using limits. Next, we try to find the bounds for x, y, z . Since $4z^2 \leq x^2 + 4z^2 \leq 4$, then $-1 \leq z \leq 1$. Since $x^2 \leq x^2 + 4z^2 \leq 4$, then $-2\sqrt{1-z^2} \leq x \leq 2\sqrt{1-z^2}$. And $x^2 + 4z^2 \leq y \leq 4$.

$$\begin{aligned} \iiint_D \frac{1}{\sqrt{1-z^2}} dV &= \int_{-1}^1 \int_{-2\sqrt{1-z^2}}^{2\sqrt{1-z^2}} \int_{x^2+4z^2}^4 \frac{1}{\sqrt{1-z^2}} dy dx dz \\ &= \int_{-1}^1 \left(\frac{4x}{\sqrt{1-z^2}} - \frac{\frac{1}{3}x^3 + 4z^2x}{\sqrt{1-z^2}} \right) \Big|_{x=-2\sqrt{1-z^2}}^{x=2\sqrt{1-z^2}} dz \\ &= \int_{-1}^1 \left(4x\sqrt{1-z^2} - \frac{x^3}{3\sqrt{1-z^2}} \right) \Big|_{x=-2\sqrt{1-z^2}}^{x=2\sqrt{1-z^2}} dz \\ &= \int_{-1}^1 \frac{32}{3} (1-z^2) dz \\ &= \frac{128}{9}. \end{aligned}$$

- (e) $\iint_D (x+y)^2(x-2y)^2 dx dy$, where D is the triangle in $\mathbb{R}^2$ with vertices at $(0,0), (1,-1), (2,1)$.

Suggested solution:

We take $u = x+y, v = x-2y$, so $x = \frac{2u+v}{3}, y = \frac{u-v}{3}$. So $\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{vmatrix} = -\frac{1}{3}$. Then we consider the region D' after the change of variables. The image of the triangle with vertices at $(0,0), (1,-1), (2,1)$ are change to triangle with vertices at $(0,0), (0,3), (3,0)$.

$$\begin{aligned} \iint_D (x+y)^2(x-2y)^2 dx dy &= \iint_{D'} \frac{1}{3} u^2 v^2 du dv \\ &= \frac{1}{3} \int_0^3 \int_0^{3-u} u^2 v^2 dv du \\ &= \frac{1}{9} \int_0^3 u^2 (3-u)^3 du \\ &= \frac{27}{20}. \end{aligned}$$

Remark: Notice that u and v are linear functions of x and y , so when we change variables, we transfer the straight lines in region D into straight lines in region D' , the new region is still a triangle in $\mathbb{R}^2$.

- (f) $\iint_D (x^2 - y^2)e^{xy} dx dy$, where D is the region in the first quadrant and bounded by $xy = 1, xy = 4, y = x, y = x+2$.

Suggested solution:

Suppose $u = xy, v = y - x$, so the region D' after the change of variables is $(u, v) \in D' = [1, 4] \times [0, 2]$.

On the other hand, $\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} y & x \\ -1 & 1 \end{vmatrix} = y + x \geq 0$ for any (x, y) in the first quadrant.

Hence,

$$\begin{aligned} \iint_D (x^2 - y^2)e^{xy} dx dy &= \iint_{D'} (x+y)(x-y)e^{xy} dx dy \\ &= \iint_{D'} -ve^u du dv \\ &= \int_1^4 \int_0^2 -ve^u dv du \\ &= 2(e - e^4). \end{aligned}$$

3. Suppose $a, b, c > 0$ are three real numbers. Evaluate the volume of the tetrahedron in $\mathbb{R}^3$ with vertices $(0,0,0), (a,0,0), (0,b,0), (0,0,c)$.

Suggested solution:

Suppose Π is the plane containing points $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$, then the equation of Π is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, then $z = c(1 - \frac{x}{a} - \frac{y}{b})$. Then the volume of the tetrahedron is the integration of a function $f(x, y) = c(1 - \frac{x}{a} - \frac{y}{b})$ over a region D , which is the triangle on the xy -plane and bounded by points $(0, 0, 0)$, $(a, 0, 0)$, $(0, b, 0)$. On the xy -plane, when $x = a$, $y = b(1 - \frac{x}{a})$.

$$\begin{aligned} V &= \int_0^a \int_0^{b(1-\frac{x}{a})} c(1 - \frac{x}{a} - \frac{y}{b}) dy dx \\ &= \int_0^a \frac{bc}{2} (1 - \frac{x}{a})^2 dx \\ &= \frac{1}{6} abc. \end{aligned}$$

4. Use triple integrals to prove that the volume of a solid hemisphere of radius r is $\frac{2}{3}\pi r^3$.

Suggested solution:

We first try to find out the bounds for variables. Given $x^2 \leq x^2 + y^2 + z^2 \leq r^2$, then $-r \leq x \leq r$. Next, $x^2 + y^2 \leq x^2 + y^2 + z^2 \leq r^2$, then $-\sqrt{r^2 - x^2} \leq y \leq \sqrt{r^2 - x^2}$. Finally, $0 \leq z \leq \sqrt{r^2 - x^2 - y^2}$.

$$\begin{aligned} V &= \int_{-r}^r \int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} \int_0^{\sqrt{r^2-x^2-y^2}} 1 dz dy dx \\ &= \int_{-r}^r \int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} \sqrt{r^2-x^2-y^2} dy dx \\ &= \int_{-r}^r \int_{\frac{\pi}{2}}^{\pi} (r^2-x^2) \cos^2 \theta d\theta dx \quad (y = \sqrt{r^2-x^2} \sin \theta) \\ &= \int_{-r}^r \int_{\frac{\pi}{2}}^{\pi} \frac{(r^2-x^2)(\cos 2\theta + 1)}{2} d\theta dx \\ &= \int_{-r}^r (r^2-x^2) \left(\frac{\sin 2\theta}{4} + \frac{\theta}{2} \right) \Big|_{\theta=\frac{\pi}{2}}^{\theta=\pi} dx \\ &= \frac{\pi}{2} \int_{-r}^r (r^2-x^2) dx \\ &= \frac{2}{3} \pi r^3. \end{aligned}$$

5.5 Chapter 5: Systems of Linear Equations

1. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & -2 \end{bmatrix}, \quad C = \begin{bmatrix} 111 & 333 & 789 \\ 333 & 999 & 666 \\ 321 & 963 & 753 \end{bmatrix}.$$

- (a) Prove that A and B are row equivalent.
 (b) Prove that A and C are not row equivalent.

Suggested solution:

- (a) We find the RREF of A and B as follows.

$$A \xrightarrow[\begin{smallmatrix} -3r_1+r_2 \rightarrow r_2 \\ -r_1+r_3 \rightarrow r_3 \end{smallmatrix}]{\begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & -8 \\ 0 & -1 & -2 \end{bmatrix}} \xrightarrow{-\frac{1}{4}r_2 \rightarrow r_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -1 & -2 \end{bmatrix} \xrightarrow[\begin{smallmatrix} -2r_2+r_1 \rightarrow r_1 \\ -2r_2+r_3 \rightarrow r_3 \end{smallmatrix}]{\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}}$$

$$B \xrightarrow{r_1 \leftrightarrow r_2} \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ 0 & -1 & -2 \end{bmatrix} \xrightarrow{-2r_1 + r_2 \rightarrow r_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & -1 & -2 \end{bmatrix} \xrightarrow{r_2 + r_3 \rightarrow r_3} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Since the RREF of A and B are the same, they are row equivalent.

- (b) Let $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ and $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ be the columns of A and C respectively. Observe that $3\mathbf{c}_1 = \mathbf{c}_2$. If A and C are row equivalent, then $3\mathbf{a}_1 = \mathbf{a}_2$ by the column correspondence property. However, this is not the case since $3\mathbf{a}_1 = (3, 9, 3)^T \neq \mathbf{a}_2$. Thus, A and C are not row equivalent.

2. Find the value(s) of t such that the system of linear equations

$$\begin{cases} x + y + tz = 1, \\ x + ty + z = 1, \\ tx + y + z = 1 \end{cases}$$

- (a) has no solution;
 (b) has a unique solution;
 (c) has exactly 3 solutions;
 (d) has infinitely many solutions.

Suggested solution:

We first transform the augmented matrix $A = \left[\begin{array}{ccc|c} 1 & 1 & t & 1 \\ 1 & t & 1 & 1 \\ t & 1 & 1 & 1 \end{array} \right]$ using EROs as follows.

$$\left[\begin{array}{ccc|c} 1 & 1 & t & 1 \\ 1 & t & 1 & 1 \\ t & 1 & 1 & 1 \end{array} \right] \xrightarrow{\begin{array}{l} -tr_1 + r_3 \rightarrow r_3 \\ -r_1 + r_2 \rightarrow r_2 \end{array}} \left[\begin{array}{ccc|c} 1 & 1 & t & 1 \\ 0 & t-1 & 1-t & 0 \\ 0 & 1-t & 1-t^2 & 1-t \end{array} \right] \quad (5.1)$$

If $t \neq 1$, we further apply EROs as follows.

$$\left[\begin{array}{ccc|c} 1 & 1 & t & 1 \\ 0 & t-1 & 1-t & 0 \\ 0 & 1-t & 1-t^2 & 1-t \end{array} \right] \xrightarrow{\begin{array}{l} \frac{1}{1-t} r_3 \rightarrow r_3 \\ \frac{1}{t-1} r_2 \rightarrow r_2 \end{array}} \left[\begin{array}{ccc|c} 1 & 1 & t & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1+t & 1 \end{array} \right] \xrightarrow{\begin{array}{l} -r_2 + r_3 \rightarrow r_3 \\ -r_2 + r_1 \rightarrow r_1 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & t+1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2+t & 1 \end{array} \right] \quad (5.2)$$

- (a) When $t = -2$, the matrix in (2) has a leading one in the last column, there is no solution.
 (b) When $t \neq 1, -2$, we can further reduce the matrix in (2) to the RREF $\left[\begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array} \right]$. There is a unique solution.
 (c) It is impossible to have exactly 3 solutions.
 (d) When $t = 1$, the matrix A is row equivalent to $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$ by (1). There are infinitely many solutions.

3. Prove that we can use Type II and Type III EROs to interchange two rows. In other words, a Type I elementary matrix can be represented as a product of Type II and Type III elementary matrices.

Suggested solution:

Suppose we want to interchange row i and row j . To have more observations on the entries, we focus on $(i, 1)$ th entry a and $(j, 1)$ th entry b .

- $\mathbf{r}_i + \mathbf{r}_j \rightarrow \mathbf{r}_j$: the entries become a and $a + b$
- $-\mathbf{r}_j + \mathbf{r}_i \rightarrow \mathbf{r}_i$: the entries become $-b$ and $a + b$
- $\mathbf{r}_i + \mathbf{r}_j \rightarrow \mathbf{r}_j$: the entries become $-b$ and a
- $-\mathbf{r}_i \rightarrow \mathbf{r}_i$: the entries become b and a

The two entries are interchanged. As the same holds for other columns, we have interchanged row i and row j using this sequence of operations.

4. Let A be an $m \times n$ matrix, and let $\mathbf{b}$ be an $m \times 1$ column vector. Suppose $A\mathbf{x} = \mathbf{b}$ is consistent, and $\mathbf{x} = \mathbf{p}$ is a particular solution. Prove that $\mathbf{u}$ is a solution to $A\mathbf{x} = \mathbf{b}$ if and only if $\mathbf{u} = \mathbf{v} + \mathbf{p}$ for some $\mathbf{v}$ which is a solution to the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

Suggested solution:

"If" part: Since $\mathbf{u} = \mathbf{v} + \mathbf{p}$ with $A\mathbf{v} = \mathbf{0}$ and $A\mathbf{p} = \mathbf{b}$, we have

$$A\mathbf{u} = A(\mathbf{v} + \mathbf{p}) = A\mathbf{v} + A\mathbf{p} = \mathbf{0} + \mathbf{b} = \mathbf{b}.$$

This shows $\mathbf{u}$ is a solution to $A\mathbf{x} = \mathbf{b}$.

"Only if" part: Since $A\mathbf{u} = \mathbf{b}$ and $A\mathbf{p} = \mathbf{b}$, by defining $\mathbf{v} = \mathbf{u} - \mathbf{p}$, we have $\mathbf{u} = \mathbf{v} + \mathbf{p}$ and

$$A\mathbf{v} = A(\mathbf{u} - \mathbf{p}) = A\mathbf{u} - A\mathbf{p} = \mathbf{b} - \mathbf{b} = \mathbf{0}$$

This shows $\mathbf{v}$ is a solution to $A\mathbf{x} = \mathbf{0}$.

5. Let A be an $m \times n$ matrix with $m > n$, prove that there does not exist an $n \times m$ matrix B such that $AB = I_m$.

Suggested solution:

We prove by contradiction. If there exists B such that $AB = I_m$, then $\text{rank}(AB) = m$. Note that $\text{rank}(A) \leq n < m$ and all columns of AB are linear combinations of columns of A , then $\text{rank}(AB) \leq \text{rank}(A) < m$. We arrive at a contradiction, so such B does not exist.

Remark: We can prove $\text{rank}(AB) \leq \text{rank}(A)$ by consider the column space of AB is a subspace of the column space A , so the dimension cannot be larger. Column space will be learned in Chapter 7.

6. For an $m \times n$ matrix A , prove that $\text{rank } A = \text{rank } A^T$.

Suggested solution:

Let $A = \begin{bmatrix} \mathbf{a}_1^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix}$, then $A^T = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_m]$. Let $r = \text{rank}(A^T)$, then there exist r row vectors of A such that

they are linearly independent. Without loss of generality, we assume that $\mathbf{a}_1, \dots, \mathbf{a}_r$ are linearly independent. Next, we claim that $A\mathbf{a}_1, \dots, A\mathbf{a}_r$ are also linearly independent.

Prove the claim: Suppose that there exist $c_1, \dots, c_r$ such that $c_1 A\mathbf{a}_1 + \dots + c_r A\mathbf{a}_r = \mathbf{0}$. It is equivalent to $A(c_1 \mathbf{a}_1 + \dots + c_r \mathbf{a}_r) = A\mathbf{v} = \mathbf{0}$, where $\mathbf{v} = c_1 \mathbf{a}_1 + \dots + c_r \mathbf{a}_r$. It also implies that $\mathbf{a}_i^T \mathbf{v} = 0$ for all $i = 1, \dots, r$. Then $\mathbf{v}^T \mathbf{v} = 0$ and thus $\mathbf{v} = \mathbf{0}$. Since $\mathbf{a}_1, \dots, \mathbf{a}_r$ are linearly independent, we have $c_1 = \dots = c_r = 0$ and thus $A\mathbf{a}_1, \dots, A\mathbf{a}_r$ are linearly independent.

Note that $A\mathbf{a}_1, \dots, A\mathbf{a}_r$ are linear combinations of columns of A , we have $\text{rank}(A) \geq r = \text{rank}(A^T)$. We do a similar analysis to A^T , we have $\text{rank}(A^T) \geq \text{rank}(A)$. Since the two inequalities hold simultaneously, we arrive at $\text{rank}(A) = \text{rank}(A^T)$.

5.6 Chapter 6: More on Matrices and Determinants

1. Using elementary row operations, find the inverse of $A = \begin{bmatrix} -3 & -2 & 2 \\ 4 & 1 & -4 \\ -2 & 3 & 5 \end{bmatrix}$.

Suggested solution:

$$\begin{aligned} (A|I) &\xrightarrow{r_2+r_1 \rightarrow r_1} \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 1 & 1 & 0 \\ 4 & 1 & -4 & 0 & 1 & 0 \\ -2 & 3 & 5 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{2r_1+r_3 \rightarrow r_3 \\ -4r_1+r_2 \rightarrow r_2}} \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 1 & 1 & 0 \\ 0 & 5 & 4 & -4 & -3 & 0 \\ 0 & 1 & 1 & 2 & 2 & 1 \end{array} \right] \\ &\xrightarrow{r_2 \leftrightarrow r_3} \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 1 & 1 & 0 \\ 0 & 1 & 1 & 2 & 2 & 1 \\ 0 & 5 & 4 & -4 & -3 & 0 \end{array} \right] \xrightarrow{\substack{-5r_2+r_3 \rightarrow r_3 \\ r_1+r_2 \rightarrow r_1}} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 3 & 3 & 1 \\ 0 & 1 & 1 & 2 & 2 & 1 \\ 0 & 0 & -1 & -14 & -13 & -5 \end{array} \right] \\ &\xrightarrow{-r_3 \rightarrow r_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 3 & 3 & 1 \\ 0 & 1 & 1 & 2 & 2 & 1 \\ 0 & 0 & 1 & 14 & 13 & 5 \end{array} \right] \xrightarrow{\substack{-r_3+r_2 \rightarrow r_2 \\ r_1+r_3 \rightarrow r_1}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 17 & 16 & 6 \\ 0 & 1 & 0 & -12 & -11 & -4 \\ 0 & 0 & 1 & 14 & 13 & 5 \end{array} \right] \end{aligned}$$

$$\text{Then } A^{-1} = \begin{bmatrix} 17 & 16 & 6 \\ -12 & -11 & -4 \\ 14 & 13 & 5 \end{bmatrix}.$$

2. Determine whether each of the following is true or false and briefly explain the reasons.

- (a) Let A, B, C and D be invertible matrices of the same size. Then the block matrix defined by $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ is invertible.
- (b) If S_1 and S_2 are finite subsets of $\mathbb{R}^n$ such that S_1 is contained in $\text{Span } S_2$, then $\text{Span } S_1$ is contained in $\text{Span } S_2$.
- (c) Some systems of linear equations have exactly two solutions.
- (d) If a system of linear equations has more variables than equations, then it must have infinitely many solutions.
- (e) If $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $U : \mathbb{R}^n \rightarrow \mathbb{R}^n$ are linear transformations such that $T \circ U$ is the identity transformation, then U is the inverse of T .
- (f) If $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an invertible linear transformation, then T^k is the identity transformation for some positive integer k , where $T^k = \underbrace{T \circ T \circ \cdots \circ T}_{k \text{ times}}$.

Suggested solution:

- (a) False. Consider the 1×1 matrices $A = B = C = D = [1]$ which are invertible. But $M = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ is non-invertible.
- (b) True. If S_1 is contained in $\text{Span } S_2$, then the vectors in S_1 can be represented as linear combinations of vectors in S_2 . $\text{Span } S_1$ is the linear combinations of vectors in S_1 , which can also be represented by linear combinations of vectors in S_2 .
- (c) False. A linear system can only have no solution, unique solution, and infinitely many solutions.
- (d) False. If the last row of the RREF of $(A|\mathbf{b})$ is $[0 \ \cdots \ 0 \ 1]$, then the linear system has no solution.
- (e) True. Suppose A and B are standard matrix of T and U respectively. Then $T(\mathbf{x}) = A\mathbf{x}$ and $U(\mathbf{x}) = B\mathbf{x}$ for any $\mathbf{x} \in \mathbb{R}^n$. Given $T \circ U$ is the identity transformation, we have $T \circ U(\mathbf{x}) = A(B\mathbf{x}) = AB\mathbf{x} = \mathbf{x}$ for any $\mathbf{x} \in \mathbb{R}^n$, then $AB = I$, $B = A^{-1}$. Then $U(\mathbf{x}) = B\mathbf{x} = A^{-1}\mathbf{x} = T^{-1}(\mathbf{x})$, U is the inverse of T . (Remark: From $AB = I$, we know that A and B are invertible, so do T and U , hence T and U have inverse.)
- (f) False. Consider an invertible linear transformation defined on $\mathbb{R}^2$ by $T(x, y) = \begin{bmatrix} x+y \\ y \end{bmatrix}$. Then $T(0, 1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $T^2(0, 1) = T(1, 1) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, and inductively, $T^k(0, 1) = \begin{bmatrix} k \\ 1 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ for any positive integer k . Therefore, T^k cannot be the identity transformation.

3. Compute the determinant of following matrices:

(a) $\begin{bmatrix} 2 & 1 & 0 & 1 \\ -2 & -3 & -1 & -2 \\ 4 & 4 & -8 & 0 \\ 3 & -7 & -2 & 4 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 9 & 0 & -3 & 0 \\ 1 & -4 & 0 & 2 & 0 \\ 3 & -2 & 8 & -4 & 5 \\ -6 & 2 & -7 & 3 & 0 \\ 0 & 2 & 0 & 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 5 & 7 & 9 \\ 1 & 4 & 7 & 10 & 13 \\ 1 & 5 & 9 & 13 & 17 \end{bmatrix}$

(d) $A = \begin{bmatrix} 2 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & \cdots & 1 \\ 1 & 1 & 2 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 2 \end{bmatrix}$ is an $n \times n$ matrix

(e) **Vandermonde matrix** is defined by $V = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ 1 & x_3 & x_3^2 & \cdots & x_3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{bmatrix}$

Suggested solution:

(a)

$$\begin{aligned} \begin{vmatrix} 2 & 1 & 0 & 1 \\ -2 & -3 & -1 & -2 \\ 4 & 4 & -8 & 0 \\ 3 & -7 & -2 & 4 \end{vmatrix} & \begin{array}{l} -2c_4 + c_1 \rightarrow c_1 \\ -c_4 + c_2 \rightarrow c_2 \\ \hline \end{array} \begin{vmatrix} 0 & 0 & 0 & 1 \\ 2 & -1 & -1 & -2 \\ 4 & 4 & -8 & 0 \\ -5 & -11 & -2 & 4 \end{vmatrix} = - \begin{vmatrix} 2 & -1 & -1 \\ 4 & 4 & -8 \\ -5 & -11 & -2 \end{vmatrix} \\ & \begin{array}{l} 2c_2 + c_1 \rightarrow c_1 \\ -c_2 + c_3 \rightarrow c_3 \\ \hline \end{array} \begin{vmatrix} 0 & -1 & 0 \\ 12 & 4 & -12 \\ -27 & -11 & 9 \end{vmatrix} = - \begin{vmatrix} 12 & -12 \\ -27 & 9 \end{vmatrix} \\ & \begin{array}{l} \frac{1}{12}r_1 \rightarrow r_1 \\ \frac{1}{9}r_2 \rightarrow r_2 \\ \hline \end{array} -108 \begin{vmatrix} 1 & -1 \\ -3 & 1 \end{vmatrix} = 216. \end{aligned}$$

(b) Expanding along some column or row with some entries are 0, we have

$$\begin{vmatrix} 2 & 9 & 0 & -3 & 0 \\ 1 & -4 & 0 & 2 & 0 \\ 3 & -2 & 8 & -4 & 5 \\ -6 & 2 & -7 & 3 & 0 \\ 0 & 2 & 0 & 0 & 0 \end{vmatrix} = 5 \begin{vmatrix} 2 & 9 & 0 & -3 \\ 1 & -4 & 0 & 2 \\ -6 & 2 & -7 & 3 \\ 0 & 2 & 0 & 0 \end{vmatrix} = -35 \begin{vmatrix} 2 & 9 & -3 \\ 1 & -4 & 2 \\ 0 & 2 & 0 \end{vmatrix} = 70 \begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix} = 490.$$

(c) We have

$$\begin{vmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 5 & 7 & 9 \\ 1 & 4 & 7 & 10 & 13 \\ 1 & 5 & 9 & 13 & 17 \end{vmatrix} \xrightarrow{r_1 + r_3 \rightarrow r_3} \begin{vmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 6 & 8 & 10 \\ 1 & 4 & 7 & 10 & 13 \\ 1 & 5 & 9 & 13 & 17 \end{vmatrix} \xrightarrow{-2r_2 + r_3 \rightarrow r_3} \begin{vmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 4 & 7 & 10 & 13 \\ 1 & 5 & 9 & 13 & 17 \end{vmatrix}.$$

Since there is a zero row, the determinant is 0.

(d) Adding each row 2, row 3, $\dots$, row n to row 1, we obtain

$$\det A = \begin{vmatrix} n+1 & n+1 & n+1 & \cdots & n+1 \\ 1 & 2 & 1 & \cdots & 1 \\ 1 & 1 & 2 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 2 \end{vmatrix} = (n+1) \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & \cdots & 1 \\ 1 & 1 & 2 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 2 \end{vmatrix}.$$

Next, we subtract the first row from each of the other rows. This implies

$$\det A = (n+1) \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{vmatrix} = n+1,$$

since the last matrix is a triangular matrix with determinant of 1.

(e) For every column from the second column, subtract x_1 times the previous column.

$$V \rightarrow \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & x_2 - x_1 & x_2^2 - x_1x_2 & \cdots & x_2^{n-1} - x_1x_2^{n-2} \\ 1 & x_3 - x_1 & x_3^2 - x_1x_3 & \cdots & x_3^{n-1} - x_1x_3^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n - x_1 & x_n^2 - x_1x_n & \cdots & x_n^{n-1} - x_1x_n^{n-2} \end{bmatrix} := \tilde{V}$$

then

$$\det V = \det \tilde{V} = \det \bar{V},$$

where

$$\bar{V} = \begin{bmatrix} x_2 - x_1 & x_2^2 - x_1x_2 & \cdots & x_2^{n-1} - x_1x_2^{n-2} \\ x_3 - x_1 & x_3^2 - x_1x_3 & \cdots & x_3^{n-1} - x_1x_3^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ x_n - x_1 & x_n^2 - x_1x_n & \cdots & x_n^{n-1} - x_1x_n^{n-2} \end{bmatrix}.$$

Note that

$$\det \bar{V} = \prod_{i=2}^n (x_i - x_1) \cdot \det V_{n-1},$$

where

$$V_{n-1} = \begin{bmatrix} 1 & x_2 & x_2^2 & \cdots & x_2^{n-2} \\ 1 & x_3 & x_3^2 & \cdots & x_3^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-2} \end{bmatrix}.$$

Similarly, we have

$$\det V_{n-1} = \prod_{i=3}^n (x_i - x_1) \cdot \det V_{n-2},$$

where

$$V_{n-2} = \begin{bmatrix} 1 & x_3 & x_3^2 & \cdots & x_3^{n-3} \\ 1 & x_4 & x_4^2 & \cdots & x_4^{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-3} \end{bmatrix}.$$

Moreover,

$$\det V_2 = (x_n - x_{n-1}) \cdot \det V_1$$

and

$$\det V_1 = 1.$$

Therefore, we have

$$\det V = \prod_{a \leq i < j \leq n} (x_j - x_i).$$

5.7 Chapter 7: Vector Spaces

1. Let V and W be subspaces of $\mathbb{R}^n$. Which of the following must be a subspace of $\mathbb{R}^n$?

- (a) $V \cup W = \{\mathbf{v} : \mathbf{v} \in V \text{ or } \mathbf{v} \in W\}$
- (b) $V \cap W = \{\mathbf{v} : \mathbf{v} \in V \text{ and } \mathbf{v} \in W\}$
- (c) $V + W = \{\mathbf{v} + \mathbf{w} : \mathbf{v} \in V, \mathbf{w} \in W\}$

Suggested solution:

(a) It may not be a subspace. For example, consider the subspace $V = \{(x, 0)^T : x \in \mathbb{R}\}$ and $W = \{(0, y)^T : y \in \mathbb{R}\}$ in $\mathbb{R}^2$. We have $(1, 0)^T, (0, 1)^T \in (V \cup W)$, but their sum $(1, 1)^T$ does not belong to $V \cup W$.

(b) It is a subspace. We check the following conditions:

- Since $\mathbf{0} \in V$ and $\mathbf{0} \in W$, we have $\mathbf{0} \in (V \cap W)$.
- For any $\mathbf{u}, \mathbf{v} \in (V \cap W)$, we have $\mathbf{u}, \mathbf{v} \in V$ and $\mathbf{u}, \mathbf{v} \in W$. Since V and W are subspaces, $\mathbf{u} + \mathbf{v}$ belongs to both V and W . Thus, $\mathbf{u} + \mathbf{v} \in (V \cap W)$.
- For any $\mathbf{u}, \mathbf{v} \in (V \cap W)$ and $c \in \mathbb{R}$, we have $\mathbf{v} \in V$ and $\mathbf{v} \in W$. Since V and W are subspaces, $c\mathbf{v}$ belongs to both V and W . Thus, $c\mathbf{v} \in (V \cap W)$.

(c) It is a subspace. We check the following conditions:

- Since $\mathbf{0} \in V$ and $\mathbf{0} \in W$, we have $\mathbf{0} \in (V + W)$.
- For any $\mathbf{x}, \mathbf{y} \in (V + W)$, we can write $\mathbf{x} = \mathbf{v} + \mathbf{w}$ and $\mathbf{y} = \mathbf{v}' + \mathbf{w}'$ such that $\mathbf{v}, \mathbf{v}' \in V$ and $\mathbf{w}, \mathbf{w}' \in W$. Note that $\mathbf{v} + \mathbf{v}' \in V$ and $\mathbf{w} + \mathbf{w}' \in W$. So

$$\mathbf{x} + \mathbf{y} = (\mathbf{v} + \mathbf{v}') + (\mathbf{w} + \mathbf{w}') \in (V + W).$$

- For any $\mathbf{x} \in (V + W)$ and $c \in \mathbb{R}$, we can write $\mathbf{x} = \mathbf{v} + \mathbf{w}$ such that $\mathbf{v} \in V$ and $\mathbf{w} \in W$. Note that $c\mathbf{v} \in V$ and $c\mathbf{w} \in W$, so

$$c\mathbf{x} = c\mathbf{v} + c\mathbf{w} \in (V + W).$$

2. (a) Let A and B be matrices. Show that there exists a matrix C for which $A = BC$ if and only if $\text{Col } A$ is a subset of $\text{Col } B$.
- (b) Let X be an $m \times n$ matrix with $m \neq n$. Using part (a), prove that there does not exist a matrix Y such that $XY = I_m$ and $YX = I_n$.

Suggested solution:

(a) Suppose $A = BC$ for some C . For any $\mathbf{v} \in \text{Col } A$, we have $\mathbf{v} = A\mathbf{x}$ for some $\mathbf{x}$. Thus, we have

$$\mathbf{v} = A\mathbf{x} = (BC)\mathbf{x} = B(C\mathbf{x}) \in \text{Col } B,$$

This implies $\text{Col } A \subset \text{Col } B$.

Conversely, suppose $\text{Col } A \subset \text{Col } B$. Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ be the columns of A . As each $\mathbf{a}_j \in \text{Col } A$, we have $\mathbf{a}_j \in \text{Col } B$. Thus, $\mathbf{a}_j = B\mathbf{c}_j$ for some $\mathbf{c}_j$. Define $C = [\mathbf{c}_1 \ \mathbf{c}_2 \ \dots \ \mathbf{c}_n]$. Then

$$BC = [B\mathbf{c}_1 \ B\mathbf{c}_2 \ \dots \ B\mathbf{c}_n] = [\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n] = A.$$

(b) Without loss of generality, we assume $m > n$. If $XY = I_m$, then $\mathbb{R}^m = \text{Col } I_m \subset \text{Col } X$ by part (a). This shows the columns of X generate $\mathbb{R}^m$. But this is impossible since X only contains $n < m$ columns. So $XY \neq I_m$ for any Y .

3. Let V and W be two subspaces of $\mathbb{R}^{m+n}$ such that $\dim V = m, \dim W = n$, and $V \cap W = \{\mathbf{0}\}$. Let β and γ be bases for V and W respectively. Prove that $\beta \cup \gamma$ is a basis for $\mathbb{R}^{m+n}$.

Suggested solution:

Note that $\beta \cap \gamma = \emptyset$ since $V \cap W = \{\mathbf{0}\}$ (and $\mathbf{0}$ cannot belong to β or γ). So $\beta \cup \gamma$ contains $m + n$ vectors in total. If we can prove that $\beta \cup \gamma$ is linearly independent, then it must be a basis for $\mathbb{R}^{m+n}$.

Let $\beta = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ and $\gamma = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. Consider

$$a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + \dots + a_m\mathbf{u}_m + b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \dots + b_n\mathbf{v}_n = \mathbf{0}.$$

We can rewrite this as

$$a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + \dots + a_m\mathbf{u}_m = -(b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \dots + b_n\mathbf{v}_n).$$

Note that the left-hand side belongs to V , while the right-hand side belongs to W . As $V \cap W = \{\mathbf{0}\}$, both sides are $\mathbf{0}$. As β and γ are linearly independent, this implies all a_i 's and b_j 's are 0. Thus, $\beta \cup \gamma$ is linearly independent, and we are done.

4. Let V be a vector space with dimension k and let $T : V \rightarrow V$ a linear transformation such that $T(2v_1 - 3v_2) = 5v_1 + 3v_2$ and $T(-3v_1 + 5v_2) = 4v_1 + 3v_2$ for some vectors $v_1, v_2 \in V$. Given that v_1 and v_2 are linearly independent, write $T(v_1)$ and $T(v_2)$ in terms of v_1 and v_2 .

Suggested solution:

Suppose A is the standard matrix of the linear transformation T , then A is a $k \times k$ matrix. We know that

$$A \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix} = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 3 & 3 \end{bmatrix},$$

and thus

$$A \begin{bmatrix} v_1 & v_2 \end{bmatrix} = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix}^{-1} = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} 37 & 23 \\ 24 & 15 \end{bmatrix}.$$

So $T(v_1) = 37v_1 + 24v_2$, $T(v_2) = 23v_1 + 15v_2$.

5. Let $\beta = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an ordered basis for $\mathbb{R}^n$. Suppose another ordered basis for $\mathbb{R}^n$ is $\gamma = \{\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2, \dots, \mathbf{v}_1 + \mathbf{v}_2 + \dots + \mathbf{v}_n\}$. If $\mathbf{v} \in \mathbb{R}^n$ satisfies $[\mathbf{v}]_\beta = (c_1, c_2, \dots, c_n)^T$, find $[\mathbf{v}]_\gamma$.

Suggested solution:

Since $[\mathbf{v}]_\beta = (c_1, c_2, \dots, c_n)^T$, we have $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n$. Now we have

$$\begin{aligned} \mathbf{v} &= c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n \\ &= (c_1 - c_2)\mathbf{v}_1 + (c_2 - c_3)(\mathbf{v}_1 + \mathbf{v}_2) + \dots + (c_{n-1} - c_n)(\mathbf{v}_1 + \mathbf{v}_2 + \dots + \mathbf{v}_{n-1}) + c_n(\mathbf{v}_1 + \mathbf{v}_2 + \dots + \mathbf{v}_n). \end{aligned}$$

Therefore, $[\mathbf{v}]_\gamma = (c_1 - c_2, c_2 - c_3, \dots, c_{n-1} - c_n, c_n)^T$.

6. Let β be an ordered basis for $\mathbb{R}^n$, and let $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \in \mathbb{R}^n$. Prove that $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are linearly independent if and only if $[\mathbf{u}_1]_\beta, [\mathbf{u}_2]_\beta, \dots, [\mathbf{u}_k]_\beta$ are linearly independent.

Suggested solution:

Let B be the matrix whose columns are the vectors in β . We have

$$\begin{aligned} c_1[\mathbf{u}_1]_\beta + c_2[\mathbf{u}_2]_\beta + \dots + c_k[\mathbf{u}_k]_\beta &= \mathbf{0} \\ \Leftrightarrow c_1(B^{-1}\mathbf{u}_1) + c_2(B^{-1}\mathbf{u}_2) + \dots + c_k(B^{-1}\mathbf{u}_k) &= \mathbf{0} \\ \Leftrightarrow B^{-1}(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k) &= \mathbf{0} \\ \Leftrightarrow c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k &= \mathbf{0} \end{aligned}$$

Therefore, the first equation only has the trivial solution if and only if the last equation only has the trivial solution. This shows $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are linearly independent if and only if $[\mathbf{u}_1]_\beta, [\mathbf{u}_2]_\beta, \dots, [\mathbf{u}_k]_\beta$ are linearly independent.

7. Determine whether each of the following is true or false.

- (a) If β and γ are ordered bases for $\mathbb{R}^n$ such that $[\mathbf{v}]_\beta = [\mathbf{v}]_\gamma$ for all $\mathbf{v} \in \mathbb{R}^n$, then $\beta = \gamma$.
 (b) Let β and γ be two ordered bases for $\mathbb{R}^n$, and let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation. Then $[T]_\beta$ and $[T]_\gamma$ are similar.

Suggested solution:

- (a) True. Let $\mathbf{v}_k$ be the k th vector in β . Then $[\mathbf{v}_k]_\beta = \mathbf{e}_k$. By the given condition, we also have $[\mathbf{v}_k]_\gamma = \mathbf{e}_k$, which means k th vector in γ is $\mathbf{v}_k$. Therefore, $\beta = \gamma$.
 (b) True. Let B and C be the matrices whose columns are the vectors in β and γ respectively, and let A be the standard matrix of T . By assignment 7 Q4(b), we have $[T]_\beta = B^{-1}AB$ and $[T]_\gamma = C^{-1}AC$. Therefore,

$$[T]_\beta = B^{-1}AB = B^{-1}(C[T]_\gamma C^{-1})B = (B^{-1}C)[T]_\gamma(B^{-1}C)^{-1}.$$

This implies $[T]_\beta$ and $[T]_\gamma$ are similar.

5.8 Chapter 8: Eigenvalues and Eigenvectors

- Let J be the $n \times n$ matrix whose every entry is 1. Find all eigenvalues and eigenvectors of J , and determine if J is diagonalisable.

Suggested solution:

Consider the equation $J\mathbf{v} = \lambda\mathbf{v}$ where $\mathbf{v} = (v_1, v_2, \dots, v_n)^T$. This implies

$$\begin{bmatrix} v_1 + v_1 + \dots + v_n \\ v_1 + v_1 + \dots + v_n \\ \vdots \\ v_1 + v_1 + \dots + v_n \end{bmatrix} = \begin{bmatrix} \lambda v_1 \\ \lambda v_2 \\ \vdots \\ \lambda v_n \end{bmatrix}.$$

Therefore, we need $v_1 + v_1 + \dots + v_n = \lambda v_j$ for each j .

- If $\lambda = 0$, we need $v_1 + v_1 + \dots + v_n = 0$. This allows all nonzero vectors $(v_1, v_2, \dots, v_n)^T$ satisfying $v_1 + v_1 + \dots + v_n = 0$ are eigenvectors corresponding to the eigenvalue 0.
- If $\lambda \neq 0$, we must have $v_1 = v_2 = \dots = v_n$. Then each equation becomes $nv_j = \lambda v_j$, and hence $\lambda = n$. This shows all nonzero vectors $(t, t, \dots, t)^T$ are eigenvectors corresponding to the eigenvalue n .

Note that the eigenspace $E_0 = \{\mathbf{v} : v_1 + v_2 + \dots + v_n = 0\}$ has dimension $n - 1$. It is easy to find n linearly independent eigenvectors, for example,

$$(1, -1, 0, 0, \dots, 0)^T, (1, 0, -1, 0, \dots, 0)^T, \dots, (1, 0, 0, \dots, 0, -1)^T, (1, 1, \dots, 1)^T.$$

Then J is diagonalisable since for each eigenvalue, the algebraic multiplicity and geometric multiplicity are equal.

- Let $A = \begin{bmatrix} -1 & 2 \\ -3 & 6 \end{bmatrix}$.

- Find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.
- Compute A^n where n is a positive integer.

Suggested solution:

- The characteristic polynomial is

$$\begin{vmatrix} -1-t & 2 \\ -3 & 6-t \end{vmatrix} = (-1-t)(6-t) - (-3)2 = t^2 - 5t.$$

The eigenvalues are $\lambda = 0, 5$.

For the eigenvalue $\lambda = 0$, we solve $A\mathbf{x} = \mathbf{0}$, i.e. $\begin{bmatrix} -1 & 2 \\ -3 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0}$. A solution is $(x, y) = (2, 1)$.

For the eigenvalue $\lambda = 5$, we solve $(A - 5I)\mathbf{x} = \mathbf{0}$, i.e. $\begin{bmatrix} -6 & 2 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0}$. A solution is $(x, y) = (1, 3)$.

Let $P = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and $D = \begin{bmatrix} 0 & 0 \\ 0 & 5 \end{bmatrix}$. Then we have $A = PDP^{-1}$.

- We have

$$\begin{aligned} A^n &= PD^nP^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 5^n \end{bmatrix} \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} 0 & 5^n \\ 0 & 3 \cdot 5^n \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -5^{n-1} & 2 \cdot 5^{n-1} \\ -3 \cdot 5^{n-1} & 6 \cdot 5^{n-1} \end{bmatrix}. \end{aligned}$$

- Let A and B be diagonalisable matrices. Which of the following matrices must be diagonalisable?

- A^3
- $A + B$

(c) AB

Suggested solution:

(a) Yes. If $A = PDP^{-1}$ where P is invertible and D is diagonal, then $A^3 = PD^3P^{-1}$ where D^3 is diagonal.

(b) No. Both $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$ are diagonalisable as their algebraic multiplicities of all eigenvalues are 1. However, $A + B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is not diagonalisable.

(c) No. Both $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$ are diagonalisable as their multiplicities of all eigenvalues are 1. However, $AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is not diagonalisable.

4. Suppose that A, B are $n \times n$ matrices and diagonalisable, and A has distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Prove that $AB = BA$ if and only if A and B can be simultaneously diagonalized, i.e., there exists an invertible matrix P such that $P^{-1}AP$ and $P^{-1}BP$ are both diagonal.

Suggested solution:

If A and B can be simultaneously diagonalized, we have $D_1 = P^{-1}AP, D_2 = P^{-1}BP$ for some invertible matrix P and diagonal matrices D_1 and D_2 . Then we have $A = PD_1P^{-1}$ and $B = PD_2P^{-1}$. Then $AB = (PD_1P^{-1})(PD_2P^{-1}) = PD_1D_2P^{-1}, BA = (PD_2P^{-1})(PD_1P^{-1}) = PD_2D_1P^{-1}$. Since D_1 and D_2 are diagonal matrices, then we have $D_1D_2 = D_2D_1$. So $AB = BA$.

If $AB = BA$, we try to show that if $P^{-1}AP$ is diagonal for some invertible matrix P , then $P^{-1}BP$ is also diagonal.

Method 1: If $A\mathbf{x} = \lambda\mathbf{x}$ for some eigenvalue λ and eigenvector $\mathbf{x}$ of A ($\mathbf{x}$ is a column vector of P), then $AB\mathbf{x} = BA\mathbf{x} = B(\lambda\mathbf{x}) = \lambda(B\mathbf{x})$. If $B\mathbf{x} \neq \mathbf{0}$, the $B\mathbf{x}$ is an eigenvector of A , and $B\mathbf{x} \in E_\lambda$. Since A has n distinct eigenvalues and is diagonalisable, we have $E_{\lambda_i} \cap E_{\lambda_j} = \{\mathbf{0}\}$ if $i \neq j$, and $\dim E_{\lambda_i} = 1$ for any $i = 1, 2, \dots, n$. So we have $B\mathbf{x} = \mathbf{0}$ or $B\mathbf{x} = \beta\mathbf{x}$ for some $\beta \neq 0$. Then $B\mathbf{x} = \beta\mathbf{x}$ for some $\beta \in \mathbb{R}$. So $\mathbf{x}$ is also an eigenvector of B , hence, $P^{-1}BP$ is also diagonal.

Method 2: Suppose $P^{-1}AP = D$ and D is diagonal, $P^{-1}BP = C$ for some matrix C . Then

$$P^{-1}ABP = P^{-1}AP \cdot P^{-1}BP = D \cdot C$$

$$P^{-1}BAP = P^{-1}BP \cdot P^{-1}AP = C \cdot D.$$

Here, $D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$, and $\lambda_i \neq \lambda_j$ for $i \neq j$. Then

$$(D \cdot C)_{ij} = \lambda_i \cdot C_{ij}$$

$$(C \cdot D)_{ij} = \lambda_j \cdot C_{ij}$$

and $\lambda_i \cdot C_{ij} = \lambda_j \cdot C_{ij}$. If $i \neq j$, we have $C_{ij} = 0$, so C is a diagonal matrix.

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